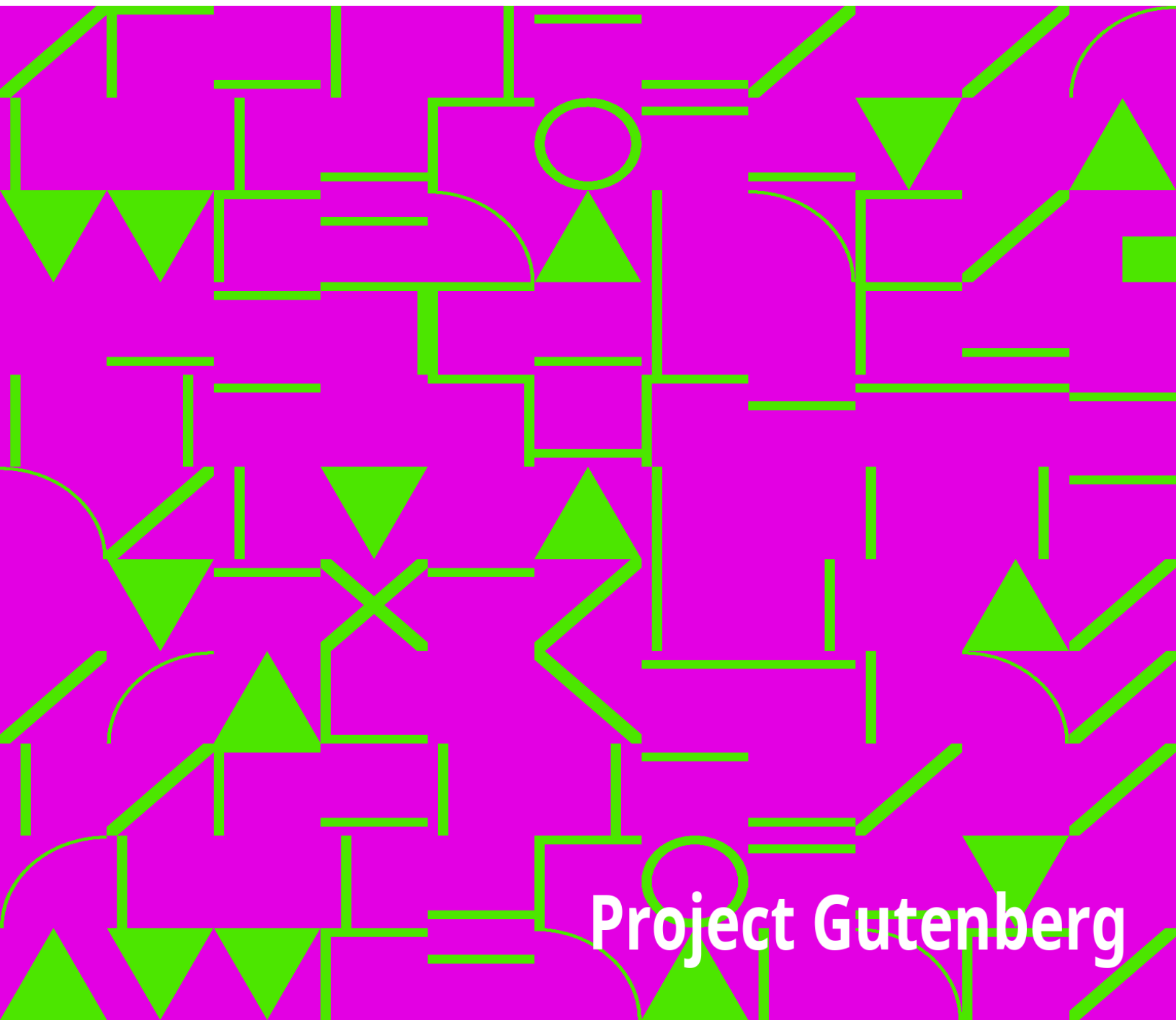


Northern Nut Growers Association, Report of the Proceedings at the Seventh Annual Meeting

Northern Nut Growers Association



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Title: Northern Nut Growers Association, Report of the Proceedings at the Seventh Annual Meeting

Editor: Northern Nut Growers Association

Release date: May 25, 2008 [eBook #25597]

Most recently updated: January 3, 2021

Language: English

Other information and formats: www.gutenberg.org/ebooks/25597

Credits: Produced by Marilynda Fraser-Cunliffe and the Online Distributed Proofreading Team at <https://www.pgdp.net>

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THE SEVENTH ANNUAL MEETING ***

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NORTHERN NUT GROWERS ASSOCIATION

REPORT OF THE PROCEEDINGS AT THE SEVENTH ANNUAL MEETING

WASHINGTON, D. C. SEPTEMBER 8 AND 9, 1916.

PRESS OF The Advertiser-republican, ANNAPOLIS, MD.

CONTENTS

PAGE

Officers and Committees of the Association	4
Members of the Association	5
Constitution and By-Laws	10
Proceedings of the Seventh Annual Meeting	13
Report of the Secretary-Treasurer	14
Notes on the Chinquapins, Dr. Robert T. Morris, New York	15
The Black Walnut, T. P. Littlepage, Washington, D. C.	25
Discussion on the Almond	33
Discussion on the Hazel	37
The Chestnut Bark Disease, Dr. Haven Metcalf, Washington, D. C.	41
Discussion on Quarantine for Chestnut Nursery Stock	49
Hybrids and Other New Chestnuts for Blight Districts, Dr. Walter Van Fleet, Washington, D. C.	54

President's Address, Dr. J. Russell Smith, Roundhill, Va.	58
Diseases of the Persian Walnut, S. M. McMurrin, Washington, D. C.	67
Discussion on Winter Killing	72
Address of Col. C. A. Van Duzee, Cairo, Georgia	75
Resolutions on Chestnut Blight Quarantine	80
Resolution on Investigations in Nut Tree Propagation	84
Discussion on the Growth and Fruiting of Pecans in the North	86
Top Working Pecans on Other Hickories	91
Appendix:	
Letter from W. C. Reed, Vice-President	98
The Food Value of Nuts, Dr. J. H. Kellogg, Battle Creek, Mich.	101
Letter from J. C. Cooper, McMinnville, Oregon	114
List of those present at the meeting	117

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Canada G. H. Corsan 63 Avenue Road, Toronto

Connecticut Charles H. Plump West Redding

Delaware E. R. Angst 527 Dupont Building, Wilmington

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Illinois H. A. Riehl Alton

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Kentucky A. L. Moseley Calhoun

Maryland C. P. Close College Park

Massachusetts James II. Bowditch 903 Tremont Building, Boston

Michigan. Miss Maude M. Jessup 440 Thomas St., Grand Rapids

Minnesota L. L. Powers 1018 Hudson Ave., St. Paul

Missouri P. C. Stark Louisiana

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North Carolina W. N. Hutt Raleigh

Ohio Harry R. Weber 601 Gerke Building, Cincinnati

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Johnson, Chet, R. D. 1, Biggs

Tucker, T. C., Manager California Almond Growers' Exchange,
311 California St., San Francisco

CANADA

Corsan, G. H., University of Toronto

Dufresne, Dr. A. A., 1872 Cartier St., Montreal

Sager, Dr. D. S., Brantford

CONNECTICUT

Barnes, John R., Yalesville

Deming, Dr. W. C., Georgetown

Deming, Mrs. W. C., Georgetown.

Goodwin, James L., Box 447, Hartford

Hungerford, Newman, Torrington, R. 2, Box 76, for circulars, Box 1082,
Hartford, for letters

Ives, Ernest M., Sterling Orchards, Meriden

Lay, Charles Downing, Wellesmere, Stratford

Lewis, Henry Leroy, Stratford

Mikkelsen, Mrs. M. A., Georgetown

*Morris, Dr. Robert T., Cos Cob, R. 28, Box 95

Plump, Charles II., West Redding

Sessions, Albert L., Bristol

Staunton, Gray, R. D. 30, Stamford

White, Gerrard, North Granby

Williams, W. W., Milldale

DELAWARE

Augst, E. R., 527 DuPont Building, Wilmington, Del.

Lord, George Frank, care of DuPont Powder Company, Wilmington

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Goddard, R. H., States' Relations Service, Washington

*Littlepage, T. P., Union Trust Building, Washington

Reed, C. A., Nut Culturist, Department of Agriculture, Washington

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Bullard, Wm. P., Albany

Van Duzee, C. A., Judson Orchard Farm, Cairo

Wight, J. B., Cairo

ILLINOIS

Casper, O. II., Anna

Poll, Carl J, 1009 Maple St., Danville

Potter, Hon. W. O., Marion

Riehl, E. A., Alton

INDIANA

Hutchings, Miss Lida G., 118 Third St., Madison

Lukens, Mrs. B., Anderson

Reed, M. P., Vincennes
Reed, W. C., Vincennes
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Woolbright, Clarence, R. D. 3, Elnora

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KENTUCKY

Matthews, Prof. C. W., Horticulturist, State Agricultural Station,
Lexington
Moseley, A. L., Bank of Calhoun, Calhoun

MARYLAND

Campbell, George D., Lonaconing
Darby, R. U., Suite 804, Continental Building, Baltimore
Hayden, Chas. S., 200 E. Lexington St., Baltimore
Keenan, Dr. John N., Brentwood
King, W. J., 232 Prince George St., Annapolis
Kyner, James H., Bladensburg
Littlepage, Miss Louise, Bowie
Murray, Miss Annie C., Cumberstone
Stabler, Henry, Hancock
White, Paul, Bowie

MASSACHUSETTS

*Bowditch, James II., 903 Tremont Building, Boston
Cleaver, C. Leroy, Hingham Center
Cole, Mrs. George B., 15 Mystic Ave., Winchester
Hoffman, Bernhard, Overbrook Orchard, Stockbridge
Smith, Fred A., 39 Pine St., Danvers

Vaughan, Horace A., Peacehaven, Assonet
White, Warren, Holliston

MICHIGAN

Copland, Alexander W., Strawberry Hill Farm, Birmingham
Jessup, Miss Maud M., 440 Thomas St., Grand Rapids
Johnson, Franklin, Munising
Kellogg, J. H., Battle Creek
Staunton, Gray, Muskegon, Box 233

MINNESOTA

Powers, L. L., 1018 Hudson Ave., St. Paul

MISSOURI

Bauman, X. C., Ste. Genevieve
Darche, J. H., Parkville
Funston, E. S., 1521 Morgan St., St. Louis
Phelps, Howe, Pine Hurst Dairy, Carthage
Stark, P. C., Louisiana (Mo.)

NEBRASKA

Kurtz, John W., 5304 Bedford St., Omaha

NEW JERSEY

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Childs, Fred., Morristown, R. D. 2
Jaques, Lee W., 74 Waverly St., Jersey City Heights
Lovett, J. T., Little Silver
Marston, Edwin S., Florham Park, Box 72
Mechling, Edward A., Wonderland Farm, Moorestown
Ridgeway, C. S. Floralia, Lumberton, N. J.

Roberts, Horace, Moorestown
Young, Frederick C., Palmyra, Box 335

NEW YORK

Abbott, Frederick B., 419 Ninth St., Brooklyn
Atwater, C. G., The Barrett Co., 17 Battery Place, New York City
Baker, Dr. Hugh P., Dean of State College of Forestry, Syracuse
Baker, Prof. J. Fred, Director of Forest Investigations, State College
of Forestry, Syracuse
Baker, Wm. A., North Rose
Bixby, Willard G., 46th St. and 2nd Ave., Brooklyn
Brown, Ronald J., 320 Broadway, New York City
Ellwanger, Mrs. W. D., 510 East Ave., Rochester
Fullerton, H. B., Director Long Island Railroad Experiment Station,
Medford, L. I.
Haywood, Albert, Flushing
Hickox, Ralph, 3832 White Plains Ave., New York City
Holden, E. B., Hilton
*Huntington, A. M., 15 W. 81st St., New York City
Jackson, Dr. James H., Dansville
Loomis, C. B., East Greenbush
Miller, Milton R., Batavia, Box 394
Morse, Geo. A., Fruit Acres, Williamson, N. Y.
Nelson, Dr. James Robert, 23 Main St., Kingston-on-Hudson
Olcott, Ralph T., Ellwanger & Barry Building, Rochester
Palmer, A. C., New York Military Academy, Cornwall-on-Hudson
Pannell, W. B., Pittsford
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Stuart, C. W., Newark

Teele, A. W., 30 Broad St., New York City
Teter, Walter C., 10 Wall St., New York City
Thomson, Adelbert, East Avon
Tuckerman, Bayard, 118 E. 37th St., New York City
Ulman, Dr. Ira, 213 W. 147th St., New York City
Wile, M. E., 37 Calumet St., Rochester
Williams, Dr. Charles Mallory, 48 E. 49th St., New York City
*Wissman, Mrs. F. de R., Westchester, New York City

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Van Lindley, J., J. Van Lindley Nursery Company, Pomona
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OHIO

Dayton, J. H., Storrs & Harrison Company, Painesville
Evans, Miss Myrta L., Briallen Farm, Oak Hill, Jackson County
Miller, H. A., Gypsum
Thorne, Charles E., Wooster, Agric. Exp. Sta.
Weber, Harry E., 601 Gerke Building, Cincinnati
Yunck, E. G., 710 Central Ave., Sandusky

PENNSYLVANIA

Druckemiller, W. C., Sunbury
Fagan, Prof. P. N., Department of Horticulture, State College
Grubbs, H. L., Fairview, R. 1
Hall, Robt. W., 133 Church St., Bethlehem
Harshman, U. W., Waynesboro
Heffner, H., Highland Chestnut Grove, Leeper
Hile, Anthony, Curwensville National Bank, Curwensville

Hoopes, Wilmer W., Hoopes Brothers and Thomas Company, Westchester
Hutchinson, Mahlon, Ashwood Farm, Devon, Chester County
Jenkins, Charles Francis, Farm Journal, Philadelphia

*Jones, J. P., Lancaster, Box 527

Kaufman, M. M., Clarion

Leas, F. C., 882 Drexel Building, Philadelphia, Mountain Brook Orchard
Company, Salem, Va.

Middleton, Fenton H., 1118 Chestnut St., Philadelphia

Murphy, P. J., Vice-President L. & W. R. R. R. Company, Scranton

O'Neill, Wm. C., 1328 Walnut St., Philadelphia

Rheam, J. F., 45 N. Walnut St., Lewistown

Rick, John, 438 Pennsylvania Sq., Reading

Rife, Jacob A., Camp Hill

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Webster, Mrs. Edmund, 1324 S. Broad St., Philadelphia

*Wister, John C, Wister St. and Clarkson Ave., Germantown

Wright, R. P., 235 W. 6th St., Erie

SOUTH CAROLINA

Shanklin, Prof. A. G., Clemson College

TENNESSEE

Marr, Thomas S., 701 Stahlman Building, Nashville

TEXAS

Burkett, J. H., Nut Specialist, State Dept, of Agric., Clyde

Trumbull, R. S., Agricultural Agent, El Paso & S. W. System, Morenci
Southern Railroad Company, El Paso

VIRGINIA

Crockett, E. B., Monroe

Engleby, Thos. L., 1002 Patterson Ave., Roanoke

Lee, Lawrence R., Leesburg

Miller, L. O., Miller & Rhodes, Richmond

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* Life member.

CONSTITUTION

ARTICLE I

Name. This society shall be known as the NORTHERN NUT GROWERS ASSOCIATION.

ARTICLE II

Object. Its object shall be the promotion of interest in nut-bearing plants, their products and their culture.

ARTICLE III

Membership. Membership in the society shall be open to all persons who desire to further nut culture, without reference to place of residence or nationality, subject to the rules and regulations of the committee on membership.

ARTICLE IV

Officers. There shall be a president, a vice-president and a secretary-treasurer, who shall be elected by ballot at the annual meeting; and an executive committee of five persons, of which the president, two last retiring presidents, vice-president and secretary-treasurer shall be members. There shall be a state vice-president from each state, dependency or country

represented in the membership of the association, who shall be appointed by the president.

ARTICLE V

Election of Officers. A committee of five members shall be elected at the annual meeting for the purpose of nominating officers for the following year.

ARTICLE VI

Meetings. The place and time of the annual meeting shall be selected by the membership in session or, in the event of no selection being made at this time, the executive committee shall choose the place and time for the holding of the annual convention. Such other meetings as may seem desirable may be called by the president and executive committee.

ARTICLE VII

Quorum. Ten members of the association shall constitute a quorum, but must include a majority of the executive committee or two of the three elected officers.

ARTICLE VIII

Amendments. This constitution may be amended by a two-thirds vote of the members present at any annual meeting, notice of such amendment having been read at the previous annual meeting, or a copy of the proposed amendment having been mailed by any member to each member thirty days before the date of the annual meeting.

BY-LAWS

ARTICLE I

Committees. The association shall appoint standing committees as follows: On membership, on finance, on programme, on press and publication, on nomenclature, on promising seedlings, on hybrids, and an auditing committee. The committee on membership may make recommendations to the association as to the discipline or expulsion of any member.

ARTICLE II

Fees. The fees shall be of two kinds, annual and life. The former shall be two dollars, the latter twenty dollars.

ARTICLE III

Membership. All annual memberships shall begin with the first day of the calendar quarter following the date of joining the association.

ARTICLE IV

Amendments. By-laws may be amended by a two-thirds vote of members present at any annual meeting.

Northern Nut Growers Association

SEVENTH ANNUAL MEETING

SEPTEMBER 8 AND 9, 1916

WASHINGTON, D. C.

The seventh annual meeting of the Northern Nut Growers Association was called to order in rooms 42-43 of the new building of the National Museum at Washington, D. C., on Friday, September 8th, at 10 a. m., the president, Dr. J. Russell Smith, presiding.

THE PRESIDENT: It is often customary to start meetings of this sort with a considerable amount of eloquence, such as an address of welcome by some high city or state official, a response to the address of welcome by some one else high in authority, and so on, during which the visitors are told of the many privileges they may enjoy, "the keys of the town" are handed over to them, and a good deal of high-flown oratory is indulged in. We suppose that the people in attendance at this meeting are so well acquainted with Washington that those preliminaries are unnecessary, and I have been informed by the members of the local committee that we can dispense with the frills in this case and proceed with the business of the meeting, which we think is going to rather crowd our time if we get said all that we want to say. We are going to devote this morning's programme first to a paper by Dr. Robert T. Morris on the chinquapin, and then to the discussion of a comparatively newly considered member of our nut family, namely, the American black walnut. We have been heretofore much interested in sundry exotics and talking far too little about this great tree nearer home.

Before taking up the technical programme we have a few matters of business to put through. First, we will have the report of the secretary and treasurer.

REPORT OF THE SECRETARY-TREASURER

Balance on hand date of last report \$ 140.24

Receipts:

Dues 292.75

Advertisements 21.00

Contributions 5.50

Sale of report 34.75

Contributions for prizes 10.00

Miscellaneous .65

\$504.89

Expenses:

Printing report \$ 142.56

Envelopes for report 9.00

Miscellaneous printing 32.50

Postage and stationery 49.26

Stenographer 26.35

Express and freight 2.77

Prizes 18.00

Checks, J. R. S. expenses and circulars 180.00

Lantern operator 3.00

Litchfield Savings Society 20.00

\$483.44

Balance on hand \$21.45

Receipts from all sources, except sale of reports, have fallen off markedly, as have new members, 31 less than last year, though we have now 154 paid up members, one more than last year. 10 members have resigned and 42 have been dropped for non-payment of dues. We have lost one member by death, Herbert R. Orr, of Washington.

The committees on membership and on finance should be more active.

Our annual report constitutes the minutes of the last meeting. Our nut contest and other matters of interest have been reported through the columns of the American Nut Journal, our official organ.

[Accepted.]

THE PRESIDENT: Next in order of business is the first step toward the election of officers for the ensuing year. It is our custom to have a nominating committee elected at an early session. They deliberate and bring forward a slate which is voted on at a later session. This morning is a suitable time for the election of a committee, and tomorrow morning will be a suitable time for their report. Are there any nominations for the Nominating Committee?

MR. M. P. REED: Mr. President, I move that Dr. Morris, Mr. C. P. Close, Mr. C. A. Reed, Prof. Stabler and Dr. Ira Ulman be appointed as the Nominating Committee.

THE PRESIDENT: Are there any other nominations?

MR. C. A. REED: Mr. President, I would like to ask that Mr. Littlepage's name replace my name on that committee.

THE PRESIDENT: Will the nominating member accept that amendment?

MR. M. P. REED: Yes, sir.

THE PRESIDENT: Are there any other nominations? Do I hear a second to the nominations?

A MEMBER: Second it.

[Carried.]

THE PRESIDENT: Are there any other committees to report at this time?

THE SECRETARY: There is a Committee on Incorporation.

MR. T. P. LITTLEPAGE: Mr. President, the Committee on Incorporation has done some investigating as to the desirability of incorporating the Association, and also, if desirable, under what laws, but that committee has not yet made any final report nor come to any final conclusion, and I would suggest, as a member of the committee, that the committee be continued and instructed to report the following year.

THE PRESIDENT: I think that it is unnecessary to vote on the continuance of the committee, as it was appointed with indefinite tenure. We will proceed with the programme and first have the pleasure of listening to Dr. Morris.

NOTES ON THE CHINQUAPINS.

DR. ROBERT T. MORRIS, NEW YORK

According to Sargent the chinquapin (*Castanea pumila*) occupies dry sandy ridges, rich hillsides and the borders of swamps from southern Pennsylvania to northern Florida and the valley of the Neches River in Texas. He states that this chestnut is usually shrubby in the region east of the Alleghany Mountains, and assuming the tree form west of the Mississippi River. Most abundant and of largest size in southern Arkansas and eastern Texas.

Curiously enough there are chinquapins also in northeastern Asia which occur as understudies of the larger chestnuts, very much as they do in America.

The indigenous range of the chinquapin in America is limited northward by a plan of nature for checking distribution of the species. This plan is manifested in a habit which the nuts have of sprouting immediately upon falling in the early autumn. They proceed busily to make a tap root which may become several inches in length before frost calls a halt. In the north where the warm season is not long enough to allow the autumn sprout to lignify sufficiently for bearing the rigors of winter it is killed. If we protect the small autumn plants, or if we transplant older seedlings from their natural habitat, they may be grown easily far north of their indigenous

range. Thrifty chinquapins are happy in the Arnold Arboretum at Jamaica Plain in Massachusetts, and no one knows but they might be cultivated in Nova Scotia and Minnesota.

The American chinquapin is one of the many beautiful and valuable plants which have not as yet been taken up by horticulturists for extensive development. It promises to become one of our important sources of food supply for tomorrow. If we were to develop all of our plant resources at once it would be an unkindness to the horticulturists of two thousand years from now, who would be left moping around with nothing to do. Chinquapin nuts borne in heavy profusion by the plants are delicious in quality, but usually too small to attract customers aside from the wood folk. The wood of the chinquapin of tree form (*C. pumila* var. *arboriformis*) is valuable for purposes to which wood of the common American chestnut is put, and some of the tree chinquapins acquire an earned increment of two or three feet diameter of trunk, and a height of more than fifty feet. The bush chinquapin on the other hand feels rather exclusive when attaining a height of as much as fifteen feet.

I present for inspection a freshly cut branch from an ordinary bush chinquapin, loaded with burs, indicating the prolific nature of the variety. The nuts in this particular specimen are small. The next branch exhibited is from a similar bush, but with nuts quite as large as those of the average common chestnut. The horticulturist has only to graft or bud his ordinary run of chinquapin stocks from some one bush which bears large nuts, and he will then have a valuable graded market product. The larger the nut the less prolific the plant is a rule which holds good with the fruiting of almost any plant.

Look at this branch from a tree chinquapin. It is not remarkable in any way, but the leaves seem to be a little larger than those of the bush

chinquapin. My tree chinquapins came from Stark's nursery in Missouri. The first two which came into bearing had nuts quite as large as those of the common chestnut and I imagined that a discovery of value had been made, but other trees of this variety later bore very small nuts, and all of the tree chinquapin nuts, large and small, were much duller in color than those of the bush chinquapin. My final conclusion is that so far as nuts alone are concerned we may plant and cultivate either the tree variety or the bush variety of the species and then bud or graft any number of stocks from some one plant which bears the best product.

DR. AUGUSTUS STABLER: Is it a somewhat finer grained wood than the ordinary chestnut?

DR. MORRIS: I think it is. All the chestnuts have rather coarse wood. It is strong, hard, durable, and valuable. This chinquapin wood is somewhat coarse grained, but, for comparison with the American chestnut, I don't know. I imagine it is finer grained.

DR. AUGUSTUS STABLER: I know that the chinquapin wood is very much tougher than the American chestnut.

DR. MORRIS: Oh, yes. You cannot break the branches so easily.

Here is a branch from a hybrid between a chinquapin and a common American chestnut (*Castanea dentata*). The leaves and bark, you will observe, are very much like those of the larger parent. The burs are borne singly or in small groups like those of the common chestnut, instead of being crowded in dense clusters like chinquapin burs. There are two or three nuts to the bur, while the chinquapin has normally, but one nut to the bur. This particular hybrid tree showed an interesting peculiarity. During the first two seasons of bearing it had but one nut to the bur, and this was of chinquapin character. In the third year its nuts were still borne singly, but

they were lighter in color than before and oddly corrugated at the base. As the tree became older its chestnut parentage influence pre-dominated, and the tree began to bear two or three nuts to the bur, and more like chestnuts in character, becoming smooth again at the base.

I have a number of hybrids between chinquapins and various species and varieties of other chestnuts, but none of these as yet has produced nuts of marked value. There seems to be a tendency for the coarseness of the larger nuts to prevail in the hybrids, a certain loss of gentility beneath a showy exterior.

The next branch which I present for inspection is from a most beautiful member of the chestnut family, the alder-leaved chestnut (*Castanea alnifolia*). It is classed among the chinquapins in Georgia where the plant is nearly if not quite evergreen. At Stamford it is deciduous very late in the autumn, but sometimes a green leaf will be found in February, where snow or dead leaves on the ground have furnished a protecting covering. The notable value of this species is perhaps in its decorative character for lawns, although the nuts are first rate. The dark green brilliant leaves are striking in appearance, and the shrub is inclined toward a trailing habit, much like that of some of the junipers. This species is one of my pets at Merribrooke, and a perennial source of wonder that nurserymen have not as yet pounced upon it for purposes of exaggeration and misstatement in their annual catalogues.

All of these specimens shown today are from my country place at Stamford, Connecticut, where the mercury in the thermometer leads one to make quotations relating to the Eve of Saint Agnes; five or ten degrees below the zero of Fahrenheit occasionally, and once down to twenty degrees below without injury to any kind of chestnut so far as I could observe.

I cannot make an exhibit of the golden-leaved chinquapin, from the Pacific slope, because tragedy came to all of my little trees of this species, and like most of the Pacific slope plants they are not very joyous in the east. One lot lived through one winter at Merribrooke, but they were the first green things that my cows saw in the springtime, and further comment would be surplus. A single specimen took courage in its root and grew finely until autumn, but it was near a path and somebody pulled it up and left it lying stark naked on the ground. Botanists have recently made two species of the golden-leaved chinquapin, one of the species attaining a height of more than one hundred feet. If horticulturists will secure specimens of *Castanopsis chrysophilla* from the region of Mount Shasta in California I presume that this beautiful evergreen chinquapin may be taught to grow in some of our gardens. It is cultivated in the gardens of temperate Europe. In our north it should be planted close to a running brook, where the roots of young trees can carry water in plenty to the evergreen top while the ground is frozen hard in winter.

Our common chinquapin of the east is perhaps the one that will be cultivated most profitably in the region between the Rocky Mountains and the Atlantic coast. The beauty of freshly picked bush chinquapin nuts is not rivalled by that of any other kind of nut that I have ever seen. The exquisitely polished mahogany color comes out of a light downy cloud near the apex of the nut, dark as midnight for a moment and then shading through glows of lively chestnut until it dawns in a dreamy cream color at the base, with just enough suggestion of green to temper the reds.

If any gourmet with a color soul could serve each one of his friends to a plate of twenty freshly picked bush chinquapins along with two Bennett persimmons, and all resting upon late September leaves of tupelo or of sweet gum the friends would remain and live at his expense while the combination lasted.

Furthermore, the children must always be taken into consideration along with chinquapin questions. According to authorities on the subject of decadence, we do not care very much about the children in these days. If some old-fashioned folks still remain, and if these old-fashioned folks do not take any particular personal interest in the beautiful garden and lawn trees that America has held out towards us in the chinquapins, they may at least plant a few of them because of the social standing that will follow. How so? Well, you see, it's because the parents of elite children will run over for a little visit in order to find out why the children do not come home. Then again, we are kind to dumb animals when raising chinquapins. Squirrels and white-footed mice, crows and blue jays are full of enthusiasm for the nuts, and they will assume the responsibility of gathering the crop if the matter is left in their charge.

This is really a funny country; something of a joke of a country when you come to think of it. Instead of setting out trees that will become both useful and beautiful, in accordance with the old Greek ideal of combining beauty and utility we set out Norway spruces that will make people hate evergreens in general. We set out poplars and all sorts of bunches of leaves in our parks and along the highways, instead of trees still more beautiful that would yield tons of coupon dollars every autumn. *De gustibus non est disputandum!*

When experimenting with hybridization of chinquapins, I ran across a phenomenon of new interest to botanists, and quite accidentally. A number of clusters of pistillate flowers of the bush chinquapin had been covered with paper bags, but not pollenized because of a shortage of pollen. An active man with a good sense of neatness and order would have removed those bags merely for the sake of appearance, but I was lazy and allowed the bags to remain for two or three weeks. When they were finally removed, it was found that the branches had set quite full of fruit, although not so full

as other branches that had been pollenized from oaks. We were evidently dealing with an instance of parthenogenesis. The flowers that had received oak pollen did not show any oak parentage later in their progeny, and it was observed in other experiments in other years that almost any cupuliferous pollen would start cells of the chinquapin ovary into division and into the development of fertile nuts, but without inclusion of the pollen cell in a gamete. For purposes of convenience in thinking I have temporarily called this phenomenon "stereochemic parthenogenesis." Apparently the propinquity of foreign pollen serves to stimulate a female cell into division, although the pollen cell retains fixed molecular identity, and does not fuse with the female cell. I need not bring up abstruse questions of chromatin or of subatomic influence here.

At Stamford the bush chinquapins begin to blossom regularly about the twelfth of June, irrespective of weather conditions. The tree chinquapins blossom a little later, but the alder-leaved chestnut may not blossom until July, later than the common American chestnut. The bush chinquapins begin to open their burs very regularly about the fifteenth of September; earlier than any other chestnuts. They bear at an early age, sometimes in their fifth summer.

Grafting and budding is easily done among all of the chestnuts as a rule, and this year I employed for the first time a large chinquapin bush for top-working with the choice Merribrooke variety of the common chestnut. Every one of the grafts caught, and some of them have grown tremendously. This introduces an interesting question. May we graft the common American chestnut upon bush chinquapin stocks and secure precocious bearing? In that case we shall have trees like the dwarf apple and pear trees that are readily pruned and sprayed.

The chinquapin is practically immune to the blight (*Endothia parasitica*.) Easily blighting varieties of choice American chestnuts may be grafted upon these blight resistant stocks in orchard form if my experiment proves to be a success. It will not lessen the vulnerability of the American chestnut, but dwarf trees will be within reach of the horticulturist's pruning knife and spray outfit. Orchards of fine varieties of the common chestnut may perhaps be maintained in this way until the present epidemic of *Endothia* has expended its protoplasmic energy, or until it has succumbed to microbic parasites of its own.

THE PRESIDENT: Are there any questions to put to Dr. Morris?

THE SECRETARY: I venture to say that a good many people have tried, in the north, to raise the chinquapin, and I would like to have Dr. Morris tell us what to do to get it to grow best, whether to buy the trees from the nurserymen, or to plant the nuts, and just how to do it.

PROF. C. P. CLOSE: I would like to ask Dr. Morris about those chinquapins that set without the application of pollen, whether they fill well and whether they sprout at planting?

MR. LITTLEPAGE: With us out in Maryland it isn't a question of producing the chinquapin; we cut the bushes down every year by the thousands; we have nothing at all against it, but we have found that the weevil has been absolutely unsurmountable with us. It is the only discouraging thing about it in this part of the country. Around Washington the chinquapin is a weed tree, and if you gather a peck of chinquapins you will find that the whole peck, in two weeks, have turned to weevils. Perhaps Dr. Morris can tell us what to do about that, and put us on the road to success.

THE PRESIDENT: I should like to ask Dr. Morris two questions, first, as to the possibility of utilizing the western tree chinquapins as stocks for the larger eastern chinquapins with nuts of chestnut size. Is there a possibility thus of getting a larger tree?

The second question is akin to that—utilization of the western tree as a stock for a hybrid chinquapin which might have arboreal possibilities and enough chinquapin qualities to be blight-resistant.

DR. STABLER: I am very much interested in Dr. Morris' proposition to produce dwarf chestnut trees by grafting on chinquapin stocks. Now, the difficulty I would expect to encounter is the same as when pecans are grafted on hickory, and when sweet cherries are grafted on Mahaleb, namely, that the root is not sufficiently vigorous to support the top. The fact that his grafts grew so tremendously when put on the chinquapin roots would look as though that might occur.

THE PRESIDENT: The audience seems to have run out of questions.

DR. MORRIS: All right, sir. First, how are we to grow chinquapins? Plant as soon as the nuts have fallen. Put them in a cage. I have wire cages that are about eight inches high, and about two feet wide and three feet long. I plant all the nuts there. They have wire mesh tops to keep out the rodents; that is the important thing. All nuts, I find, are best planted under conditions which simulate the normal conditions. Our nut trees are not as yet domesticated. They haven't learned bad habits, and they depend upon peculiarly favorable conditions of moisture, warmth and light. You plant a nut two inches below the surface, but nature doesn't do anything like that. Consequently, that nut is surprised, doesn't know what to do, and stays down there looking for something to happen. But if you put that nut so it is about half buried in the sand, so that it is damp on one side and the sun strikes it on the other side, and the frost and snow affect it naturally, the nut

does just what you want it to do. It gets out of that uncomfortable condition and begins to grow. (Laughter.) When planting any nuts, I place them in the sand and leave one side exposed to light, moisture, frost, and the observation of visitors. When I have sprouted chinquapins in the north and there is danger of their not lignifying when the ground begins to freeze, I put a lot of little sticks upright amongst them, so that my mulch will not bear too heavily upon the chinquapins, and then cover them with several inches of oak leaves, or any good, strong leaves that will not pack too tightly. That mulch of loose leaves will protect the sprouted nuts perfectly during the winter in Connecticut, so they all start growing in the next spring.

Another way is to buy chinquapin stocks from any of the nurserymen, stocks two or three years old, which begin to bear when four or five years of age.

Professor Close, I think it was, who asked if the nuts were fertile, both the ones that developed without fertilization by any pollen and the ones that developed by stereochemic parthenogenesis—by the influence of neighboring pollen. Both sorts are fertile, and I presume that the effect of that would be similar to the effect of close inbreeding. In other words, we would have intensification of characteristics of some one parent. If you get parthenogenesis through two or three generations I presume that same peculiar feature of the original parent would become so intensified as to become a marked feature of the progeny. This offers a new line of cleavage for horticultural investigation. I am very glad that you raised that question.

Answering Mr. Littlepage, I have apparently managed to get some crosses back and forth between chestnuts, and oaks, and beech, this year. I have a number of those crosses now under way that are apparently good hybrids.

DR. STABLER: A cross between a chestnut and a beech?

DR. MORRIS: Yes, I think so. You see, I have got to wait a year or so until the plants develop later characteristics. All of these parent trees are pretty closely related, you see. The blooming period between the different ones may be as much as two or three weeks, or three months apart, in fact. I have cross pollinated hazels and oaks, this year. The way to do that is to find correspondents at the extreme limit of the blossoming range of the species, who will send pollen. For instance, Professor Hume, in Florida, sends me chestnut pollen in time to cross my oaks, and Professor Conser, of the University of Maine, has some beeches that blossom in time for me to cross with chinquapins and oak trees. That is one way to do it.

Another way is to put your pollen in cold storage with some sphagnum moss, just put a little damp moss in your box with the pollen and put it in cold storage, and keep it at just about forty, above the freezing point. Another way is to put branches with dormant flower buds in cold storage. Hazels, for instance, may be kept for six months in this way. Put them in water, in the sun, and you soon have flowers furnishing pollen. I would take up the whole session of two days here if you were to ask too many questions along that line. (Laughter.)

Mr. Littlepage's question about the weevils. The question may be settled very easily where there are not many chinquapin trees. That is the case in Connecticut. Collect all the nuts, and you collect all of the weevil larvae. Curiously enough, the common chestnut weevil, that had become very abundant, has disappeared locally with the disappearance of our American chestnut, and has not attacked our chinquapins. If you have an orchard of chinquapins and collect all the nuts you will soon dispose of the weevils. That is the only way that I know of for disposing of the weevils. Eat them up. (Laughter.) You can pick out the weevil chestnuts fairly well if you toss

all of the nuts into a cup of water and pick out the ones that float. Pound them up with a mallet and throw them into the chicken coop.

Dr. Smith asks if the use of the tree chinquapin as a stock for the American chestnut would give good-sized trees. Undoubtedly, and, besides that, if it is used for hybridizing purposes, we shall probably find that we have, now and then, an individual that is very much larger than the American chestnut or the tree chinquapin. It is a peculiarity of hybrids to show eccentricities, and many hybrids that occur are very thrifty and larger than either parent. That is the case with the Royal walnut that they have said so much about in California. It grows so rapidly there that even Californians do not dare to tell about it. (Laughter.)

Another question, the last one—will the effect of using a bush chinquapin stock for the American chestnut be like that of growing sour cherries upon stocks which do not carry them well? Now, we have there what the lawyers call "a question of fact," and we shall have to work that out. Some tops will exhaust a root. Some tops will grab a root by the back of the neck and drag it right along. Some tops will adjust themselves philosophically to almost any sort of unusual conditions, and go on and bear fruit like true philosophers. We have an instance of that in the dwarf apple, which is a success. We have an instance of failure in some of the cherries which exhaust themselves. We have an example of dragging the smaller stock along when we graft the Royal walnut upon the common black walnut. The Royal walnut just drags the black walnut along where it doesn't want to go at all. So there we have three instances of grafting a foreign visitor upon another stock.

I have taken more than my share of time, Mr. Chairman, but the discussion has been very interesting, indeed. (Applause.)

THE PRESIDENT: I am going to take the liberty of asking Dr. Morris one more question, which, perhaps, is of interest to others. In your experience with the golden-leafed chinquapins, from how far South have you secured stock, and how far North will the golden-leafed chinquapin grow?

DR. MORRIS: My specimens I got from a dealer in Portland, Oregon, and they grew pretty far North. The tree ranges from Oregon and Washington down through the lower extremities of the Coast range, but we had better get the northern forms, and there is one man, Carl Purdy, of Ukiah, California, who has the golden chinquapin for sale.

THE PRESIDENT: The next subject on the programme is the American black walnut. We have sent to the membership a series of questions about the black walnut which I will read for the benefit of those who haven't this programme.

First. What evidence is there to show that the black walnut may become a valuable nut commercially?

Second. Is quality important with the black walnut, and is there much difference in the quality of different nuts?

Third. What varieties of black walnut are most promising?

Fourth. Is the Thomas black walnut better than many others that have been brought to notice?

Fifth. What are the best methods of propagating?

Now, we have no set paper on that subject. I will call on ex-President Littlepage to make a few sallies concerning the black walnut.

MR. LITTLEPAGE: Mr. President, the black walnut ought to be the easiest subject in the world to talk about. It is a question of how much one ought not to say, however, in a limited time. The pecan tree was my first love. I shall always stick to the pecan. But if I were called upon today, to point out to this Association or to any prospective grower who actually wants to make money raising nuts, and who wants something that will pay the grocery bill and his sixty or ninety day notes, I think I should tell them to plant the black walnut. And I don't think, either, that that is treason, because I think, as we go through with this programme, the pecan will be properly taken care of.

In the first place, the black walnut is a native tree. I have seen it growing, too, on the Gulf of Mexico, and in the Dominion of Canada. Most native trees are immune to fungous and bacterial diseases that destroy so many trees. The black walnut is a hardy tree, and a fine timber proposition. In the second place, it is a fast growing tree. I don't know just how quickly one could actually produce a black walnut orchard, but, outside of a few trees, such as the black locust and a number of others that do not produce nuts, the black walnut is one of the fastest growers. If you will feed a young walnut tree a small application of wood ashes and some stable manure it will commonly make a growth of six or seven feet a year. Therefore, you don't have to wait a long time for walnut trees to come into bearing.

It is easy to propagate the black walnut. Cleft grafting is one of the simplest methods in the spring. Dormant wood, cut in February or March and put in cold storage, and cleft grafted in the spring, ought to give from sixty to seventy per cent of success. I haven't had experience budding, but those who have say it is easy. Mr. Roper says it is, but grafting is easy and simple. The walnut, like other nut trees, must be propagated by budding or grafting in order to come true. It will not come true from seed.

Up until a few years ago I seldom saw a whole half of a black walnut. The ordinary black walnut cracks about like this (showing picture). Here is a black walnut cracked with two halves, and you can't even see the kernel. The two upper pictures show very beautiful walnuts, but they defy you to get out a whole kernel.

Now, then, when you come to a black walnut like this (showing picture), where you can crack out anywhere from fifty to seventy-five per cent of whole halves, and many entirely whole kernels, the most important problem is solved, and the black walnut has come into the competition.

This variety was discovered by Henry Stabler, and I named it after him. Perhaps one out of every ten of these nuts furnishes a whole, solid, undivided kernel. The other walnut is the ordinary field walnut that has little commercial value for the reason that you can't get the kernels out. It wouldn't make any difference if the nuts grew as big as pumpkins and a million of them on a tree if you couldn't get the meat out of them. I suppose no one will question that the black walnut will grow and bear almost anywhere. It is a weed tree in this part of the country. On President Smith's farm last year I saw them growing everywhere. They grow and bear all over the fields. And, as I said, the question of propagation is rather simple. I think the great trouble we are up against on the farm in America is labor, and that is because you cannot afford to pay good labor. You want a superabundance of laborers in the summer time for two or three months, and expect them to loaf all winter. The farm proposition isn't a profitable one, very largely because of the question of labor, and the farmers of this country must produce something profitable enough to enable them to hire and pay high-grade labor the year round, or they will go broke. They must raise such crops as Alfalfa that they can feed to their dairy cattle, and tree crops that they can use their labor on in the winter time. Nine men are leaving the farm today for every one going there. If you don't believe it,

read the census statistics. The reason is labor and because you can't afford to pay it. I don't think there is any profit in selling the black walnut as a nut, but there will be profit in gathering that nut, storing it, and, when your farm crops are all in and you are ready to discharge the labor, put up an ordinary cheap cracking shed and let them crack the nuts for you, and sell the meats. That solves the question of what to do with farm labor in the winter time. The walnuts return about ten pounds of meat to a bushel, and a good cracker ought to crack from four to six bushels of nuts a day. Suppose you get only twenty-five cents a pound for the meats and your men crack only three bushels a day, each there is \$7.50 a day coming in from each cracker, and, besides, you have made a valuable employment for your labor through the winter, and you can afford to pay them for their work. That is why I say the black walnut is, to my mind, one of the best commercial propositions.

I don't know how soon you can bring a black walnut orchard into bearing. Here is a picture of a tree probably seven or eight years old, loaded with nuts. That is a seedling tree. I should think a budded tree would bear sooner than that.

I don't know much about walnut varieties. The Rush and Thomas are excellent nuts. But this Stabler walnut, in my opinion, is in a class by itself in cracking possibilities. It is simply a cracking proposition with the black walnut, and that is, to my mind, about all there is to it. Perhaps, other good varieties will be discovered. Then, suppose we find, after a while, an English walnut much better and more profitable than we have at present, and one that is blight resistant. If you have an orchard of black walnuts you have an ideal stock to top-work to English. I will show you one on my farm with a larger top than I cut off grown in two summers, and it set some nuts last spring. So, if you want a foundation for an English walnut orchard, you can't make any mistake in planting the budded or grafted varieties of these black walnuts.

The black walnut is a beautiful roadside tree. There are different types, the same as with the pecan tree. Here is a picture of curly black walnut wood. The logs were cut from a tree in Kentucky. It took three wagons to haul this one tree to market, and it brought thirty-five hundred dollars.

THE PRESIDENT: I wish to present Mr. Stabler as the original propagator of the tree that bears his name. The nuts of the Stabler black walnut have been pronounced by a good many authorities as the best variety thus far discovered.

MR. HENRY STABLER: Dr. Smith has just introduced me as the discoverer of this walnut. This is hardly fair to Mr. Littlepage, who first introduced and, probably, first propagated this walnut. It was discovered by my grandfather a little over forty years ago. Nothing was done with it at that time for the reason that nothing could be done, but I was not the first one to get the idea of propagating it, because my father, who is here today, attempted to graft these walnuts, and every cion failed.

It seems to me that Mr. Littlepage strikes the key-note in his article in *The Country Gentleman* last spring when he says that:

"Through the efforts of the Northern Nut Growers' Association there was recently discovered a black walnut tree in Howard County, Maryland, producing nuts that crack out seventy-five to eighty per cent of whole halves. The meat of this variety, the Stabler, weighs forty-seven per cent of the whole nut."

That's it, gentlemen. I did not discover this walnut, and without the organization of the Northern Nut Growers' Association I could not have done any more with it than my grandfather was able to do forty years ago, but, as it was, we just took up several samples and the Northern Nut

Growers did the rest. The walnut has been attracting more and more attention ever since.

Considering the black walnut as timber, here is a picture of a black walnut log, published in Farmers' Bulletin No. 715, of the Department of Agriculture. The original owner, a farmer, sold the whole tree, standing, for fifty dollars; the buyer felled it at a cost of fifteen dollars, and sold it there for \$138.26. It was resold, without being removed, for \$164.84, and later sold (the last price is not published) to a large sewing machine factory, but it certainly brought more than that last price which is printed, of \$164.84. We occasionally hear of prices of \$100 or so being paid for black walnut trees on the stump. The reason we don't hear of such prices being paid more frequently is because the farmer in not more than one case out of twenty gets real value for his black walnut trees. There is a very highly organized and efficient system in the United States of gathering up the black walnut trees which are large enough to use for furniture and other purposes and paying for them as little as possible; but they make a practice of getting them even if they do have to pay more. There was a man living not so far from where I live, up in our country, who had a very fine black walnut tree standing in his yard. One day a man came around and entered into conversation with him, and said, "Mr. Harder, what will you take for that tree in your yard?" "It isn't for sale," said Mr. Harder. "Well," said the man, "I'll give you a hundred dollars for it." Mr. Harder merely shook his head. The buyer dickered along a little bit more and after a while said, "I'll tell you what I'll do; I'll give you \$150 for that tree." Mr. Harder said "If you don't get off this place, sir, immediately, I'll shoot you."

I am prepared to say that if you are going to plant trees for timber there is no other tree which will give such a yield as the black walnut, with the exception of the catalpa, and, perhaps, the black locust. It is the most valuable tree we have, and it is the most valuable wood grown in the North.

I don't believe, either, the black walnut will ever be less valuable than it is. I know positively that the Stabler tree is not over sixty-five years old, perhaps, not over sixty, and yet that tree, judging from the prices I have seen paid for other trees of similar size, is worth from \$125 to \$150 on the stump. From the time that tree started until now, it has increased in value at the rate of two dollars a year, for timber alone, to say nothing of the nut. Suppose the tree had been purchased sixty years ago at two dollars from the nurseryman. It would have paid one hundred per cent annually on the investment. It bears, as a regular thing, a crop every other year.

As to what Mr. Pomeroy said about the black walnut not cracking well and crumbling up when it gets to be old, I have some specimens here of the Stabler walnut I cracked this morning, which are of the 1915 crop.

The kernel of these old nuts keeps its flavor and sweetness wonderfully. There is hardly any change in quality within one year, whereas some other nuts, as the hazel and some varieties of the pecan, become rancid after keeping six months.

DR. MORRIS: I would like to say one word about the curly walnut. In Maine, not long ago, I saw a young man who had bought a bird's eye maple, perhaps fifty years of age, that he paid \$1,500 for. I asked him why he didn't graft one million ordinary maples from that tree and sell the stock at \$200 per tree, and then he would have \$200,000,000 at just about the time of life when he could enjoy it. Well, that hadn't occurred to him. Now, if Mr. Littlepage will hunt up this curly black walnut stump that sold for \$3,500, and if he will graft a million trees from that he will be able to raise a family of ten children (Laughter.)

DR. STABLER: Mr. President, I just want to call attention to an omission in the little talk that my son gave about the characteristics of this Stabler tree, namely, its beauty as a shade tree. He didn't mention that, and I don't

think any one has mentioned it in connection with the black walnut. Now, the black walnut trees, as we meet them along the roadsides, vary exceedingly in habit of growth. The majority of them have very few main limbs, perhaps not over half a dozen main limbs on a tree, and they will be gaunt, ungainly things, stretched out straight, like great arms reaching out with very little beauty. Now, if you plant seedlings, that is what you are likely to get on your lawn. You may have something that is not pretty except as a trunk, but the tree that produces these very remarkable nuts is also one of the most beautiful in its conformation. It is shaped just like an umbrella, rather low, very spreading, and very frequently with a very much larger number of limbs than almost any black walnut tree that I have ever seen, and its habit of growing in the nursery confirms that opinion—that it produces a very large number of buds and branches from each graft.

Mr. Littlepage has in his fence row, uncultivated and surrounded by bushes of every kind, a small seedling walnut that he grafted this year with the Stabler walnut. When he grafted the seedling it was a little over an inch in diameter. I measured the growth from that graft recently, and five shoots measured over five feet long, and others over four feet long. Four month's growth—five shoots over five feet long! Now, I don't know of any other walnut or any other nut tree that would have produced that many shoots from a single graft. It makes a very beautiful shade tree and has a top which is capable of producing very large crops of fruit.

THE PRESIDENT: It sometimes makes me feel ashamed of my race when I realize our limitations in comparison with the trees. We run across a valuable type of tree genus, and we can make millions like it in a short time. But when a remarkable specimen of the genus homo, arises, he stays with us but a short while before we cart him off to the cemetery, and that is the last of him. Does any one else wish to make a contribution to the black walnut?

MR. M. P. REED: Mr. Littlepage made the remark that it is very easy to propagate the black walnut. We haven't found it so. We have made almost a complete failure of both budding and grafting.

MR. LITTLEPAGE: Well, I was speaking of my experience in grafting this spring. I think I remarked that my personal experience in budding had not gone far enough to tell definitely what the results are going to be. But I put in about fifty-five grafts, and I had fifty of them to grow, and of that fifty there were probably ten or twelve knocked out—thrown out at the first cultivation—and probably thirty-five are growing there yet. I don't know what Mr. White's experience was in Indiana. I think it was, perhaps, not as good as he expected, because of the fact that a lot of the bud-wood dried out, but I think Mr. McCoy can give some experience. Now, Mr. Roper here, has had experience in budding the black walnut, haven't you?

MR. W. N. ROPER: We only put in about a dozen buds a short time ago. I think half of those are growing.

MR. LITTLEPAGE: Well, we budded, perhaps, two or three hundred this summer, and I don't know really how they are coming out, but, from the way these grafts behaved this spring, I don't see any reason why it is going to be very difficult. What do you know about it, Mr. McCoy?

MR. R. L. MCCOY: Mr. Stabler's grafts didn't take very well, but so far as budding the black walnut is concerned, it is just as easy as handling the peach; there is nothing to it when you get the bud-wood; but first you have got to have the bud-wood. You can't jump on to any old tree and get buds that will give satisfactory results. Now, if Mr. Reed and his father had to go into Wisconsin and Michigan to get their bud-wood, and cut it from some old cherry trees, we'll say, and came back to Indiana and tried to produce trees from those buds in the nursery, they would fail.

THE PRESIDENT: Well, the net result, apparently, of the discussion on propagation seems to be that Mr. McCoy, in Indiana, has had great success with buds; Mr. Littlepage, in Maryland, has had great success with grafts; I also had great success with grafts put in by a man who could neither read nor write, but who was taught the technique as taught by this Society. Is there any further discussion?

MR. LITTLEPAGE: Mr. President, Professor Hutt ought to know something about the black walnut. He knows something about everything else I have ever talked to him about. I believe he wrote me, in connection with some of his tests, that forty-seven per cent of the Stabler nuts were meat.

PROF. W. N. HUTT: I think so. I think it was pretty close to a half. There were no broken halves at all, and some of them came out entirely whole.

THE PRESIDENT: We want to hear from Dr. Deming.

THE SECRETARY: I just want to call attention to one of the questions on our list. "What can we do to cheapen nuts and nut meats in the retail market so as to make this valuable food available to persons of small means?" It seems to me that we are going to do that with such nuts as the black walnut. I think we ought to work for the time when the black walnut can be sold in quantity in New York City, and in all the larger cities for around a dollar a bushel. Perhaps the shellbark hickory is also going to be a nut of the same kind, a nut that can be put on the market in large quantities at a small price, for the man of limited means to buy and crack out himself. Dr. Morris, speaking of some tough nut, once said it was so tough that it was only of interest to squirrels and men out of work. That expression about "men out of work" made me think, as do so many things that Dr. Morris says. If a man out of work can buy a bushel of black walnuts for a dollar, and if he can crack out several bushels a day, or even only one bushel a day,

he can make more wages just cracking out that bushel of black walnuts than at ordinary laboring work. I think that we ought to get on the market a supply of cheap nuts for the man of limited means and that we ought to educate the people to a knowledge of the value of such nuts.

THE PRESIDENT: It is always well to put the brakes on. I haven't heard a thing about this black walnut except virtues. I believe Mr. McMurren, of the Department of Agriculture, is present, and I think he has been giving particular attention to the black walnut, and perhaps he will tell us of some of its enemies, either animal or vegetable.

MR. S. M. MCMURREN: Well, Mr. President, unfortunately, I haven't given much attention to the black walnut. My time has been given to the pecan until this summer, when I worked on the persian walnut to some extent, but I can say, generally, that the black walnut hasn't got any very serious enemies. Everything it has got is right here now. There isn't any reason to suppose that it would have any serious disease if we cultivated it on an extensive scale.

As to the insects, I am not able to state. I have never noticed any particularly on the nut since a boy.

MR. LITTLEPAGE: Mr. President, I think Mr. McMurren has covered the diseases of the black walnut. I think the observation of every one will confirm what Mr. McMurren has said.

THE PRESIDENT: The chair will deviate from parliamentary practice for a moment by dismissing the question. I wish to contribute three small facts. One is with reference to the special growth of the black walnut under fertilization. The men on my place have to cut bushes around apple trees, and some stray black walnuts planted by nature under those trees have been cut for 10 years but for the last two seasons have been left alone. They have

promptly come up through those apple trees, under the influence of nitrate of soda, like a steer going through a bush. They have grown five or six feet each season.

Another point is the great variation, apparently, of the black walnut with regard to its keeping qualities. I recall putting away in a garret, in 1894, a number of bushels of a nut of particular merit, and they were perfectly sweet and edible as much as seven years later. Now it is only occasionally that you will find one that will keep as long as that, but with the trees bearing every two years, it is quite possible that the fruit would be marketable for two or three, or even four years afterwards, if kept properly.

There is no reason to think that the Stabler is the best nut growing in the United States. It merely grew within reach of the eyes of observing men.

The filbert and the almond we hope to cover briefly before adjourning. I will ask Mr. Reed to give us a short contribution on the almond.

MR. M. P. REED: This almond (exhibiting specimens) we received scions of from Mr. C. A. Reed, of the Department a few years ago. It was three years ago this summer that we top-worked it, and we picked almost half a bushel of almonds from it this summer. The almond has a thick shell, kernel of good flavor, but I don't think it will amount to anything very much except for home use.

MR. LITTLEPAGE: How old was the tree that bore them?

MR. M. P. REED: Top-worked three years ago this summer.

MR. LITTLEPAGE: And bore how many?

MR. W. C. REED: Bore a half a bushel this last summer.

MR. LITTLEPAGE: If any one here would like bud-wood of that almond I will be glad to send it to them.

THE PRESIDENT: Mr. Littlepage offers to send those present bud-wood of that tree, which can be, with great ease, top-worked on the peach by the ordinary process of shield budding.

DR. IRA ULMAN: I have grafted scions of this nut on *Amygdalus Davidiana*, the new Chinese peach of the Department of Agriculture, and the growth is marvelous. It does just exactly as Mr. Reed told you.

DR. STABLER: I would like to ask whether the almond is attacked by the same insects and diseases that affect the peach, whether it is affected by peach yellows and whether it is affected by the peach borer. I understand that the apricot is, in a measure, immune to the peach borer at least, and possibly also to the peach yellows. If the almond is to be short-lived like the peach tree, it may not be nearly as valuable as if it were a hardy tree. If you place it upon peach stock it seems to me you must expect it to be attacked by the peach borer.

MR. M. P. REED: I believe that the original tree of this variety is something over sixty years old. Not very many peach trees live to be that old, and in the nursery it is a very vigorous grower.

THE PRESIDENT: The commercial almond is a rather long-lived tree in the countries where it is grown. Of course, here is a question of technique and individual behavior which only experience can answer. We ought to take some of these nuts home that Mr. Reed has given us. I should like to know why Mr. Reed so deprecates a tree which bears so much fruit in so short a time. If the fruit is good, why can't it be handled commercially?

MR. M. P. REED: It is the cracking quality. It has a very thick shell.

THE PRESIDENT: Is that a problem that machines cannot solve?

MR. M. P. REED: No, sir.

MR. LITTLEPAGE: How is the flavor?

MR. M. P. REED: The flavor is good.

MR. LITTLEPAGE: I was just going to say, Mr. President, that I visited Mr. Reed's place this summer, and it is utterly surprising how fast and beautifully this hardy almond grew. He took me out at the edge of the garden where he has them growing, and I could hardly realize that they were only three-year-old trees. They were as full of little almonds as the peach trees were of peaches, only they were much longer and with very red leaves. Vincennes, Indiana, is on the thirty-ninth parallel, which is the northern boundary of the District of Columbia, and it gets much colder there than here, and those trees haven't the slightest sign of winter-killing. I don't know anything about the quality of the meat, but they are certainly wonderful bearers.

DR. MORRIS: I find that in the region of Stamford, Connecticut, hard shelled almonds do pretty well if you look after them pretty closely, but they take all your time. They have so many different blights on them that I am glad mine died a long time ago. They bore heavily, but they were too much trouble. They blossom so early in our locality that the blossoms are apt to be caught by frost. You may overcome that if you set the trees on the north side of a stone wall where the ground retains the frost for from one to two weeks later than on the south side. I find, that by doing this you can retard their time of blossoming sufficiently to materially lessen the danger of their being caught by spring frosts.

MR. HARRY R. WEBER: Will you get the same results if you put a mulch under the tree? Won't that prevent thawing and hold the tree for a week or two?

DR. MORRIS: Yes, sir.

THE PRESIDENT: Have you used this particular almond?

DR. MORRIS: One very much like it, and it was a mighty good almond—hard to get at but good.

THE PRESIDENT: I would like to ask Mr. Reed as to the blooming time of this particular tree in comparison with some standard peach like the Elberta.

MR. M. P. REED: It bloomed about a week earlier than the Elberta, and the peach crop is light.

MR. HENRY STABLER: I have been associated for the past three or four years occasionally with Mr. M. B. Waite, of the Department of Agriculture, and I have had a good chance to study the effect of spraying on peaches in preventing brown rot and curculio. At Mr. Littlepage's I observed an almond tree that started, I should think, with twenty-five or thirty almonds on it this spring. Those almonds gradually succumbed to the curculio and brown rot until, at last, only one was left, and it seems to me that, if this almond is to be grown commercially in this climate, we will have to use the same methods of growing as with peaches, and we will have to spray them.

THE PRESIDENT: I think the chief benefit of the discussion of the almond would be to get more of us to try it, and the fact that we have one which is only one week earlier than the Elberta peach in blooming shows

that we have a good chance, possibly, of even exceeding the possibilities of the peaches.

MR. MCCOY: Mr. President, I notice a good many almonds bloom about the same time as Elberta peaches. I have probably twenty-five trees of this almond that Mr. Reed spoke of, and I think they were in bloom at the time the peaches were. It is very productive, just as he says. I have noticed some of the old trees around in our neighborhood have borne good crops for several years, and I don't notice much disease on them either.

DR. STABLER: I asked the question whether anybody knows whether the almond is affected by peach yellows, and nobody seems to know, but peach yellows is something connected with climate. There is a yellows line that has remained definite and distinct for the last twenty-five years, and you can describe that line on the map, and it stays right where you put it. All north of that line the peach trees are affected by yellows, and south they are not. That line runs through Mount Vernon and Annapolis, and across Chesapeake Bay to Chestertown. Now, below that isothermal line there is a little peninsula south of Chestertown, in Kent county, a little peninsula there—a little long neck that runs out into the bay below Chestertown—where they have never had any peach yellows, and yet at Chestertown the trees have always been affected by peach yellows, and it is probable that it will be found, if the almond is affected by peach yellows, that the same laws apply to it. That is, south they will have the yellows, and north they will not. Now, at Vincennes, I suppose that they are north of the yellows line for peaches. Do your peach trees have peach yellows?

MR. M. P. REED: No, sir.

DR. STABLER: Perhaps you are north of it, then. If so, the almond hasn't been tried out as to yellows.

THE PRESIDENT: This association is greatly indebted to Dr. Morris, who helped to get it together, for his indefatigable searching of the corners of the earth for specimens, species and varieties of trees in his ambition to get to his Stamford place all of the varieties of nut-bearing trees. Several of our members have taken a little interest in the question of the hazel-filbert family. Dr. Morris has taken a lot of interest. Last year he gave us a most exhilarating presentation of the subject, and he is this year going to give us some brief notes on the progress of his knowledge concerning the hazels and filberts. Dr. Morris.

DR. MORRIS: Just a word, in order to start the discussion. I have tried to work out during the past year two or three points that came up for discussion last year. I stated that in Connecticut the common American hazel would probably not become a horticultural proposition for the reason that the main stock seldom lives more than seven or eight years, and then dies. New stolons, starting from the root, make abundant new stocks. In that way, dying at the center, and growing at the periphery, like a ring worm, one plant may extend so widely as to drive cows out of the pasture lot. (Laughter). Dr. Deming understood me to say that it spread so "rapidly" as to drive the cows out of a lot. I said "widely," not "rapidly." (Laughter). For that reason a plant of our common hazel bears a few nuts about the third year; it bears a good crop about the fourth year and sometimes in the fifth year. It then begins to die and is gone by the seventh or eighth year, while new stolons, coming up on all sides, are ready to perpetuate that rotation. That, at least, is ordinary hazel history in my part of Connecticut. So I doubt if this species will ever be a good horticultural proposition.

This year, for the first time, I have budded the European hazel upon our common stock for the purpose of observing whether the character of the guest will change the character of the host.

Now another point. Many of the European hazels that have been brought to this country, I find, do not bear for the reason that they flower so early that the staminate flowers are caught by frost—not the pistillate. The pistillates will hold out against frost for a long time and make good. There are two or three ways for overcoming this difficulty. We may select for cultivation those kinds which bloom a week or two, or even three weeks later than others, as in the case of the Bony Bush variety.

There is hardly any more valuable tree in Central Europe than the purple leafed hazel. I never have seen one bearing in this country. Its staminate flowers come out too early in Connecticut. I have now some in which I have grafted the Bony Bush, which flowers so much later that I hope to have my purple hazels bearing nuts at Merribrooke.

On the whole, most of the points have been simply confirmatory of points previously considered. We need not fear hazel blight because it is very easily controlled, and many of the European hazels will furnish an immensely valuable crop for almost all parts of temperate America. We may develop, by breeding and by cultivation, types which will be hardy, which will give us large, valuable, marketable crops, and which will be desirable from the market man's point of view.

DR. STABLER: Can you get stocks that are free from blight?

DR. MORRIS: Last year I showed specimens of blight. The blight, fortunately, begins upon a fairly large stem—upon a part of the stem that is in plain sight. It takes from two to four years for a patch of that blight to encircle a limb. If one will go over his hazel orchard once a year and, where a bit of blight appears, cut it out with his jack-knife and later paint the spot with a little white paint, one can very readily control hazel blight. It is so easily done that we need not fear it at all.

THE PRESIDENT: Dr. Ulman, I believe, is a hazel enthusiast.

DR. ULMAN: I have attempted to gather as much information as I could by seeking out the failures with hazel because I had found no one reporting success. In answer to a large number of letters which I sent out I received some 290 replies which reported failures with the European hazel. Dr. Morris tells us that blight can be readily controlled. So far, that does not seem to be the experience of others, but it is only fair to say that they do not know how to get rid of it in the way that Dr. Morris has told us.

THE PRESIDENT: Dr. Ulman, I should like to ask if it is not true that the hazels growing at Rochester could be added to your collection of 290 and change this complexion a little bit. Certainly last year we saw hazel trees almost the diameter of this room which appeared to be perfectly healthy.

THE SECRETARY: Can we recommend the hazel to be planted commercially?

THE PRESIDENT: If the hazel propagates by underground stolons automatically, why can we not take the stolons and plant them in the places that the trees have abandoned, letting them run on elsewhere?

DR. MORRIS: In regard to Dr. Deming's question, green European hazel nuts are now selling in New York, out of cold storage, at seventy-five cents a pound. Green hazel nuts like green almonds are prized by the gourmet. All of the European hazels will eventually furnish a good commercial proposition provided that the market is large enough, and the market will probably grow, is growing in fact. Ripe filberts bring, approximately, from ten to fifteen cents a pound. The trees bear well, and I don't know of any reason why the hazel proposition should not be a first rate one right now. The thing to do is to select kinds which we know are valuable here. One may go through the seedling orchards at Rochester and select one parent

which bears large nuts prolifically, and then propagate any number of European hazels from that one stock. My Bony Bush is probably a desirable hazel.

In regard to the question of breeding from stolons, if we can keep that thing going it would be all right, but it requires so much work I doubt if we shall do anything in that way with the American hazel. The European hazels don't travel by stolons. That is the advantage. So I have given up the common American hazel as a commercial proposition. A number of European and Asiatic hazels will be commercially profitable, unquestionably, just as soon as we care to develop them.

MR. WEBER: What do you know about the hazels of the Western coast?

DR. MORRIS: Very profitable in parts of Oregon and Washington. They have a large, good crop, which sells locally, but, like most Pacific Coast fruits, the nuts lack flavor and quality. They have size and beauty, but lack quality. The fruits and nuts grown on the Pacific Coast all lack a certain fineness of character, for some reason yet unknown.

You have got to look after your European hazels, and not neglect the orchard. I remember seeing some very beautiful apple trees in central Maine not very long ago—no blight—no codling moth, and the trees free from almost everything in the way of insects or fungous troubles—beautiful, cultivated trees, and beautiful apples on them. I asked another man, one of my acquaintances there, an old farmer, why he didn't set out a lot of similar trees and make a good income. He said, "Well, it won't go." He had a pasture about eight miles north of there, and, said he, "I spent thirty dollars for apple trees to put into an orchard, and I had great ideas about those apples. I set the trees out in that orchard about three or four years ago, and last year when I went up to look at them, there were hardly

any apple trees left." He hadn't looked at them for three or four years. (Laughter.) You can't raise hazels that way.

THE SECRETARY: The reason I asked about the commercial value of the hazel is that my own experience has been very unsatisfactory. I got some hazels from Gillet, on the Pacific Coast six or seven years ago, set them out around my place, and they have grown beautifully. I haven't been able to detect any blight on them anywhere. Some of them are fifteen feet high, have grown luxuriantly, and blossom every year, but I haven't seen one nut yet. On the other hand, the other day I visited a man near my home, who told me that he had raised some trees from nuts which he had bought from an Italian grocery on the corner. He gets the nuts when the crop first comes in, and stratifies in wet sand all winter, and he says they all grow. He had some beautiful hazel trees. One I estimated to be twenty feet high. I never saw a hazel tree which approached it. He said it was only five or six years old. Last year he had a fine crop of nuts from it. This year, however, he said that during a warm spell in the winter the staminate bloom came out and was killed, and there were no nuts on the trees. Now it seems to me that there is great uncertainty about the hazels, and I don't know exactly what to advise people to do. People ask me for advice as to what nuts to plant commercially. I don't know whether to advise them to plant hazels or not, and I don't know what varieties to advise people to plant. I don't know how to advise them to overcome this difficulty of the early staminate bloom and the winter killing. I can't now conscientiously recommend people to plant hazel nuts commercially.

DR. MORRIS: Go to Rochester and get some there that bloom every year and that bloom later. My Bony Bush blossoms some three weeks later than the others, I presume. It is a bush that bears well every other year, apparently.

THE PRESIDENT: Are there any further questions about this large family of nut trees of which we have but a small corner of knowledge? If not, we may look to an adjournment.

First, I wish to announce that this afternoon we are going to devote to an excursion around the city, to see some of the most remarkable Persian walnut trees which I think may be found anywhere.

I am asked by Prof. Close to say that the Department of Agriculture has an exhibit of nuts on the fourth floor at 220 Fourteenth Street, Southwest.

Meeting adjourned.

FRIDAY, SEPTEMBER 8TH, 2 P. M.

Meeting called to order by the President.

THE PRESIDENT: Dr. Metcalf, Chief of the Bureau of Forest Pathology, of the Department of Agriculture, has been in charge of the investigations concerning the chestnut blight for a number of years.

DR. HAVEN METCALF: Mr. Chairman and Members of the Society: I will present, first, a few general facts regarding the present status of the chestnut bark disease, and, for the greater part of the information you desire, I will rely on you to ask me questions.

The chestnut bark disease is getting to be an old story, but that plant hyphenate, that objectionable imported disease, is more of a live issue today than it ever has been before. All my attention during recent months has been taken up with another imported plant disease, the white pine blister rust, of which you have heard, and which does not concern the special subject matter in which this Association is interested, unless, perhaps, you may be interested in the pinon nut as the pinon pine may ultimately be subject to attack by blister rust. However, this disease, like the chestnut blight, is an example of what a relatively harmless, or at least, not serious disease in a foreign country can do when it is permitted to get into the United States.

This brings us to the question of the origin of the chestnut bark disease, which, although the story has been told many times before, has been the subject of so much dispute that I probably had better recapitulate that matter. It has been proved beyond question that the chestnut bark disease is a native of eastern Asia, China, Japan and Korea; that it was introduced into this country in the '90's, upon diseased chestnut nursery stock. It was not critically observed until 1904, but the condition of trees which were observed at that time shows conclusively (provided the disease progressed in those early years as it has since) that it was introduced into the country as early as the late 90's. The final demonstration of the fact that the disease is a foreign disease and a native of Asia we owe to Mr. Frank Meyer, of the Office of Seed and Plant Introduction, of the Department of Agriculture. Mr. Meyer's observations are so interesting that I will pass around a few pictures illustrative of his observations in China, the first picture showing the country that is the home of the chestnut bark disease. The second picture shows a chestnut orchard in China where the trees have, with characteristic thrift, been planted around human burial mounds. The remaining pictures show how the chestnut blight acts in China—very differently from the way it acts in this country. In China, it produces, as the pictures show, definite cankers, which do not girdle the tree, which kill young trees occasionally, mutilate old trees, kill branches, but the cankers do not girdle the trees. That disease has been known in China we have no idea how many years, and, while it does a certain amount of harm, is said by Mr. Meyer not to be really serious in China. You can readily see, upon examining these pictures, that there is a sharp contrast in the behavior of the disease as observed in China and its behavior as observed in this country, where it will girdle a comparatively large tree and the fungus spread all through the bark, completely covering it, and doing that in a very short time. Of course, then, the chestnut blight is one of those cases of which we have so many, where a

disease, passing to a new country, finds new surroundings, hosts more favorable to its development, and progresses rapidly.

The natural range of the chestnut bark disease at the present time—that is, I mean, its range on the native chestnut and the range through which it is now spreading by non-human agencies, is, on the north, practically co-extensive with the range of the native chestnut. The disease is found in Maine, New Hampshire, Vermont, as far south as Virginia, and as far west as western Pennsylvania and eastern West Virginia. Throughout this area it is spreading by what I may call natural means, and the disease has been shown to be unusually well provided with means of dissemination. I will speak a little later about the spread of the disease outside of this area—that is, west and south, since in the West and in the South it is being spread, as far as we know, exclusively by human agencies.

The question is often asked me, "What is the future of the chestnut—that is, the native chestnut—in this country? What is the course of the disease going to be?" The only way in which we can answer that is to look in the parts of the country where the disease has been present longest—Long Island, for example; Westchester county, New York; Bergen county, New Jersey; Fairfield county, Conn. Upon a recent examination of those areas I found no chestnut trees surviving in a healthy condition. We have, of course, from the beginning, hunted, and hunted hard, to find individual chestnut trees that might be immune to the disease—native American chestnuts. We expected to find such trees, but up to date we have not found them. It is a very extraordinary fact and an almost unparalleled fact, because with the majority of plants affected, by any given disease, we can find some individuals that are not only resistant, but immune.

Now, in these old areas, particularly on Long Island, in 1907, when the disease first came under my observation, I marked certain trees in order to

observe how long the stumps of these trees or the dead trees would continue to send up sprouts from the ground. It is an interesting fact that some of those trees which were dead in 1907 are still putting up sprouts. The sprouting capacity of the chestnut tree is indeed marvelous, but I am sorry to say that I haven't been able to find any healthy sprouts over three years old. I haven't been able to find any living sprouts more than four years old. The disease seems to be following up the sprouts as it followed up the original stem.

Right there, in the behavior of the disease toward the sprouts, we have an interesting fact. During the first year of its life the chestnut tree or the chestnut sprout is immune to this disease, or practically so. You can rarely find a seedling or sprout of the first year that is attacked by the disease, and even in the second or third years a comparatively small per cent of them are attacked. It is thus possible to produce chestnut nursery stock that for several years does not show the disease.

So far as I can see, the chestnut blight is not stopping naturally in its course anywhere. I cannot get a particle of reliable evidence that it is. In this part of the country and to the south of here, in Virginia, for example, the parasite has more months in the year during which it can grow, it appears to be utilizing that time in spreading more rapidly, at least killing trees more quickly, than to the north of this area. From the standpoint of the grower of nuts, the important question is, of course, whether the disease can be controlled. I think your Secretary, in a recent article, summed the situation up as clearly and briefly as can be done. He said, in an article entitled "The Progress of Nut Culture in the East:"

"Of the chestnut we have excellent varieties such as the Rochester, Boone and Paragon, but all development in the culture of this nut is being held up by the blight. Everybody is awaiting the results of the government work in

breeding immune hybrids. There may be great opportunities, nevertheless, in chestnut growing outside its native area, where the blight can be controlled."

There is no doubt that in an orchard tree, in chestnut orchards, the disease can be controlled within reason by the cutting out method that has long been advocated, but the point is that the margin of profit on the chestnut is not sufficient to make that method pay, and whenever members of this Association or others interested in the propagation of chestnuts have written to me for advice I have simply advised them not to plant chestnuts at present. I cannot see at the present time, that any attempt at control is profitable. That is a very different thing from saying that it can not be done, or that it may not later become profitable.

A few words regarding the method of spread of the disease. In 1908, when the office of which I have charge was first organized, Professor Collins, who has addressed this Association a number of times regarding this disease, visited a number of orchards and nurseries in the Eastern States, going as far as southern Virginia to the south, and west as far as York county, Pennsylvania, Although that was comparatively early in the progress of the disease, wherever he went, without exception, where there was a nursery, he found the disease present and spreading onto the native trees. There were, however, several established orchards which he visited where that was not the case, where the disease was not present. It has been brought out repeatedly that, while the chestnut blight is marvelously adapted to spread by natural means—wind, birds, insects, rain, all the ways in which a plant disease ordinarily spreads—the way in which it spreads over great areas and through great distances is on chestnut nursery stock.

In that connection, then, I may briefly discuss the present range of the disease so far as we know it, outside of the natural range of the chestnut

tree. South of Virginia, so far as we know, the disease is present at only one point (Greensboro, N. C.), where it was introduced in a nursery and spread to native trees. In stating this area of distribution, I ought to say that for about a year and a half we have made no special effort to determine the range of this disease. I mean we have not gone out of our way to do it. We have simply collected such evidence as has come to hand casually, and so it may be that there are now other points of infection in North Carolina, or south of there, but, if so, we do not know of them. In Ohio, the disease is present at three points, of which one is a large and serious infection at Painesville. In Iowa, it is present at one point, Shenandoah, in a nursery. In Indiana, it is present at five points; in Nebraska, at two points. In Michigan, one point has been reported. In all of these cases it is in nurseries, or on very recently planted trees. There is, or was, an interesting point of infection in British Columbia. Probably the trees there are all dead by this time, but that point is very interesting as being probably an independent importation from the Orient.

There needs to be little said as to how the disease is spreading in this area. Perhaps the best thing I can do is to read some letters that have come to my attention:

"MICHIGAN AGRICULTURAL COLLEGE,
"EAST LANSING,
"Aug. 18, 1916.

"Dr. Haven Metcalf,
"U. S. Dept. of Agriculture,
"Washington, D. C.

"Dear Mr. Metcalf:

"Last December, the Forestry Department of this College ordered of Glen Bros., Glenwood Nurseries, Rochester, New York, five 6-foot trees of the Sober Paragon chestnut. These were shipped to them April 4th and were almost immediately planted in the Forestry Nursery here. About six or eight weeks ago, the Forestry Department noticed that these trees were dying and called our attention to this matter about four weeks ago. I examined the trees in company with Mr. J. H. Muncie, one of our assistants, and found all the external appearances of Chestnut Blight with, however, only a very few imperfectly developed pycnidia. We brought pieces of the bark of these trees into the laboratory and made cultures and obtained the typical mycelium of chestnut blight. The trees have been removed and we now have them in our laboratory.

"I am calling this to your attention as the trees were doubtless infected when shipped. I feel that you ought to know that this firm is sending out diseased trees.

"Very truly yours,
"(Signed,) ERNEST A. BESSEY,
"Professor of Botany."

The following is an extract from a letter from Frank N. Wallace, State Entomologist of Indiana, dated July 13, 1916:

"My Dear Sir:

"Under separate cover I am sending you some samples of chestnut blight which I secured from some trees shipped by Mr. C. K. Sober, Lewisburg, Pa. Mr. Sober doubts that we have even seen a case of chestnut blight and wanted some samples and I sent him the other half of the samples which I am sending you.

"I have been trying to check up on some of Mr. Sober's trees and so far I have found nearly fifty per cent of them have died from chestnut blight disease."

The samples sent with this letter showed typical chestnut blight.

Some months ago Dr. W. H. Long, of Albuquerque, New Mexico, became interested in the possibility of growing chestnuts in that country and communicated with Glen Brothers, of Rochester, N. Y., to secure certain information regarding them. He secured the information he wanted and also some that was slightly gratuitous. I will read extracts from the two letters:

"In regard to the blight, which you call the Eastern Chestnut Canker, would say that this tree is practically immune from this disease, and you would stand no more chance of having your chestnut trees infected with the blight should you plant them, than you would if you planted apple trees, of having them infected with the San Jose Scale or peach trees, of the Peach Blight.

"There are over half a million trees at the famous Sober orchard in Paxinos, Pa., none of which have the blight, and yet the blight rages all around them in the American Sweet Chestnut groves that are all through the mountain. Further evidence of its immunity from this disease we cannot guarantee. We think this speaks for itself.

"We believe that if you would investigate this variety that you would plant an orchard of Sober Paragon Chestnut trees, even if not a very large one. We should like very much, indeed, to serve you and shall give our personal attention to the selection and shipment of such trees as you may require.

"Very truly yours,
"GLEN BROS., INC.
GM-AB "(s) JOHN G. MAYO."

* * * * *

THE PRESIDENT: Would you mind giving us the date of that last letter?

DR. METCALF: That is October 20, 1915.

The other letter signed by Mr. Mayo is as follows, and is dated Oct. 29, 1915:

"Replying to your October 25th letter we do not think that you or your friend need have the least anxiety on account of the chestnut blight reaching your section. This disease seems to be confined to a very small area in northeastern New Jersey, southeastern New York, and southwestern Connecticut. The disease has been in existence in this country since 1842, it has made very little progress, and the highest authorities now state that it seems to be on the wane." (Laughter.)

* * * * *

MR. LITTLEPAGE: Do the experiments of the Department show any possibility of control of the disease?

DR. METCALF: I don't think that there are any methods of control which can profitably be applied to orchard trees under present commercial conditions. If a man has a few orchard trees which he regards as novelties and to which he is prepared to give very careful attention, I think the disease can be controlled. So far as I can see, the only hope of commercial control lies in none of the present varieties, but in Dr. Van Fleet's hybrids, possibly in the Chinese chestnut, and, aside from the objectionable qualities

of the Japanese nut in certain strains of Japanese. With the rapid withdrawal of the wild chestnuts from the market, however, the price of chestnuts may rise, and control methods in orchards become practicable.

MR. LITTLEPAGE: Mr. McCoy has been in Pennsylvania and has come back with the very optimistic idea that the chestnut blight was under control up there. I took him out on my farm in Maryland and showed him my trees, and that the only thing that could destroy the trees faster than the blight is a forest fire.

DR. METCALF: Exactly.

THE PRESIDENT: I believe, Dr. Metcalf, you conducted a series of spraying experiments recently, and I understand that others have done the same thing. Mr. P. A. Dupont, I believe, on his fine estate near Wilmington, tried to spray a few chestnut trees with Bordeaux mixture, and I understand he gave it up as a physical failure, to say nothing of the cost. Am I right about that?

DR. METCALF: That is my understanding, that he was dealing with large trees and failed.

A MEMBER: Well, did you succeed with small ones?

DR. METCALF: In the line of spraying? That is a long story, and I suggest that Mr. Hunt answer that.

MR. HUNT: In the spraying work conducted on Dr. Smith's place at Bluemont, Va., we had 2500 numbered trees under observation; about 1500 of them being sprayed. Equal numbers of trees were sprayed with Bordeaux and with lime-sulphur. The number of sprayings given different lots of trees varied, but even trees sprayed as often as every fifteen days blighted in a

number of instances. While I did not get a greatly reduced percentage of blight (approximately 50 per cent) among the sprayed trees taken as a whole, the difference between individual plots seemed to depend rather on location in the orchard, as some blocks of unsprayed trees showed practically no blight and some blocks of sprayed trees showed considerable blight. I might say that the grafted trees did not blight nearly so heavily as the ungrafted trees. So far as any real success is concerned there was none. It would cost over one hundred dollars per acre per year to spray as often as some of the trees were sprayed, and it wouldn't control the blight. So I wouldn't consider it at all practicable.

THE SECRETARY: What is the reason that the grafted trees blighted less than the ungrafted?

MR. HUNT: Well, I wouldn't pretend to say as to that, except that it is so. I had each tree numbered and kept an individual record of all the trees, and I found—I have forgotten the exact figures—but there was about three-fifths as much blight among the grafted trees as among the ungrafted trees. Of course, they are an imported variety, I believe, and it may be that on that account they may have developed some resistance. But Mr. Van Fleet may know more about that.

DR. METCALF: There seems to be some evidence that the imported European varieties have a slight degree of resistance, not enough to count, but enough to show in that fraction that Mr. Hunt gave.

THE SECRETARY: It is only a varietal condition, then, not from the fact of grafting, but simply because of a different variety?

MR. HUNT: Oh, yes, I think so.

MR. LITTLEPAGE: Mr. President, in view of this information about the chestnut, is there the slightest use in the world for this Association to encourage anybody to plant chestnuts anywhere in the United States?

THE PRESIDENT: Dr. Kellerman is here, and I wish to refer to him Mr. Littlepage's question with a slight addition. Is there, first, any prospect of any place staying immune? Second, would it not be to the advantage of the country if the sale of chestnut stock were stopped?

DR. KARL F. KELLERMAN: Mr. President, to answer those questions involves a rather large contract on my part. (Laughter). In the first place, the problem of growing and marketing chestnuts, I think, is one that I could hardly be expected fairly to discuss. I am here rather to explain the attitude and action of the Federal Horticultural Board than to try to give any constructive advice to the nut growers.

The Federal Horticultural Board is a board of five men to advise the Secretary of Agriculture in establishing plant quarantines, either on the introduction of plant material into the United States, or on the movement of plant material inside the United States within the quarantined areas. The Horticultural Board, therefore, has to deal more with actual conditions than with outlining such policies as your chairman has asked me to outline.

THE PRESIDENT: Excuse me, Dr. Kellerman, but we wish to know if there is, in your opinion, any prospect of any region remaining immune?

DR. KELLERMAN: Well, even that is going rather further than I would like to go, and yet the negative answer to that question is practically the basis on which the Federal Horticultural Board decided that it was impracticable to quarantine infected areas at the present time. The evidence at hand appears to indicate conclusively that if the trees that are to be grown are distinctly susceptible to the disease they will almost certainly have an

opportunity to become infected, no matter what part of the United States they may be grown in. Now, whether that infection would be a matter of a few months, or a few years, or a few decades, of course, would be altogether a matter of chance, but, with the wide distribution of nursery stock that is infected, with native chestnuts rather generally infected and continuing to be infected, and with practically no chance of preventing the continuation of the disease in the native chestnuts, abundant sources for infection of susceptible material appear to exist. For that reason, it appears to be, from an economic standpoint, inadvisable to attempt to check the disease through the establishment of quarantines.

THE PRESIDENT: Dr. Kellerman, you have answered my first question to perfection, and now I want to ask the second one. If this blight is practically a sure-kill, isn't it wrong to permit people to spend money in the hope that, in some way, they are going to escape it? And if that is the case, why shouldn't the whole traffic in chestnut trees be stopped, with the possible exception of experimental things, which might be allowed with the direct permission of some governmental board?

DR. KELLERMAN: That is a question that is very much harder to answer. There might be favored regions where orchard culture of the chestnut could go on for a considerable term of years before infections became general and before the industry would be stifled because of this disease. That is merely a matter of conjecture, as I see it. We have so little evidence as to the speed with which a paying orchard business can be developed in a new locality, so little evidence as to how the disease may act under widely separated climatic conditions, that I don't feel that we are prepared to say definitely that the industry is bound to fail in every place where it is tried. Personally, I think that it ought to be considered only on an experimental basis, but that represents merely my personal opinion, and I doubt whether there is any effective means for establishing a policy of that

sort. It might be possible for the general advice to be given that there was danger in any orchard planting of chestnuts, no matter where it might be undertaken, and of a comparatively rapid loss through the chestnut blight. I doubt whether more than that would be feasible.

THE PRESIDENT: I have been enthusiastic over the chestnut for twenty years this season, and these are matters in which I am greatly interested. As I see it, the problem is one that is really much bigger than the chestnut. The whole field of nut growing, which is now on the edge of great accomplishments, is likely to be seriously injured, because the most conspicuous thing in nut growing is the taking advertisement of the firm whose bad trees have been referred to by Dr. Metcalf. I think we do not appreciate the seriousness of the situation. The firm Dr. Metcalf referred to is selling trees that are diseased in places where they are sure to die quickly. Other men are similarly selling trees, with less skillful advertising, perhaps, but probably no less diseased. Most of these nurserymen may be honest in their belief that they are putting out stock that is not diseased. But in the infant trees it is almost impossible to detect the blight, so that the tree goes out looking like a perfectly good one. It may be two or three seasons before it dies.

Now, the economic aspects are these: Who should stand the loss, the man in the nursery or the man in the orchard? It is a toss-up, it seems to me at present, with the results apparently in favor of the nurseryman rather than in favor of the citizen. The people who have an interest in nut growing are going to have that interest lessened or destroyed by beginning with a bad kind of tree. There are possibilities of a great national injury, as I see it, if we let this thing go on.

DR. KELLERMAN: Well, as a constructive policy for aiding in the establishment of nut culture, I think your policy is sound, but as a question

of economics of operation, I doubt whether any plan of that sort can be established, beyond the plan of merely giving the general advice that such planting is attended with very grave risks.

THE PRESIDENT: Have you not authority, or does not authority exist, to prohibit shipment?

DR. KELLERMAN: The plant quarantine act gives the Department authority to quarantine infected areas and to place certain restrictions on shipments. To place any such restriction, however, it must be plainly established that beneficial results are going to result, not to a particular industry necessarily, but to the general public. The difficulty in establishing a quarantine on the shipment of nursery stock is the apparent impossibility of saying that that is going to stop the spread of the disease. That is one question. The other problem is the difficulty of determining what is infected territory and what is not. We have very serious difficulty in making regulations, excepting as between definitely infected territory and definitely clean territory.

THE PRESIDENT: And you don't have the authority to make a sweeping, blanket prohibition of the shipment of a certain thing?

DR. KELLERMAN: No, we haven't that authority.

MR. M. P. REED: We put a clause in the printed matter that goes out with all of our shipments saying that chestnuts are subject to blight, and that we don't recommend their planting. I think if nurserymen all followed that principle everybody would buy with their eyes open.

THE PRESIDENT: I am sorry you are so lonely in the business.
(Laughter.)

MR. LITTLEPAGE: As regards the possibility, or the impossibility, of doing any good to the chestnut industry by quarantining it, I fully agree with Dr. Kellerman. I think any attempt of the Board to quarantine, so far as benefit to the prospective chestnut grower is concerned, is perfectly useless.

DR. ROBERT T. MORRIS: It seems to me it may be resolved into a very simple proposition. Now, chestnuts may be raised in orchard form if we spray with Bordeaux mixture, and cut out blight when it appears. I do it. They live. Those that are not sprayed die, unless given tiresome attention. That settles that question for my part. Chestnuts may not be raised in forest form because it does not pay to spray and cut to that extent. But chestnuts may be raised profitably in orchard form by people who are willing to take the trouble to spray them, and to cut out blight early. It seems to me that people should be properly warned that they may plant chestnuts in orchard form provided they are willing to look after them, otherwise we ought to guard against the public buying chestnut trees, unwarned.

MR. LITTLEPAGE: Dr. Morris, were you in when Mr. Hunt made his statement?

DR. MORRIS: I got in late.

MR. HUNT: I sprayed fifteen times, every two or three days during the blossoming season.

DR. MORRIS: I used arsenate of lead with my Bordeaux mixture for the reason that it is convenient. That makes it stick ever so much tighter. Now, that may be a feature of my confidence. Three or four heavy storms will not wash off my Bordeaux mixed, applied in that way, with arsenate of lead.

MR. HUNT: Well, my trees are dying right along.

DR. MORRIS: I am right in the midst of the worst chestnut blight conditions. The only kinds I have that are not blighted are sprayed trees, and chestnuts of kind that resist the blight. I had twenty-six kinds from different parts of the world to test out in the blight question. One kind from Manchuria is very blight-resistant. I find that our American chinquapin, both our eastern form and the western tree form are both blight-resistant. Also the alder-leaf chestnut. That is my experience. Those four chestnuts are practically immune, and on my property American chestnuts dying all around them.

I have one particular variety of American chestnut that I think a great deal of. It was one of the first trees to go down from the blight. Stump sprouts from this tree I have grafted on other stocks, on the common American, and recently on chinquapin. The sprayed ones are all alive; the unsprayed ones are not alive. Now, that is a matter of locality, perhaps.

THE PRESIDENT: Dr. Morris, I detect a possible explanation for difference of results. Mr. Hunt's trees were sixteen or seventeen years old. Dr. Metcalf tells us, however, that young trees are relatively immune. How old are yours?

DR. MORRIS: Not over twelve years. No grafts on them over four years. That would make a difference.

MR. LITTLEPAGE: Well, Mr. Hunt did a good job of spraying. I saw his trees, and they were saturated.

MR. WEBER: Do they ever use a sticker in the Bordeaux?

A MEMBER: What preparation of Bordeaux mixture do you use, Dr. Morris?

DR. MORRIS: I use a commercial preparation called Pyrox.

MR. CARL J. POLL: Will the chestnut blight attack any other trees besides the chestnut?

DR. METCALF: Outside of the chestnut genus, that is, the genus *Castanea*, the disease goes on to a few other trees. A curious fact is that it will go on to the sweet gum, a tree not related at all, and it will go on to a few oaks, in no case enough to seriously damage them as it does the chestnuts, but enough so that those trees can easily be carriers of the disease.

THE PRESIDENT: I think we might pass from this funeral. We have a paper by Dr. Van Fleet, whose work, I suppose, is known to everybody here. The paper has been prepared by Dr. Van Fleet and will be read by the Secretary.

HYBRIDS AND OTHER NEW CHESTNUTS FOR BLIGHT DISTRICTS.

DR. WALTER VAN FLEET, WASHINGTON, D. C.

The sinister spread of chestnut blight, as the bark disease caused by the fungus *Endothia parasitica* is popularly called, within little more than 10 years, from its place of apparent origin near New York City into 13 states, practically reaching the eastern and northern limits of our native chestnut stands, and sparing in its course no individual trees exposed to infection, has about convinced even the most optimistic observers that without the intervention of natural checks the American chestnut as a forest asset will soon pass away. There is no present indication of diminution in the virulence of the fungus parasite and little reason to hope its progress as a timber destroyer can be stayed by any agency in the control of man. Already the losses, direct and indirect, occasioned by chestnut blight are computed as high as \$50,000,000, about half of the estimated value of the entire stand.

With the very reasonable assumption that our native chestnut is doomed to virtual extinction it is well to consider in time if it can be replaced as a timber and nut-producing tree by other chestnut species or combinations of species less subject to injury by this disease-producing organism. The *Endothia* fungus, as a destructive parasite, is apparently confined to the chestnut, rarely if ever harmfully affecting genera even as closely allied as

the oak (*Quercus*) or *Castanopsis*. Of the various species of chestnut or *Castanea* those native to Japan and Central China appear most resistant, probably having been for ages accustomed to the presence of the fungus, while the European chestnut, *Castanea sativa*, our native *C. Americana*, and our chinquapin, fall easy victims when exposed to infection. Of the Asiatic forms *Castanea crenata* of Japan and Eastern China and *C. molissima* of the interior are most promising in this respect, though the latter is still an almost unknown quantity as regards cultivation in this country.

Castanea crenata, commonly known as Japan chestnut, in its more typical forms is highly resistant, so seldom showing material injury that, for practical purposes, it may be regarded as immune. Japan chestnut seedlings raised from nuts grown in proximity to our native chestnut and exposed to the influence of its pollen are at times more seriously affected, but are rarely destroyed by the bark disease. The Japan chestnut is of comparatively low growth, of small value for timber purposes, but as a nut-producer is very fruitful and precocious, bearing great crops at an early age. The nuts are often very large but usually of poor quality. The species, however, proves quite plastic in the hands of the plant breeder, being readily modified in the directions most desired by the ordinary methods of cross-pollination and selection. It freely hybridizes with all other chestnut species and varieties that have been tried, and forms the basis of the most hopeful work in breeding for disease-resistance that has yet been attempted.

Castanea molissima is of much taller growth and bears nuts of moderate size, but of really good quality in the types that have reached this country. It can be infected by *Endothia parasitica* but the disease progresses slowly and in some instances results in little harm. The species has been so recently established in America that practically nothing is known of its breeding capabilities, but if its disease-resistance under our climatic conditions is assured it would appear most hopeful material for replacing our vanishing

native species. Explorers report there is a still more promising chestnut in China, reaching nearly 100 feet in height under forest conditions, but it has not yet been secured for trial in this country.

Castania sativa, the commercial chestnut of Europe, in many varieties has long been cultivated in America and for nut production is without doubt the best of the well-known exotic species. It has no great timber value, however, and its disease-resistance, though higher than *C. Americana*, is scarcely great enough to warrant extended use as breeding material.

The native chinquapin, *Castanea pumila*, in its bush and tree forms remains as the only promising chestnut not found in the Orient. While readily inoculated by artificial means, the chinquapins, especially varieties of the northern bush forms, quite often escape natural infection, doubtless because of their small size, smooth bark, and less liability to insect attacks.

Chestnut breeding for nut improvement, chiefly by selection of native European and Japanese species, has been carried on in several diverse localities in the United States, with distinctly promising results but inter-pollinations have also been effected between most species and varieties, the outcome indicating that rapid improvement along the desired lines may be expected from crossing the really desirable types.

In 1903 and succeeding years the writer made many careful pollinations of the native chestnut and the bush chinquapin with European and Japanese chestnuts in many varieties. Some hundreds of seedlings resulted, mostly showing a high level of promise as judged by their initial thrift and vigor of growth, but the appearance in 1907 of the Endothia disease among the plantings soon put an end to the work with the native and European chestnuts, as, with scarcely an exception, they quickly became infected. The crosses of chinquapin and Japan chestnut, however, showed considerable resistance as a whole, and a number of individuals have resisted infection

until the present time, though constantly exposed to the disease, both at their locality of origin in New Jersey and since at Arlington farm, to which they were transferred in the second and third years of growth. Others have been attacked in greater or less degree, but show great powers of recuperation, sending up suckers that often fruit well by the third year. The resistant varieties show great promise as nut producers, coming into bearing when three or four years old from seed and producing abundant crops of handsome nuts, of excellent quality, four to six times as large and heavy as those borne by the chinquapin parent, ripening in early September before chestnuts of any kind have appeared in the market. These nuts have thicker shells than other chestnuts, are much less subject to attacks of the chestnut weevil and preserve their fresh and inviting appearance longer when gathered. The flavor varies somewhat according to the particular pollen parent of the different varieties, but is always agreeable in the fresh state when the nuts are properly cured. When boiled or roasted they are particularly sweet and pleasant to the taste.

The trees are quite vigorous in growth, considering their rather dwarf type, reaching 10 or more feet in height at 6 to 8 years from the germination of the seeds and with scarcely an exception bear regular and increasing crops after the third year. Propagation of the most promising varieties has been effected by grafting and budding on *Castanea molissima* seedlings as resistant stocks, but it cannot be said that these processes, when performed under greenhouse conditions, give ideal unions. It is hoped to make fairly extensive trials of *C. molissima* and *C. crenata* as stocks for field grafting the coming season.

But the most encouraging feature of these chinquapin-crenata crosses is the excellence of their seedlings as grown from chance or self-pollinated nuts. Fifteen direct or second generation seedlings and one of the third generation have fruited to date. All have retained in growth and fruitage the

characters of their immediate parent and it almost appears as if the good qualities of these hybrids may be perpetuated from seeds, thus dispensing in a great measure with vegetative propagation—always costly and uncertain with nut trees.

Several hundred of these seedlings are under observation and it scarcely appears too much to hope that they may inherit the disease-resisting character of their parents as well as other desirable qualities.

Selection work with a precocious strain of Japan chestnuts of apparently pure type has been continued through 4 generations of seedlings after an initial cross-pollination of two particularly desirable varieties had been made in 1903. These seedlings show greater range of variation than the hybrids with chinquapin, but all bear nuts of marketable value in 2 to 4 years from germination. None have been attacked by the *Endothia* fungus, though many have constantly been exposed to infection. Notwithstanding their extreme precocity trees of this Asiatic strain grow steadily and if thickly planted in favorable localities may in time produce timber of local value, but it is to the taller growing species of middle China that we must look for material to replace our vanishing native forest stands. The preservation in this country of the chestnut as a nut-bearing tree appears assured in view of the progress already made and it should not be too much to hope that resistant strains of the timber type may yet be developed by systematic breeding experiments.

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THE PRESIDENT: Inasmuch as the author of the paper is not present to answer questions, the only thing that may be done is to ask further contributions of knowledge in the same field. Has anyone any contribution to make?

MISS LOUISE LITTLEPAGE: I would like to ask how long the chestnut tree has been able to live with the blight?

DR. METCALF: Do you refer to the Asiatic ones or to the ones that grow here in America?

MISS LITTLEPAGE: The American.

DR. METCALF: It is almost impossible to answer that question because you have to define just what you mean by "living." If the chestnut tree is attacked first or early on the trunk, it is girdled and dies shortly, but if it is attacked first on the top there develop conditions like what is shown in this picture (showing photograph). I am not certain that you can see these bunches of suckers a little way up the tree. Now those trees will sometimes exist four or five years. I can say safely that I have seen trees last five years.

DR. MORRIS: I can add three years to that.

THE PRESIDENT: If there is no further discussion, we may adjourn.

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FRIDAY, SEPTEMBER 8TH, AT 8.15 P. M.

Meeting called to order by the President.

THE PRESIDENT: To my mind nut growing is part of a larger field, a field of conservation, one which is going to develop a whole new series of tree crops, of which the nuts are but a part.

PRESIDENT'S ADDRESS.

DR. J. RUSSELL SMITH.

Agriculture is usually symbolized by a picture showing a man, a plow, and a sheaf of wheat. I would make the symbolization double by adding to it some kind of a nut tree in fruit. I have long had a vision of waving, sturdy, fruitful trees yielding nuts and other valuable fruit, and standing on our hilly and rocky land where now the gully and other signs of poverty, destruction and desolation gape at us. This vision of the fruitful tree also extends to the arid lands, there also vastly increasing our productive areas. Beyond a doubt the tree is the greatest engine of production nature has given us, and in its ability to yield harvests without soil injury on rough, rocky, and steep lands, and on arid lands, carries the possibility of the approximate doubling of the area of first-class cropping land in the United States, also probably in many other countries.

Twenty-one years ago this spring I began in a small way to bring into reality this vision of the tree-covered fruitful hills, although my interest in the matter goes back at least four years further to the time when I filled my pockets with the large grafted European chestnuts grown along his lanes by the late Edwin Satterthwaite, of Jenkintown, Pa.

My first essay at nut tree cropping was short but not sweet. I planted an acre and a half of Persian walnuts, seedlings being the only things then to

be found. There being no one within my reach to guide me much, if any, I bought such seedlings as were to be had from a New Jersey nurseryman. I mulched them, and saw them each year grow less and less until the third season they disappeared. I have, however, some survival from this attempt in the form of black walnuts, which I had the foresight to plant as nuts immediately beside the Persian walnuts when they were planted as trees. Some of these walnuts are now quite sturdy young trees ready to be top-worked to some good strain.

My second attempt was the Paragon chestnut. In 1897 I started in on a 100-acre tract on the Blue Ridge Mountains, near Bluemont, Va., much of it too rocky for any cultivated crop, but admirably fitted to native chestnuts, and covered with a perfect stand. I had a good many acres well established, when, in 1908, the chestnut blight convinced me that further extension was perilous. My orchard has since been given over to the Department of Agriculture as the scene of their experiments in fighting the chestnut blight, but they have given it up, withdrawn their efforts, and half the orchard is now cut down and planted to Winesap and Grimes Golden Apples, which ten year's experience has shown me can be grown on such land without cultivation if mulched with the weeds and bushes that grow around them, and given some commercial fertilizer. I have a number of such young trees planted in 1907 in land of this character that are now full of fine quality fruit.

My third nut-growing attempt was with more select strains of seedling English walnuts than the miserable chancelings with which I began. One tree from the magnificent specimens at 3115 O street, N. W., Washington, D. C., and several from Pomeroy, promptly perished, apparently from winter-killing, and my nut hopes were at a very low ebb when the Northern Nut Growers' Association came upon my intellectual horizon. From it I have learned how to graft the walnut, the pecan and other hickories, and I

have again started in on the English walnut, using the Mayette, Franquette, and several of the eastern seedlings. After the usual disastrous failures at top-working, I was this June in such a large condition of hope that I was in serious need of being hooped to keep myself down to normal size. Such artificial aids to the maintenance of normal size are, however, no longer necessary after this summer's experiences, during which the bud-worm has cut the ends of my Persian walnut shoots and the blight apparently has withered up my young grafts so that an 18 inch shoot of July 1st is now 17 inches black and 1 inch brownish green, and in other cases entirely dead. Alas what a slaughter! This apparently puts my Persian walnut hopes into a state of neutrality. I hope it is benevolent neutrality. So far as actual expecting is concerned, however, I am not doing any just now. I wait.

The grafted black walnuts, however, have met with none of these accidents, and these are a substantial and solid hope, as is the pecan, which is behaving handsomely on its own roots and also on the hickory roots.

Tree Crops Insurance.

As my experience with nut trees well shows, there is little doubt that we are now in a period of great activity of plant enemies. They are indeed a by-product of the splendid work now being done in bringing to us the crop plants of all parts of the world. Along with the Chinese and Japanese products which have already been so valuable and promise us so much more for American horticulture, we have received the San Jose scale, the chestnut blight, and probably others will follow. For the next twenty-five or fifty years while the nut industries are in what may be properly considered the experimental stage, I wish to urge the great necessity of some kind of crop insurance for the man who plants out any kind of nut tree. Say what you please, the nuts are not as well known and as reliable as the other fruits, such as the apple, and even apples are uncertain enough.

Crop Insurance Through Two-Story Farming.

By the term "crop insurance" I mean having something else on the same land that will make a profit year after year, whether the tree pays or not. If this is not feasible, there should be something else which can be quickly converted into a crop if the main hope suddenly disappears. For the man who is growing nuts on level, arable land, I believe I cannot emphasize too strongly the pastured pig. Pigs below trees (and nuts maybe above). This is merely the two-story farming that Europe was practising when Columbus was a boy. Upon all good nut growers I urge the pig for the first story. This unromantic but very practical aid to income for the nut-grower has had the great honor to be accepted by a president of the Northern Nut Growers' Association, Mr. Littlepage, and by a president of the National Nut Growers' Association, Colonel Van Duzee. Colonel Van Duzee, from the financial standpoint, really does not have to have his pecan trees either to live or bear. He is making money out of the oats, cowpeas, crimson clover, vetch, soy beans, velvet beans, and other forage crops which he is growing between the pecan trees, and which the pigs are harvesting for him and converting into salable products. Of course this makes the pecan trees grow like weeds, but I am now talking about the crop insurance aspect of it. This crop insurance aspect of Colonel Van Duzee's last planting cannot be too strongly emphasized. He has planted the trees 100 feet apart, practically four and one-quarter trees to the acre, and has then proceeded to the hog farming business as though the trees were not there. This may sound somewhat fantastic to the man of the North. Perhaps it sounds well-nigh criminal to the man who is trying to sell pecan tree land to schoolmarms, talking fifty pecan trees to the acre. When a tree has the habit of spreading two or three or four feet per year when well fed, and keeping it up an indefinite time, the question of ultimate size is one to be reckoned with. That the pecan tree can attain great size in the North, as well as in the South, is attested by the record of a tree in northern Maryland on Spesutia

Island, near the head of Chesapeake Bay. The tree is described by one of our members, Mr. Wilmer P. Hoopes, as being eighty-four years old, hale and hearty.

"This tree is 106 feet tall, with a spread of 110 feet, has two limbs, respectively 57 and 60 feet long and is 13 feet in circumference, 3 feet above ground, and is an annual bearer of thin shell, nuts that, though rather small now, are mighty good to eat."

If nut trees are going to grow into that size, we must plant very wide or make up our minds to a very heroic and very difficult act, one which many men in the South should do this minute, namely, cut down half or three-quarters of the nut trees on a given acre.

I wish to emphasize the health aspect from the standpoint of the tree of this very wide planting. It is generally recognized by horticultural authority that trees develop sickness and disease when crowded in large numbers. The pecan trees 100 feet apart may perhaps escape this danger and have the sun on all parts of their leaf surface, a fact, by the way, which is necessary to crop production on this and other nut trees.

This wide planting is practically the method followed in most of the important French Persian walnut districts. With very few exceptions their trees are isolated, a man having two or twenty, or thirty, scattered about his farm, usually in the midst of his fields where they can develop to perfection, take the tillage of the crops, and bring in some extra money, which one of the owners very significantly told me is "income without effort." This income without effort aspect of the matter takes the form of a man having to pay as much rent for a good walnut tree in the department of Dordogne, as he does for an acre of good wheat land alongside.

Rough Land Tree Crops Insurance.

What kind of tree crops insurance might I have had for my chestnuts grafted nineteen years ago? Had I known then as much as I now know about nut trees, excepting the chestnut blight, I should have planted that place thickly with black walnut nuts and northern pecan nuts, unless the squirrels were too quick for me, in which case I should have used little seedlings. These I would have kept in a submerged but hopeful condition by occasionally cutting them down. This would keep them from crowding the chestnut trees, but would by no means have kept me out of a stand of vigorous pecans and black walnuts ready to graft at very short notice. When the blight blew its signal of national alarm in 1908, I could have gone to grafting those trees and they would by this time probably have been in bearing and ready to replace the chestnuts which are now dying with the blight.

If any one wishes to contradict my statement about these trees living with such treatment, I will admit that I am not speaking from experience with regard to the pecan, but I believe the experience of others admirably verifies the statement I have made. I am, however, speaking literally from my own experience when I refer to the black walnut. For ten summers past I have in July and August scythed off a certain tract of stump land planted to apples. Each year black walnuts and butter nuts have been cut, and now at the end of that time the stubs are still annually throwing up vigorous shoots 2-1/2 to 4 feet in length, and if they are allowed to escape for a season, they dart past a man's head so fast he wonders what has happened.

While I hope to experiment for forty more years on my mountain side in the attempt to cover it with waving fruitful trees that are so immune to pests as not to need spraying, I shall never again be caught with only one possibility upon a given piece of land. If I should top-work my native

hickories to shagbark, which I know involves considerable waiting and considerable uncertainty, I can, with very little expense, put upon the same ground a full stand of grafted black walnuts and a full stand of budded pecans, or if I do not care to go to that much trouble, I can graft my hickories and plant my native black walnuts and merely keep them there in submerged condition as reserve trees ready to be grafted at any time. For a pecan orchard I can do exactly the same thing, using black walnuts as fillers, possible successors, or as ungrafted reserves. For the Persian walnut, the black walnut can again come in as a filler or as a reserve, and for grafted blacks of any variety, other blacks can be kept waiting for the arrival of possible better varieties which could easily become the head of the corner.

My experience with transplanting seedling pecans shows that they, too, can, without serious difficulty, be planted out in such rough land and kept waiting there for years until the day of possible utilization.

Lastly, I wish to emphasize one more possible crop insurance tree for the man who is planting nuts on land difficult of cultivation, or entirely untillable, and that is the persimmon. I have paid my respects above to the tilled crops and the pastured pig for the arable land, and for the unarable land I would still emphasize the pig and give him other sources of food to supplement pasture. Among these possible foods is the persimmon which as yet has been little appreciated in an extensive way, although hundreds of thousands of men know it is highly prized forage and of considerable fattening value. It has a crop insurance virtue, however, other than its acceptability as pig feed. That is the hardiness of the tree and the ease of establishing it. In my pasture lot the Angora goat, even when pushed with hunger, has not touched persimmon wood or leaves. The same is practically true of the black walnut and of the butternut. This fact is one of great importance, because it means that we can keep rough land in pastures, even goat pasture, during the period when we are planting out tall-headed nut

trees of almost any variety, and at the same time have a perfect stand of two kinds of crop insurance trees coming along, namely, walnuts and persimmons.

In this connection it is desirable to point out the relation of this recommendation to the actual practice in nut growing regions of Europe. They do not plant a little two or three foot tree. They plant an eight or nine-foot tree often so slight it can not hold itself up, and is kept in place by one or two stiff poles. This tall-headed fellow stands out in the middle of the wheat field, the vineyard, the hay field, the goat pasture, the cow pasture, with its head entirely out of reach of the pasturing animal, its trunk protected by one or two stout sticks, and in due time it takes hold. With the trees properly developed in the nursery, I know of no reason why the same practice cannot prevail here, and I have at least one Busseron pecan tree that has gone safely through the first summer of it.

The practice of one pecan grower in Texas, reported in the Nut Journal, is suggestive of a crop insurance practice capable of wide use in the North, namely, planting of filler trees of quick-yielding varieties. There is no reason why the northern nut trees might not be planted 40, 60, or 80 feet apart in peach or even apple orchards, as did the Texas man with his nut trees 72 feet apart, occupying every fourth place in an 18-foot spaced fig orchard. I would call attention of Northerners, however, to the desirability of the mulberry, the most rapid growing and cheapest of all our fruit trees, doing well in Carolina at a space of 30 feet, which would enable the Northerner, by a little variation of the interval between his mulberry trees, to plant nut trees anywhere from 60 to 100 feet apart.

Sod Mulch Nut Orchards.

I know that any suggestions of the production of trees without plowing is unorthodox, and therefore not likely to be heard straight, and particularly perilous in the presence of professional horticulturists in state or national employ. To such I wish to call attention to the fact that I have emphasized in this matter, first, the tillage methods, and that I am making no knock against cultivation. We all know that it works under some conditions, and we all also know that there are some conditions in which it will not work. If I lived on level, sandy loam, I'd be a furious tiller of tree crops fifteen times a year. But I was born upon a rocky hill, and now I live upon another that is higher and rockier, and I don't believe in tilling it fifteen times a year. Must I abandon it, or adopt uses to its conditions? Out of these conditions mulch orcharding has come. Despite the orthodox, I know that the growing of some kinds of fruit trees without cultivation has passed the experimental stage. At this moment millions of barrels of apples are approaching perfection in orchards in Virginia and other eastern states that have not been plowed for more than one, and sometimes for more than five seasons. The application of this method to nut trees is still in the embryonic stage, with theoretic factors favoring it.

I do not know how far the mulch-fertilizer method can go, but I am sure it may go much farther than most professional horticulturists will admit. I find that the pecan tree starts off nicely under the mulch fertilizer conditions of the apple. The walnut tree has certainly done it for ages with less aid, and I believe it is up to us to find methods of handling land and trees and moisture which will enable us to avoid the danger, costs, and difficulties of plowing rough land and still get good trees. For example, the absence of cultivation does not necessarily imply the absence of fertilizer. The way a few black walnut trees in my apple orchard have snapped their buds and grown in response to the nitrate of soda that has been put upon the apple trees beside has been little short of astounding. The way a poor little starveling persimmon wakes up when the same treatment comes along, is

equally interesting. I cannot speak definitely yet about the influence of fertilizer on the Persian walnut or the pecan.

In connection with the fertilization matter, it is well known that a crop of clover or other legumes is very important as a part of the rotation of crops in plow agriculture. Similarly I expect great value can be obtained in our pastured and fertilized nut orchards if we so treat the soil with lime, phosphorous, and whatever else is needed, to give a good mat of white clover and other legumes which are undoubtedly a good nitrogen supply for trees whose roots interlace with theirs.

Similarly I see great possibilities in the interplanting of some leguminous crop tree such as the honey locust or the Kentucky coffee bean in our nut orchards. It is true neither of these trees has yet been selected and developed to the crop point, but they are much more promising than Sargent says the wild Persian walnut was at its beginning. It is an established fact that a non-leguminous plant can take nourishment from the nitrate-bearing nodules on the roots of adjacent living legumes, to say nothing of its well-known ability to feed upon the nitrate collections of legumes that have lived in past seasons within reach of its roots. Thus the interplanting of a legume and a nut tree seems to promise a continuous supply of the all-important nitrates for the nut tree.

The Question of Moisture.

It is not necessarily true that a tree gets a low percentage of the local rainfall because it is not plowed. The last palliation, or is it provocation, that I would throw into the camp of the orthodox and the worshippers of the plow, is the water-pocket, or small field reservoir, draining a few square rods and holding hard by the roots of a tree a few gallons or a few barrels of water which would otherwise run away. I showed this association a number

of photographs of these water-pockets last year. Their most extensive American user, Dr. Mayer, considers them successful from the tree's standpoint and profitable from the economic standpoint. Since the great virtue of cultivation is the conservation of moisture, I will submit that this device, worked out and used for three centuries by the olive growers of Tunis, for twenty years by Dr. Mayer, of Pennsylvania, and about the same length of time by Colonel Freeman Thorpe, Minnesota, can from the point of theory and perhaps also from the point of practice, equal tillage on some soils, and with less labor and much greater economy in farm management, for the making of water pots is a job for odd times, the bane of agriculture, and tillage all comes in a pile—another bane of agriculture.

Upon the whole, I think my 21 years of nut loving have run me directly and indirectly into ten thousand hard earned, and as yet, partly not earned dollars. Rather a deep sting for a pedagogue. When the last of my grafted chestnut trees come down next year, I will have little to show for that ten thousand, but an experimental nursery and some experimental trees scattered about the hillside. But the experiments are still interesting. I still have hope, and I still love trees. I am still ahead.

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THE PRESIDENT: I believe every other man here today has defended his thesis. I will not claim any exemption.

DR. STABLER: The President has mentioned the combination of apple and walnut trees. I would like to ask him if he has seen any deleterious effects upon the apple from the proximity of walnut roots. Now, some of my friends in Montgomery county have the idea that an apple tree will not live within fifty feet of a walnut tree. I have, myself, seen a number of apple trees die, apparently because they were neighbors of walnut trees. I wasn't sure that that was the cause of death, but they died, and walnut trees

situated in an apple orchard will have a ring of dead apple trees around them. Now that is one case that I know of where the walnut tree acts injuriously upon the vegetation to which it is neighbor. All of the farm crops, wheat, corn, grass, and oats, and rye, etc., seem to thrive just as well under the limbs of a black walnut as they do away from it. In fact, frequently you see the grass greener and more luxuriant right up to the trunk of the tree than anywhere else, but it doesn't seem to be true of the apple. Now, I would like to hear from the President.

THE PRESIDENT: I simply made that as a suggestion and referred to this instance as an illustration of the effect of fertilization on the walnut.

DR. AUGUSTUS STABLER: Well, how are those apple trees doing?

THE PRESIDENT: I had enough trouble without looking for more by mixing walnut and apple trees. The walnut trees are small, merely the growth from stubs repeatedly cut.

The next on our program is a paper by Mr. McMurrin, of the Department of Agriculture, upon the question of diseases of the English walnut. Mr. McMurrin.

MR. S. M. MCMURRAN: I am sorry that in this, my first appearance before this Association I haven't a more optimistic and encouraging subject to talk on than diseases. You men and women who are burdened with establishing this industry have enough on you without contending with diseases, and it was not my intention to talk upon diseases at this meeting, but Mr. Littlepage, Mr. C. A. Reed, and Mr. Jones, and several others, have been urging the matter strongly, which explains my appearance at this time.

Walnut blight is a very common and serious disease on the Pacific Coast. It may be a native disease, though it has never been reported on native black walnuts, and it has proved a very serious menace to the seedling English walnut groves on the Pacific Coast.

This little piece of work I want to tell you about tonight was done through the co-operation of Mr. Jones and Mr. Rush, at Lancaster, Pa., and has just been completed within the last few days. I made a trip through New Jersey, Delaware, and Pennsylvania, about the first of August, and found a number of nuts that had all the appearance of being infected with the walnut blight germ. They had the same appearance as those nuts that you saw this afternoon in Georgetown. I brought them back here and made cultures from them in the laboratory, and after that the problem was absurdly easy. The germ was obtained without difficulty, I obtained a pure culture, and then I went up to Mr. Rush's place, at Lancaster, and made a number of inoculations, of which these few I have here are typical. This nut that you see here was inoculated from a pure culture along with a number of others, and the condition is as you see it, after about a month. Inoculations were also made into twigs, and I will pass these around for your examination.

The one marked, inoculated, has a little canker on it, and on the other you will have difficulty in finding the needle punctures, but you will see them if you look closely.

Now, I hardly know what to say about this disease at this time. As I have stated before, my work has been in the South for the past several years, and no work has been done on this disease in the East prior to this summer. That it must have been here for a long time seems almost a foregone conclusion, because of its wide distribution. Mr. Jones was a little bit conscience-stricken for fear he brought it here with him. Still it is in Delaware and Maryland as well as Pennsylvania, and you can't blame Mr. Jones for that. I

think, too, it is less actively pathogenic than on the Pacific Coast, or we would have heard of it before. That it should prove a serious menace to the development of the walnut industry in the East, is too much to assume at this time. It will undoubtedly eliminate a number of the varieties that are considered promising now, but the course that will have to be taken will be to propagate only varieties which are highly resistant or totally immune to the disease. Just what these varieties are going to be in the East we do not know as yet, of course. We should avoid the mistakes that the growers on the Pacific Coast have made of planting seedling trees, and taking the chance of their being resistant to the disease. A great many varieties will be automatically eliminated when the nurserymen bear in mind that this disease is one to be considered, and I want to say, that, in addition to this, the Department will take pleasure in making artificial inoculations and tests on all those concerning which there is any question. We have the germ in culture now and will maintain it, and anyone who discovers a new variety, or has an old one they would like to propagate, can communicate with us, and we will take pleasure in testing its susceptibility.

I think that is about all that can be said on the subject at this time.

This disease has been studied very carefully on the Pacific Coast and a number of publications issued from the California Experiment Station concerning it.

For those who are interested in looking the literature up, I have here the following references: Cal. Station Bulletins, 184, 203, 218, 231, and Circulars 107 and 131.

A MEMBER: Is spraying of any avail?

MR. MCMURRAN: It has helped somewhat, but it has not proved economical on the coast.

A MEMBER: In order to have that test made, would it be necessary to send the things to the Department?

MR. MCMURRAN: No; it would be necessary for me to come to you and test them on the trees.

MR. LITTLEPAGE: Did those walnuts in Mr. Brown's yard look to you as though they had the blight?

MR. MCMURRAN: Yes, they looked like this (showing specimen).

MR. LITTLEPAGE: Did you notice that tree just across the fence? The reason I ask the question is that if that is blight out there, then that tree right across the fence is very likely resistant, because I have noticed that those walnuts have had this on and off for six or seven years. The limbs of the two trees are within twenty feet of each other.

MR. MCMURRAN: Well, that is a very encouraging point.

MR. LITTLEPAGE: I didn't think that was blight. All those trees at Georgetown that I have observed have that condition on them, more or less, except that one tree.

MR. MCMURRAN: Yet, isn't it true that they bore pretty good crops of nuts, nevertheless?

MR. LITTLEPAGE: Oh, yes.

MR. MCMURRAN: Well, that was the point I had in mind. Of the two trees one bears every other year, and the other bears heavy crops every year.

DR. MORRIS: You see the same thing at the Experiment Station in southern California. One tree will be absolutely resistant to blight; the other

will be all killed. And down at Whittier, perhaps, seven-tenths of all the trees will be badly affected with the bacteriosis, and the others not very much affected, so that, apparently, it is largely a matter of this cynips, which introduces the bacteria, selecting certain trees. Certain walnuts are very much affected, and the involucre looks very much like that of these nuts (showing specimens), but, on examining them, I found a very large number of small larvae beneath the involucre. I sent some of them to the Connecticut Experiment Station and some to Washington, but they didn't tell me what they were. Those same larvae I found in one black walnut on my place, which is very heavily infested with them. Most of the nuts drop because of the injury to the involucre. I haven't determined the species yet. I don't know whether the larvae come first and the bacteriosis second, or whether it's the other way around.

THE PRESIDENT: Are there other persons who wish to give themselves a chance of asking Mr. McMurren a question? I have a question that is troubling me. Perhaps the house can throw light upon it. I had a number of Persian walnuts, Vrooman Franquette and Mayette, grafted on black, and by the Fourth of July they were growing nicely, with tops all the way from four to twenty-four inches long, and then the tip got black and the blackness went down. I sent a sample to Mr. McMurren. The leaves first died and then the twigs.

MR. MCMURREN: I received that, but it was so dried when I got it it was impossible to make anything out of it. I have seen the same thing on pecans, only in those cases the leaves just got black and fell off, and we never have been able to assign a reason for it.

THE PRESIDENT: Am I the only man that has had that experience?

A MEMBER: I had this year the same thing on the Vrooman Franquette, but it recovered and has made excellent growth since.

MR. MCMURRAN: Have yours subsequently lived?

THE PRESIDENT: No, they subsequently died. (Laughter.)

MR. J. F. JONES: I had that experience this summer. The new growth was very tender and took blight very readily.

THE PRESIDENT: Well, this doesn't appear to be blight.

MR. JONES: If the English walnut starts late and the tender growth comes in the hot weather, the sun will kill it.

THE PRESIDENT: You have described my conditions. These are late grafts. Have you had that same experience with late grafts and not with early ones?

MR. JONES: Yes, sir. The blight will show itself in the specks on the twig.

THE PRESIDENT: Well, do you mean on this sunburn?

MR. JONES. No, sir.

THE PRESIDENT: There were no specks in this case. Has any other member any question on the blight? I want to call attention to the fact that we have here in this room tonight nearly every one who is studying the question in the eastern United States.

MR. MCMURRAN: Mr. President, I would like to say that we would like to get all the information we can on it.

A MEMBER: According to my observation, the blight is not going to do much to the tree, because the tree here makes its growth and hardens up

before the blight comes. The blight, you see, must have moisture and heat to work, but it comes in just right to catch the nuts.

MR. R. L. MCCOY: Mr. President, some mighty strange things happen with grafted and budded English walnuts, and I believe I could ask questions that would puzzle a school of wise men. Now, none of the answers here will stand up very well. For instance, Mr. Jones says this dieing back is due to late grafting. Well, I had some Holdens that we budded this last June a year ago, that suddenly, all at once, along in July this year, proceeded to quit business, and quit clear down, and the root died, too, the black walnut root. It is a serious question in my mind whether the black is the best stock to be used or not. Mr. Jones and Mr. Reed have good success grafting the English on the black. We don't down our way. Both of those men are in regions where the land is inclined to be alkali. The land where my orchard is, and where Mr. Littlepage's and Mr. Wilkinson's orchards are, is inclined to be acid. I am of the opinion that, to make a success of the English walnut, we are going to have to use lime, and use it extensively, not only in the nursery, but until the time when the trees begin to bear.

THE PRESIDENT: It is one of the common pieces of knowledge of all the agriculturists of France that the walnut does well on lime soils, and they don't expect it to do well on acid soils.

MR. JONES: Mr. President, I think, if Mr. McCoy will examine his trees, he will find that the root dies first.

MR. MCCOY: Well, why should they rot?

MR. JONES: That is like a good many other things, Mr. McCoy. We don't know why.

MR. LITTLEPAGE: A pecan top-worked on a water hickory will sometimes kill the whole tree, top and all. It is the top that does it.

MR. M. P. REED: This year we made some observations of different varieties, as to time of leafing out, and we found the Eastern varieties leafed out about the first of April, and the Franquette and Mayette about the fifth of May, and one variety we got from the Department, No. 39,884, didn't leaf out until the twenty-fifth of May. That seemed to indicate that the French varieties were going to prove better than the Eastern varieties, because late frosts cannot hurt the blossoms.

MR. LITTLEPAGE: That is correct. I watched them this spring at Mr. McCoy's. Franquette and Mayette, over there and with us, were anywhere from ten days to two weeks later leafing out. Some of the buds were entirely dormant and some just bursting when many of our Eastern varieties were in full leaf. But my experience here in Maryland on walnut trees from all sections was that every one winter-killed except one Nebo tree and a top-worked Potomac. I have a Potomac which has made ten to twelve feet of growth, and it didn't winter-kill the slightest, and my Nebo tree hasn't winter-killed any, but the Franquette, the Meylan, the Rush, the Holden, and several others winter-killed very badly. At least, Mr. McMurrin said that was what it was, and I thought it was, too.

THE PRESIDENT: Well, Mr. Littlepage, isn't "winter-killing" rather a relative term, dependent partly upon the climate and partly upon the condition of the tree at the end of the growing season? Was there anything back of your statement, any late growing, or something of that sort?

MR. LITTLEPAGE: Well, they didn't grow any later than the Potomac grew, but that tree was top-worked about five or six feet above the ground and I think that makes them hardier.

A MEMBER: Were the winter-killed trees cultivated late?

MR. LITTLEPAGE: Yes; and fertilized heavily.

THE PRESIDENT: Haven't you answered your own question?

MR. LITTLEPAGE: Well, I won't grow trees if they do not grow better than that.

MR. MCCOY: Mr. Littlepage may think he has answered this question, and these other gentlemen may think they have answered it in a different way, but there are some rather peculiar phenomena there. I don't question the sincerity of these gentlemen, but I don't think they have answered the question. Whenever you transplant these trees and whenever you get to growing them in big quantities, you will have certain peculiar phenomena that you are not certain at first as to just what is the cause. Mr. White is just as near right when he says they kill in July as Mr. Littlepage when he says they winter-kill in December. And I will just say to people who buy walnut trees from our firm that when they transplant them under the same conditions as Mr. Littlepage, they may expect similar results.

MR. LITTLEPAGE: I have never seen a northern pecan winter-kill.

MR. MCCOY: Oh, I have.

MR. MCMURRAN: Mr. President, this term "winter-killing" is a little bit misleading, and it has been a matter of discussion in the National Nut Growers' Association for several years and a great loss to many Southern pecan growers. A very common statement that one hears down there is "Why, our trees don't winter-kill. We don't have cold severe enough to kill them." But they do. It isn't a question of severity of cold, but suddenness of change. For instance, in southern Georgia one year, we had a rainy period in

October; about November 20th there was a hard freeze. A number of orchards which had been fertilized late in the fall were almost wiped out. If it were not due to the fact that the term is too long, and we could say "damage due to sudden temperature change," it would convey the idea exactly. I saw trees injured in the fall of 1914 that didn't die until September of the following year, and I have a number of photographs in my office.

DR. STABLER: I believe, Mr. President, that the stimulation of growth late in the season has a great deal to do with the winter-killing of trees and other plants. I have noticed it in clover and alfalfa, and I have noticed it in peach trees.

THE PRESIDENT: I think Dr. Stabler has stated a very well-known principle, not only of horticulture, but also of agriculture. Last year we questioned Mr. W. C. Reed as to the condition of a certain top-worked, heavily forced, black-walnut we had seen the year before at Vincennes. We were confirmed in our belief that the tree was dead, but that another tree budded at the same time with the same bud-wood and not forced, lived. We had a dry summer that year, a wet fall, twenty degrees below zero at Christmas, dead apple trees. I suspect that Mr. Littlepage has a problem in the balance of tillage and top-working.

DR. STABLER: I think if he visits his neighbor, Professor Waite, he will find out how to manage trees so they won't winter-kill, because he knows how to fix it. (Laughter.)

MR. LITTLEPAGE: I treated the trees just like the pecan. I have never seen it possible yet to over-stimulate a pecan and winter-kill it. I don't say it isn't possible, but I have never seen it.

THE PRESIDENT: I can show you a few.

MR. M. P. REED: Mr. President, we have that condition in the nursery row.

MR. LITTLEPAGE: Well, those are grafted, are they not?

MR. M. P. REED: Grafts and buds, both.

DR. MORRIS: In regard to the grafting of the Persian walnut upon black walnut stock. In Connecticut, we have three species of mice—the common field mouse, the pine mouse, and the white-footed mouse. These mice all follow in the holes of the moles, and they are very fond of the bark of the Persian walnut, and will destroy a good many of them. Now, with the black walnut, on the other hand, when one of these mice comes along and takes a bite of that, he shuts one eye, cleans his teeth, and then goes on to something else. (Laughter.)

Now, in our country the soil is practically all acid. The black walnut will grow in pretty acid soil. The Persian walnut almost demands a neutral or alkaline soil. So, for Connecticut there is no doubt that we really need the black walnut stock for the Persian walnut.

THE PRESIDENT: Any further problems that are vexing the orchardist with regard to the Persian walnut? If not, I think this is a suitable time to bury it until next year. Col. Van Duzee, a man who has had more experience with the pecan than almost any one else in the room, has kindly consented to make his contribution at this time. Col. Van Duzee.

COL. VAN DUZEE: There is a longing on the part of a large percentage of men and women that I meet to escape from conditions which do not seem to be especially favorable in the large cities, and to get away into the safety of the country. I believe that nut tree growing offers one of the safest of those outlets. I believe that a nut orchard should be a part of a general

farming operation. I want to give you my ideas about inter-crops. Fifteen years ago the doctors gave me three months to live, drove me out of my business, and away from my home to prolong the agony for a few weeks or months, and I found, among my orchard trees, a reasonable amount of health which, to me, repays a greater value than I could reckon in dollars and cents. It has given me the privilege and the opportunity of removing myself from the turmoil of the city and the conflict of the business world to a peaceful, quiet existence, that, to me, is very much more satisfactory. Now, that is an inter-crop.

Down in Florida, when we used to get together in our citrus seminars and in our horticultural and agricultural meetings we used to try and make a man say on what class of soil his home or orchard was located, so that we might get his viewpoint. For the successful nut orchardist, in a small way, must, of necessity, be a successful agriculturist. He must understand soils. You can't have successful inter-cropping without understanding soils, and, therefore, I can't tell you definitely what would be good for your northern soils. But I can tell you this, that the first thing to do when you have an orchard problem to consider is to make an exhaustive survey of the character of the soil. If it is a fresh, recently cleared piece of fertile soil, under favorable conditions, I am satisfied you don't need very much in the way of inter-cropping. On the other hand, if you select for your orchard site a piece of land that has been worked to death, I believe it would be well to inaugurate a system of inter-cropping that would have for its object the building up of that soil and the improvement of the environment for the roots of those trees. In the South, we are favored with twelve months of growing weather. We plant our crops throughout the year. I am just about beginning now to plow for my oat planting. I am going to pasture those oats all winter with hogs and cattle. We will harvest our oats in May. We then follow them with a legume which will restore the fertility to that soil. In the present condition of the market for commercial fertilizers, I believe we have

gone beyond the point where any man can afford to use commercial fertilizers to any great extent on ordinary crops. I believe it is possible to go too far in the stimulation of a growing orchard. My opinion is that a series of inter-crops that results eventually in a large deposit of nitrogen, such as we get from several leguminous crops plowed under, will have a tendency to bring that orchard into a condition—I am speaking now, you understand, of the pecan—where it will be susceptible to disease and winter-killing.

If you have followed me as far as I have gone in this, you will begin to see at once that the men who are going to be successful in solving these problems are the men who are going to learn the game. Among the human family, you know, we have a stock phrase that we use sometimes when a man dies and we don't understand the cause of his death. We say "He died of heart failure." That is a convenient thing to hide behind. "Winter-killing," to my mind, is another such term. It is used, for instance, in a case where an individual tree, for some reason or other not quite understood, "passed away." (Laughter.)

I have been fifteen years in the growing of pecan trees in the South, and I am free to confess that the most disturbing element in my life at the present time is the fact that we "have known so many things that weren't true." We have gone ahead fully believing that our course was justified, that it was well digested, desirable in every way, and suddenly waked up to find that we were radically wrong, or, at least, that there was a very open question as to whether we were not absolutely wrong.

To the person of limited means the idea of being able to produce a nut orchard at very little expense is very attractive, and my heart goes out to people in that condition because I have been in that condition myself and passed through it. Ten years ago I bought a piece of land for forty dollars an acre, and planted seventeen pecan trees on each acre. It cost me twenty-five

dollars an acre to lay off the land, dig the holes, and plant the trees nicely, with about a half pound of bone meal mixed in the soil in each hole. I carried that nut orchard on, using some inter-crops, up to one year ago, when it finished its eighth year of growth, and, without burdening you with the minute figures, I am going to say we have sixty-five dollars charged up to it, and it will take \$185 more. Now, there is \$250, if I haven't made any mistake. I planted among those trees nursery stock, and I sold off, during the time that those trees were growing, nursery stock to the value of, we will say, \$250, making my inter-crops pay the expense of cultivation and interest on the investment up to that time. So don't forget that. Now, this is a case where we are going to balance our books, as every business man does, and every farmer ought to. I have, up to the time those trees were eight years of age, invested approximately \$250, and have received back not only that, but the interest on the investment. So, at eight years of age the orchard cost me nothing. Now, that would be the way a great many people would figure that proposition. I can't do it that way. I am going to charge that orchard with \$250 an acre for supervision. Now, above that line (indicating on black-board) it looks as though that orchard had been built up for nothing, and below the line you see a debit of \$250 charged against that orchard. There is not one man in a hundred that contemplates a proposition of this kind that is willing to charge his orchard up with the gray matter that he puts into it. But there was an inter-crop in that orchard, of health and satisfaction, which is worth more to me than my services, so I will put that in here as \$250. (Laughter and applause.) Now, I walked across this morning—I like to walk, and I came across the park. I saw a monument right over here in a little iron circular enclosure, erected in honor of Andrew Jackson Donald, a man who died several years ago, the man who was partly responsible for the magnificent landscape gardening effect of which this building is a part. It said on the monument this: "His life was devoted to the improvement of the national taste in rural art." Down below it said: "His

mind was singularly just, penetrating and original." Any man ought to be proud to have that sort of thing engraved upon his monument, and, gentlemen, any man who will go out and plant nut trees like those you saw this afternoon, ought to have a monument under those trees expressing sentiments similar to these, because he has done something which remains after him, and it is one of the most worth-while things that any human being can do. That is one of the other valuable things about a nut orchard.

Now, this nut orchard—this is no myth—this is a practical proposition. I was practically bankrupt when I went there. It is paying now in a small way, and will pay more later on, and I am going to leave it to my children as one of the safest and sanest investments that I could leave them, and I want to say, ladies and gentlemen, that the consciousness of possessing something of that sort, which can't be stolen, can't run away, is another inter-crop that is grown among those trees.

I sometimes tell a story of a little two-horse farm down in the South. I drove fourteen miles out into the wilderness to find some seed nuts to plant this nursery with years ago. I found there an old home which was the central home of a large plantation in days gone by, and there were half a dozen—perhaps seven or eight—magnificent, great pecan trees about the lot, and a vegetable garden at the back of the home. Those trees were loaded with nuts. There was a young man there—one of the most pitiful things that I ever saw in my life—a fine young man—magnificent character, and recently married, making his home in this old tumble-down house, making his start in the world there. He didn't own this land—rented this fifty or sixty acres of open land, and these trees went with the two-horse farm. I said, "My friend, you must receive quite a little income from those nuts." "Yes," he said, "I sell the nuts from those trees every year, for more money than I make from the two-horse farm."

I heard of another case down in north Florida where two girls were left absolutely dependent upon their own exertions, and they were girls who had been reared, as some of the Southern ladies have been reared, to be dependent on others. They didn't know how to go and fight the world for a place. They were a little too far along, perhaps, to take up that sort of battle. There were two pecan trees in front of that old homestead, and the old homestead was all that was left of the family fortune. It was furnished, had a cow in the back yard, and a garden, and a few Scuppernong grape vines. These two pecan trees in the front yard gave those two women approximately three hundred dollars worth of nuts per annum. They were magnificent, great, big pecan trees, and they lived from them the balance of their lives practically, with the help of the other things I have mentioned.

Inter-crops are nothing more nor less than the evidences of the master mind directing the problem of handling the soil in which the orchard is growing. Now, just simply go right down deep under everything, pay absolutely no attention to the wonderful stories that the promoters tell you (laughter), keep your money, save it, use it, and spend it—yes, but recognize this one thing, that the most important element in success in the small orchard, as part of the rural or suburban home, is a knowledge of agriculture and horticulture. It is one of the most fascinating studies in the world, and I have no doubt but what you will find that you can go right along inter-cropping with vegetables and other crops, bush fruits, strawberries, and all those things for the first few years after you plant your nut trees, and even if they all die you will have been able to break even on the commercial side of the proposition, and then you will have the additional years of experience, which no nut orchardist can dispense with. You can't buy it with money or get it out of books. You have got to dig it out of the ground yourself. (Applause.)

THE PRESIDENT: I am going to take the liberty of emphasizing one point the Colonel made. He told you about the great number of things they knew down South that were not so. I wish to give some geographical spread to his generalities. We are in the same condition in the North. If you will stop and look clear through an agricultural idea, you will be astonished, ladies and gentlemen, absolutely astonished, to see how, mostly, we don't know it. The other day I happened to be walking through an apple orchard with the official horticulturist, and in response to some remark he made I asked: "Do you know that, or do you think it?" "Has that been experimentally proven?" He answered: "No, it has not." Most of the things we read in the books and hear in this place and other places we don't know. We think we know, but when we come to a show-down we really haven't got experimental data. I know of no people to whom that thing needs to be emphasized more than to the Northern Nut Growers' Association.

SATURDAY, SEPTEMBER 9TH, AT 10.30 A. M.

Meeting called to order by the President.

THE PRESIDENT: The first order of business, I believe, will be the report of the Nominating Committee.

THE SECRETARY: The report of the Nominating Committee is the following:

For President, W. C. Reed, Vincennes, Indiana; Vice-President, W. N. Hutt, Raleigh, North Carolina; Secretary-Treasurer, Dr. W. C. Deming, Georgetown, Connecticut.

MR. LITTLEPAGE: Mr. President, I move that the report of this Nominating Committee be accepted and adopted, and these officers declared elected.

THE PRESIDENT: Do I hear a second?

A MEMBER: Second the motion.

THE PRESIDENT: The nomination of this list is moved and seconded. Is there any discussion? If not, all those in favor will say "aye;" opposed, like sign, it is carried, and they are elected.

MR. LITTLEPAGE: While I am on the floor, I want to read a resolution which I have drafted, and I will read the clauses separately:

"Owing to the fatal character, the unchecked and rapid spread of the chestnut blight," is there any question about that? If that is not true, let somebody hold up his hand.

"Owing to the fact that it has been widely disseminated through the shipment of nursery stock;" is that correct?

"Owing to the fact that in the first stages the disease cannot be easily detected;" is that true?

"Owing to the fact that the young trees apparently have a temporary immunity from the disease;"

"Therefore, the Northern Nut Growers' Association believes that the continued free shipment of chestnut nursery stock will be productive of endless destruction of property in those places where the chestnut trees haven't yet the disease." If that is unsound, why, somebody say so.

"Therefore, be it resolved, that we, the Northern Nut Growers' Association, suggest that the Secretary of Agriculture prohibit the shipment of chestnut nursery stock, except in the localities known to have the blight, and that with each permit for shipment shall go a bulletin or circular giving the important facts about the chestnut blight. The only exception to this regulation shall be the shipments for experimental purposes, and such shipments must have the above mentioned permit, and the name of the nursery from which such trees have come, and must be inspected by Federal inspectors." I assume, of course, that inspection is a general inspection. I don't mean each particular shipment. If there are any questions about that, why, I will let the chair answer them.

DR. ULMAN: Mr. Littlepage, I would like to ask a question, or, rather, offer a criticism. If I understand you rightly, you say, "except in the districts where blight is prevalent." As a matter of fact, sir, the particular nursery that advertises the chestnut tree works within a radius of possibly 250 miles of Rochester, in a district where there are many prospective horticulturists. One of the things that impressed me more than anything else in the report of the Secretary was the fact that we have lost a large number of members, and that we haven't attracted to ourselves many new members. So far as my personal experience goes, if I were to choose the one method of being most thoroughly disliked, it would be to ask my neighbors, particularly those who do not know me, to become members of any kind of a nut association. There is a glamour about planting, and it is a sort of a disease with some people, year after year, to seek for novelties. These nut tree advertisements that read so well attract many purchasers. Right here in this section people are buying nut trees that they are going to plant in a blighted district, and these people, when they see what utter failures they have, will be so disgusted with nut growing that when you approach them you cannot talk nuts to them, and you will never have them join the Association. More and more are leaving the Association, and very few new ones are coming in to take their places. So I think the resolution ought to be changed.

THE PRESIDENT: In what respect would you have it changed?

DR. ULMAN: To apply generally.

MR. LITTLEPAGE: Yes. Well, I agree very largely with what the doctor says. I have always felt that the success of this organization—the success of the nut industry as a whole—depended upon its being upon an entirely truthful, fair and honest basis. I would rather see a crooked cashier in a bank than a crooked nurseryman or tree man. The cashier you can check up at 4:30 every afternoon; you can't check up the crooked tree man for about

ten years. I think the worst of all discouraging things to people who want to go to the country to build up farms and homes is to run into alluring, but misleading, advertisements. I have an abounding faith in tree culture. I think that the pecan tree, the black walnut, varieties of the English walnut and of a number of other nut trees, are going to make it most possible and more desirable for men to go to the country, but I think the success of those things depends upon giving those people, as far as possible, facts, and not misleading them. Wherever a man sets a tree that is a failure you have a man as a failure generally as a tree man, and wherever you get a man to set a tree that succeeds, you have a living, walking advocate of the tree business. This Association has been fortunate all along in its policies. It has always stood against the fraudulent promotions; it has always stood against fraudulent nursery stock; it has always stood against fraudulent representations, and I think, for that reason, that its future is reasonably safe, assuming that is its continued policy.

THE PRESIDENT: Do you accept Dr. Ulman's amendment?

MR. LITTLEPAGE: I accept it.

MR. JONES: As I understand the resolution, it applies to nurseries in the infected areas.

MR. LITTLEPAGE: Yes.

MR. JONES: I believe it is practically impossible to grow trees in an infected area without sending out the blight, but if a man is isolated, like Mr. Riehl, at Alton, Illinois, he can grow trees without danger of sending out the blight.

THE PRESIDENT: Well, the resolution permits him to do that.

MR. M. P. REED: Did I understand Mr. Kellerman to say the Department hasn't authority to quarantine against such things?

THE PRESIDENT: No. The point brought up was the theory that lay back of the quarantine. The speaker made the point that shipment of infected trees was killing the tree aspirations of the people who ought to be developing the nut industry. Every time a man buys a chestnut tree and it dies with blight that man is chilled out of business. Now this resolution doesn't cover that man. It is based on the ground of injury to the industry. You can't very well define the limits of where the blight is not, but it can be fairly well defined as to where it is, and that is up to the Department.

MR. LITTLEPAGE: The resolutions are offered in a suggestive way, Mr. President. If the Secretary wants to turn the suggestion down, we will meet again next year any way.

PROF. CLOSE: I would like some information as to how you propose to take this matter up with the Department. I was present a year or so ago at a hearing before the Federal Horticultural Board—I don't know whether any one else present was there at the time—but the whole thing hinged largely on Colonel Sober's attitude in propagating and sending out the Paragon chestnut, and I think the Department—the Federal Horticultural Board—originated the question that you are discussing now, and Colonel Sober came there with a whole lot of pretty good information, and people to back up what he said, and the Department put up a mighty poor show, I tell you. I was ashamed of what the Department men had to say, and Colonel Sober won out hands down. Now, if this question comes up again, it will be referred, no doubt, to the Federal Horticultural Board, and you will need a good, strong representation, with plenty of facts back of you, and if you can put up a strong enough case there is no doubt but what you can establish

this quarantine. But I would hate to see the question taken up again and floored as easily as it was at that time.

THE PRESIDENT: Prof. Close, I have read part of the testimony.—I was not present at the meeting—and when one considers the number of things that were said at that meeting that are not so, and the amount of other evidence that has come up since, I think the defenders of the public will have the material to make a much stronger presentation than they did then, and, what is more, I think some of them will be there. Of course, when a man has a possibility of getting a quarter of a million dollars out of a lot of junk, he can spend money to hire people to say things, and when "the dear public" is paying nobody to go, as was the case last time, nobody goes. If that hearing comes again, I think some from this Association will be present.

PROF. CLOSE: That is just the point I want to bring up. You have got to be there with the information.

MR. LITTLEPAGE: I just want to say a word, rather endorsing what Professor Close has had to say. The Department of Agriculture, the quarantine board, or anybody else, can't go out of their limitations and get testimony. If we think there ought to be a quarantine, then, whenever there is a public invitation sent out, such as was sent out before, we ought to have the nerve to go down there before the Department officials and tell them the truth. It is very easy for us to stand up here and write papers and articles criticising the quarantine board and the Department of Agriculture. If we have anything to say about these things we ought to go down there and say it. If other people come there and present facts as a matter of record, the Board can't entirely go outside of those facts and decide a case right out of the clear sky. If this organization wants to be effective, it ought to appoint a committee to present those things before that Board.

Resolution adopted.

THE SECRETARY: I have had in mind some time the idea involved in this resolution, which I have hastily drawn up.

"Since the principles underlying the successful and economical propagation of nut trees are not yet thoroughly understood or generally known, and much effort is being wasted and much disappointment incurred in unsuccessful or partially successful efforts in propagation.

"Resolved, that it is the sense of this meeting that systematic and controlled experiments be made, under the direction of the Department of Agriculture, for the purpose of determining the principles underlying the successful propagation of nut trees in all sections of the country."

DR. ULMAN: Second that motion. (Carried.)

MR. C. A. REED: I would like to present an invitation to meet at Battle Creek.

MR. ROPER: Petersburg invites us to meet at Petersburg.

THE PRESIDENT: Those matters are settled by the Executive Committee.

MR. T. P. LITTLEPAGE: Would it not be well, Mr. President, to determine upon a meeting place now, and let it be known, so that everybody can prepare for it? Being a member of the Executive Committee, I would prefer myself that the Association take the responsibility for deciding the meeting place. If these meeting places are selected in advance, it makes it possible for a good many people to plan their vacation trips to fit in. In order to get the matter before the Association, I move that the Society determine right now the next meeting place. (Seconded.)

MR. OLCOTT: I think Mr. Littlepage's motion is of more than ordinary importance. The Association, heretofore, has left that matter, very properly, perhaps, to the Executive Committee. The result is that little or no attention is given to the place of meeting until thirty or ninety days before the date of that meeting. It would be very much better if we knew several months ahead about the meeting, and I think we would have a larger attendance and more enthusiasm. The American Association of Nurserymen names the date at the time of their meeting for the following meeting, and most other organizations do the same, and the results are quite perceptible.

(Motion carried.)

THE SECRETARY: We have an invitation from the Evansville Chamber of Commerce, one from the San Francisco Convention League, to meet at San Francisco, one from Sears, Roebuck & Company, to meet in Chicago, and enjoy a luncheon at their expense and a trip through their plant; one from Dr. Morris to meet at his place, or to meet at Stamford and spend as much time as possible on his place. We could meet in New York City and visit Dr. Morris' place very comfortably. We have an invitation from Petersburg, Va.; one to meet with Dr. Kellogg, at Battle Creek, Mich., and one from Mr. Rush to meet at Lancaster, Pa. We have had under consideration a proposition to meet somewhere in the South, possibly with the Southern Nut Growers. Those are all the invitations that I know of.

MR. C. A. REED: May I make a remark right here? It seems to me that before we decide on the place of meeting we ought to take into consideration what we are going to any of these places to accomplish, and the time of year that we want to go there. Now, if we go to Lancaster, or to almost any of these other places, we ought to have a summer meeting when we can go out and see the trees, but if we go up to Battle Creek we could just as well go there in the winter time. The purpose of going there, as I

understand it, would be to lay emphasis on the subject of nuts for food. Whether we want to take our time now for a meeting, to emphasize that, or whether we want to see nut trees growing and discuss cultural problems, is a question to be decided.

THE PRESIDENT: In the absence of a definite method of procedure on this question, which we never before handled in this way, the chair is entirely willing to receive instructions, but I suggest that we have a rising vote for one place after another, and that the place receiving the greatest number of votes gets the convention.

MR. M. P. REED: We have seen trees in nurseries for several years, and I think that now we ought to select some place where we can get other and broader ideas on nuts. I think Battle Creek would be the best place.

THE PRESIDENT: Does any other favorite son or neighbor wish to make a speech in favor of his own or nearby city?

MR. HENRY STABLER: It appeals to me very strongly to see Dr. Morris' experimental grounds at Stamford, Connecticut. As I understand it, he has the greatest collection of nut-bearing trees in the United States, and looking over this would help us in a fine way.

MR. JAMES H. KYNER: Mr. President, I am not a member of this Association, but for a number of years I have been trying to grow nuts. I am very much interested in the subject, and I would like to know if I have any rights on this floor.

MR. LITTLEPAGE: It costs you two dollars to vote.

MR. KYNER: All right, I will just give two dollars.

MR. LITTLEPAGE: I move that the gentleman be accepted as a full member now and have full authority to make speeches. (Laughter.)

THE PRESIDENT: The gentleman will proceed to the making of his speech.

MR. KYNER: I have no speech. I simply want to vote "aye" for Stamford and New York.

(Vote taken.)

THE SECRETARY: The result is as follows: For Evansville, 2; Stamford, 10; Battle Creek, 3; Petersburg, 3; Lancaster, 1; for Chicago, San Francisco and the National Nut Growers, 0.

THE PRESIDENT: We are now ready, I believe, to proceed with the technical part of the programme. The chair would like to call for information as to the relative behavior of the Northern pecans, top-worked or transplanted. Is there, for example, any evidence anywhere as to the fruiting of any Northern pecan except on the parent tree?

MR. MCCOY: Mr. Wilkinson, in Indiana, has some top-worked bearing trees.

THE PRESIDENT: Unfortunately, they are right at home. What varieties has he?

MR. MCCOY: The Major.

THE PRESIDENT: And how old was it before he top-worked it?

MR. MCCOY: Three years, I think.

THE PRESIDENT: On pecan?

A MEMBER: Yes, sir.

THE PRESIDENT: The Major bore the fourth year, three years old. I believe that is the first record we have of that sort. Has any other borne?

MR. MCCOY: A good many of my young trees bloom in the nursery, but I don't think they succeeded in setting any nuts.

MR. M. P. REED: We have had some two-year trees in nursery, grafted. Most all of them bloomed when two years old—the staminate but not the pestillate blossoms.

THE PRESIDENT: I had a staminate blossom the third season on Butterick in northern Virginia.

MR. HENRY STABLER: Is there any difference between trees budded from young trees in the nursery row which are not in bearing, which have a growth very much resembling water-sprouts, and those budded from bearing trees?

PROF. HUTT: Mr. President, I can't give any experimental data on that line, but the common practice of nurserymen in taking their bud-wood from the nursery stock has been in use for years and years, as with peaches. Very seldom do the nurserymen go to the original trees and get their buds, but it is cut from nursery stock, because it is in a fine condition to work. I think that trees propagated from young, vigorous wood, cut in the nursery, are all right. I am not so sure as to how long it is before they come into bearing.

MR. HENRY STABLER: I don't mean to say it is an undesirable practice to bud from the nursery row, but is there any difference in the time of coming into bearing?

THE PRESIDENT: I spent a very considerable amount of time and money in that belief, but at State College, they made an elaborate test, and they have found no difference between the tree from a water-sprout and one from the bearing tree.

MR. JONES: It is not practicable to propagate very largely from young trees, either fruit trees or nut trees, but there is a good deal in maturity of the wood. The plan we follow is to have mature plots and graft from these old trees. That gives the best wood for nursery propagation.

THE PRESIDENT: Keeping the same tree?

MR. JONES: Yes, right along. That costs a little more money than to propagate from the nursery, but we think it is better. We get better results.

THE PRESIDENT: How have the different varieties of the northern pecan shown up with regard to speed of growth? At the present time we are practically ignorant as to which of seven or eight named and propagated varieties to count on. Apparently, the Busseron has the record for early bearing, with the Major as second. What about the record of the trees for making wood, not in the nursery row, but after it has been transplanted and put in the field? Is there any distinct leadership of one Northern pecan over another in making wood?

MR. LITTLEPAGE: If the members who go out to my place this afternoon will observe closely they will have a chance to see something of the tree growth for the first three years. They will have a chance to observe the Indiana, the Busseron, the Kentucky, the Green River, the Major and the Posey, with three year's growth. They will see a row of Green Rivers, some trees nine feet high, and others that haven't grown two feet. That is the individual tree variation, however. They will see certain characteristics running clear through.

THE PRESIDENT: Well, Mr. Littlepage, it is a job to go and get exact results from another man's experimental ground. Which is the winner for speed, Mr. McCoy?

MR. MCCOY: Well, I know more about how they grow in the nursery than I do when transplanted. I haven't transplanted as many trees as Mr. Littlepage, but, of course, the tree will act very similarly in the nursery to what it does after you transplant it. We have learned at a glance to tell the difference in the varieties. We don't have to go to the books or to the stakes to tell each particular variety, as each variety has its distinguishing characteristics. For instance, the Kentucky and the Butterick and the Busseron are all inclined to grow up. I don't know why that should be true, but they all have the lumber characteristics. The Kentucky grows in the river bottoms surrounded by lumber trees. Now, the Posey doesn't grow very tall, but it grows a wonderful stocky, sturdy tree, and has leaf stems as long as my arm in the nursery. Of course, each particular wood has its color characteristics. But one thing I observed was that in the other nurseries they don't color up as they do in mine. For instance, at Mr. Jones', it will puzzle me sometimes to tell which variety it is by looking at the wood. Of course, after he would say "This is Butterick" or "Busseron," I could see, probably, the characteristics, but there is a little difference in the color of the wood.

THE PRESIDENT: Have you found any difference between these three trees as to attainment of height?

MR. MCCOY: Well, I suspect that the Butterick is the fastest grower of them.

THE PRESIDENT: What is the slowest?

MR. MCCOY: The Indiana, I guess.

A MEMBER: How does the Major behave?

MR. MCCOY: The Major is a very slender, tall tree. The Green River is inclined to be spreading.

THE PRESIDENT: That testimony as to the Indiana being a slow grower—does anybody verify it?

MR. LITTLEPAGE: Same thing in Maryland, Mr. President—slowest grower I have.

THE PRESIDENT: Is that sufficiently marked to make it best for us to hold up its propagation until it has shown some reason for being grown?

MR. LITTLEPAGE: I don't know. The Busseron and the Indiana, which is supposed to be a seedling of the Busseron—the Busseron outgrows the Indiana in Maryland five times. But the Indiana is a thin-husked nut. The Busseron, on the other hand, is a thick-husked nut, a fine place for the nut worm if he ever gets bad. There are a lot of such things that you have to think of.

MR. MCCOY: I visited most of these parent trees this year. They are all centered around Evansville. There is no crop on the Busseron. The Indiana will, perhaps, have a peck. In the month of May, in Kentucky, and Indiana, and Illinois, we had rains continually. I have often heard the expression from the Southern nurserymen that "the pecan is caught with the frost." Now, that is clear out of place with us. We all smile at the idea that an Illinois, Indiana or Kentucky pecan would be caught with the frost, which never affects them. But the rains always affect them. If the month of May is a beautiful, dry, clear month, you can gamble on the pecan crop. Now, this year we won't have much of a crop. The Warwick will have a gallon or two,

and the Kentucky crop is a failure. The Green River and Major we didn't get to, but I suspect that very few of our own trees will have a crop this year.

MR. M. P. REED: Mr. McCoy, I was up there last week, and the Busseron has probably four times as many nuts as the Indiana. It has a light crop, while the Indiana has a very light crop. (Laughter.)

MR. MCCOY: When were you there, Mr. Reed?

MR. M. P. REED: Last Sunday.

MR. JONES: You can't judge a pecan by the growth of the tree. You take a pecan that makes a thick head and lots of limbs, and it is very likely to be a heavy bearer. On the other hand, a nurseryman likes a variety that makes a tree, you know.

THE PRESIDENT: On your criterion of a bunched top, which of these eight varieties we are now propagating is the most promising?

MR. JONES: The Butterick appeals to me.

THE PRESIDENT: Is the Posey in the same class?

MR. JONES: The Indiana makes a thick head.

THE PRESIDENT: Does any other do that?

MR. JONES: The Green River is inclined to on the mature block, but not the first year in the nursery.

MR. LITTLEPAGE: Mr. President, in view of the fact that this meeting is reported, and that what we say will go into the official records to be read by lots of people who can't come and examine us, it might be understood that there would be some question about the bearing of these Northern pecan

trees. As a matter of fact, I am surprised that any of them bore any nuts this year when I think how hard Mr. McCoy and Mr. Reed and myself have cut them for bud-wood. As a matter of fact, our opinion is that these Northern pecan trees are all excellent bearers, as the bearing reputation goes with pecan trees. I have watched them pretty carefully, and the best evidence of what I think of them is that I am setting them in my orchard. For fear that the minutes may leave the impression with some casual reader later that these trees bear a quart, and two gallons, I just want to say that if these gentlemen put into the record the amount of nuts that they know the Green River, the Butterick, the Posey and Major have borne—for instance, six weeks ago I bought sixty pounds of Posey nuts from a certain tree. The man who counted them counted 120 pounds on the tree, and if the boys around were as active as when I was a boy, I bet he didn't get more than half of them.

THE PRESIDENT: This is the time of year when the squirrels get nuts, and
I expect they got after the trees, too.

MISS LOUISE LITTLEPAGE: Why does the rain affect the nuts, and why in that certain one month?

MR. MCCOY: In our latitude the pecan blooms somewhere near the twentieth of May, from that probably up to the twenty-fifth, and the pollen is scattered by the winds, and, if it rains at that particular time, the female bloom perishes, and we have no pecans. I think the pecan depends entirely upon the winds.

THE PRESIDENT: We have been hoping all the time that we would have a chance to hear from Prof. Hutt on the relation of the hickory stock to the pecan top. A good many persons have experimented with it, and papers are

giving, from time to time, glowing accounts of the pecan tree on hickory roots. We would like to hear from Prof. Hutt.

PROF. HUTT: We haven't much data matured on that at present, Mr. President. It takes so long to get data on those subjects. We have a lot of trees budded on the stocks of water hickory and on the pecan, and we are testing them out. My theory was that the *Hicoria aquatica*, growing in wet, sour lands, would enlarge the range of probable production of pecans on such lands, and on lands on which the pecan, on its own roots, could not normally be grown, but our data are not matured yet. I think they have been three years in the nursery and two years set in the orchard. It will probably be four or five years before we get any exact data on that subject.

THE PRESIDENT: Perhaps, Mr. C. A. Reed has investigated some of the later top-worked hickories.

MR. C. A. REED: That is an old question—pecan on hickory. It has been tried all over the South and the Southwest, and you will see some this afternoon at Mr. Littlepage's place. As a usual thing, the enthusiasm over pecan on hickory has run high while the experiment was new. The propagator has found that it was not a difficult thing to make the scions live, and, so long as the hickory stock is larger than the pecan scion, so that the feeding capacity is equal to, or even greater than, the consuming capacity of the scion, the outlook has been very satisfactory and encouraging, and while that stage has been going on a great deal has been written. A little later you hear less about it, and less and less, until, finally, you hear almost nothing. But I will say this, that there are sections in the Southwest where there is considerable enthusiasm over it just now. Just recently an article was published by Judge Frank Gwynn which was quite encouraging, and from his point of view it is. He is on high, hilly land, where he has no pecan trees, and he has been able to get nuts considerably sooner by top-working

these dryland hickories—the mocker nut, or "bull nut," as it is known down there—and so far he is getting very satisfactory crops. But it is the consensus of opinion over the entire South, so far as I have observed it, that where there are pecan trees suitable for top-working, they answer much better, and the final outcome is very much more satisfactory with pecan on pecan than with pecan on hickory. Now, with pecan on *Hicoria aquatica*, which Prof. Hutt spoke of, I can cite you one instance which is very interesting south of Morgan City, Louisiana. Mr. Frank Beadle, I believe, was the name, top-worked a number of trees that were standing in water, and he also top-worked some that he had transplanted from the wet bottom to higher land. Those that were transplanted lived and bore nuts for quite a number of years. The last I knew they were bearing quite satisfactory crops, but those that were allowed to remain in the standing water died very shortly after the pecan top began to develop. The entire tree died.

THE PRESIDENT: That is, the pecan top killed the native right in its own habitat.

MR. C A. REED: That's right.

DR. STABLER: How about the acidity of the soil on that higher land? Was that tested?

MR. C. A. REED: Well, there would be so very little difference in the level of the soil that I imagine the acidity would be about the same. When I said "high land" I meant land that wasn't over-flowed.

DR. STABLER: Oh, yes.

THE PRESIDENT: Mr. Reed, you made the qualified statement awhile ago that where a man had a choice between hickory and pecan stock for top-working, he should take the pecan. Now, in the North there are

magnificent stands of native hickory—the Appalachians are full of it from end to end. Would you advise him not to bother with that?

MR. C. A. REED: There is another question that enters there. I don't believe that you can grow good pecans on hickory stocks on uplands where there is not moisture enough in the soil to grow good pecans on pecan stocks. It takes moisture to make pecans, and if there isn't enough in the upland soil to grow pecan trees on pecan roots I don't believe there is any evidence to indicate that you can get them on hickory roots.

MR. LITTLEPAGE: Mr. President, the hickory only grows about six or eight weeks every summer, and the pecan grows all summer. I think that answers the question.

PROF. HUTT: A case came up last year in the National Nut Growers' Association that was quite interesting. Mr. Smithwick, of Americus, Georgia, brought to the meeting and exhibited a number of varieties of pecan grown on hickory—fourteen varieties, standard varieties, grafted on a hickory tree, and they were remarkable for their small size. They were remarkably small—smaller than ordinary, woods-grown seedling pecans. There were Schleys and Delmas, and various other varieties that you could recognize by the form of the nuts, but exceedingly small. I believe Mr. Reed's point is the crux of the whole situation, that if you have a good supply of moisture they will make nuts of a pretty fair size, but unless the moisture supply is very large you get diminutive nuts. These were matured in the South. The hickory is such a slow grower in comparison with the pecan—that is, the common varieties—that it can't keep up with the pecan top.

MR. C. A. REED: Some of the nuts from that tree were on exhibition where you were this morning.

THE PRESIDENT: Then you have, practically, a dwarfing, with the dwarfing manifesting itself in the fruit rather than in the wood.

MR. C. A. REED: It did in that one instance, but, on the other hand, we have seen pecans grown on top-worked hickories that you could hardly tell from typical specimens of pecans grown on pecan stocks.

THE PRESIDENT: Isn't the bitternut several times as rapid in growth as the shagbark, or some others? That is, probably, one of the best stocks for

the hickories if one wishes to experiment.

MR. C. A. REED: AS Colonel Van Duzee said last night, "there are a lot of things we don't know." This is one of them. I might quote a number of men who are right here in this audience to convince you that we don't any of us know much about nut culture today. I will quote Dr. Morris and Mr. Littlepage. We were talking about hickory nut varieties in Dr. Morris' office one night about the first of this year, when Mr. Littlepage made the remark that "the man who didn't change his mind every three years on nut culture didn't keep up with the game," and Dr. Morris replied that he had changed his mind so much in the last five years he had no respect for any man who believed what he said. Now, when you can't believe Dr. Morris, Colonel Van Duzee, or Dr. Smith, what are you going to do with the rest of us?

COL. VAN DUZEE: Mr. Reed, I didn't say what you said I did. I said: "There are so many things you already know that are not true."
(Laughter.)

MR. C. A. REED: Well, now, I will quote another man, Dr. Curtis, one of the best known pecan men in the South. It was Dr. Curtis that I went to for my initial experience in pecans. The first I ever saw were in his orchard in Florida, and I asked him quite a good many questions, and he would tell me a story and go away. And I called him up one day, went into his orchard in harvest time when he was gathering the nuts in the hulls and taking them to the packing house. And I said "What is that for?" And he said "Don't you see those shuck worms all through the hulls here? I am throwing them out there to let the chickens get them." "Well," said I, "can you say you are getting rid of the shuck worms by doing that?" And he replied, "I can see, one year with another, that they are gradually getting less." A year later I went down there before he did. He was in Maine at the time, but his orchard trees were just alive with shuck worms, every variety almost eaten up with

them. I said to him, when he came back, "I thought you were going to get rid of those shuck worms by feeding them to the chickens?" "Well, there it goes," he said, "you get a nice theory all worked out and some one comes along and asks you a simple little question that knocks it all in the head." And that is almost the unanimous experience. What you know you have got to qualify if you talk at all. I am getting to be such a pessimist I am not much good in the government any more. (Laughter.)

THE PRESIDENT: The one hope a college professor of my acquaintance has is when a student comes around and says he believes he doesn't know much. He regards that as the beginning of knowledge, and I think that Mr. Reed's confessions, and incriminations of the rest of us, show one thing, perhaps, better than anything else, and that is the great necessity of organizations of this sort in which many men who are trying many things in many ways come together and give the results of their observations. No doubt, this whole question of agriculture in general, and nuts in particular, is so complex, it is so run through and through with so many different controlling factors, and, with them, so many new things are constantly coming along, that we are all going to be handing down to our children and grandchildren a great and, perhaps, increasing host of problems to be investigated, and new realms in which knowledge can be piled up for the benefit of those who wish to use it.

COL. VAN DUZEE: Mr. President, may I talk half a minute? I can't help but feel that, perhaps, there may be some good brother or sister who may have been over-impressed with the difficulties, who might have been discouraged, who might have left this meeting, perhaps, and failed to see what this meeting is for—to stimulate the planting of nut trees. Notwithstanding the emphasis that has been put on all these things, notwithstanding the difficulties and disappointments that we are all laboring under at the present time, I feel that we have a wonderful industry ahead of

us. I can't see any reason in the world why we should not go on within our means, wisely planting nut trees. It doesn't make any difference if you are seventy-five or eighty years old, plant nut trees, because they will be a constant pleasure to you, and, ultimately, a benefit to some one else.

MR. LITTLEPAGE: Mr. President—

THE PRESIDENT: This is Mr. Littlepage, ladies and gentlemen.
(Laughter.)

MR. LITTLEPAGE: That is a very important suggestion that you just made. If you were to ask the average groceryman in Washington City whether he wanted his son to go into the grocery business he would say no. If you asked a lawyer if you should make a lawyer out of your son, if the lawyer looks back over the drudgery and years of toil that it takes to make a lawyer, he would undoubtedly hesitate to recommend it, and if you asked a doctor or a college professor a similar question, they, no doubt, would steer you clear away from a university. And so, Mr. President, if you stand back on the difficulties in these things, there would be not only no grocerymen, but no lawyers, no doctors, no dentists, and, perhaps, nobody working for the government. (Laughter and applause.)

THE PRESIDENT: I want to take the liberty of using thirty seconds in this period of exhortation and confession to come in on the same strain. After all, what is life for? How many of us want the thing that is dead easy, and how many of us want the job with nothing to do? We all, in a certain lazy mood, say we want something easy and want to rest, but if there is anything on earth that a man shuns above all else it is that little room with absolutely nothing to do, namely, a cell. When they want to break a man they don't put him at hard labor on the stone pile; they put him in a little room with nothing to do. The youngster who plays doesn't want a dead easy game. He builds a house, and, when he has done with it, bang, he doesn't

want the house he wanted to build. And I must confess that if it were perfectly plain sailing and you could plant out all these nut trees and have them grow like fury, it would not be much fun. It is a fact that men like to achieve and experiment; men like effort. Suppose everybody in this country retired and could put up his feet and do nothing, there wouldn't be a name in the paper the next morning. Mr. Hughes, President Wilson, Mr. Taft, Mr. Brandies, and all of the great men who are doing things in this world would all be gone fanning themselves quietly. This world is run by men who don't have to work; they work for fun. So I wish to submit that the tree—if a man happens to be built to love plants that grow—that the tree is one of the great avenues of fun.

MR. WEBER: Mr. President, along the same line of thought, I wish to express my views with what Colonel Van Duzee has had to say. If we were to attend a convention of surgeons and hear different diseases and ailments of the body discussed, we would probably all be disposed to think that we were standing on the tip-end of the diving board into eternity beyond. But people keep on living just the same, notwithstanding the knocking of the doctors, and the diseases to which we are subject, and trees will keep on growing just the same, notwithstanding their diseases and various other troubles, and so I think no one should be discouraged.

THE SECRETARY: I just want to add my little encouragement. In spite of all the failure that I have had, and they have been many, in spite of the reports of failures of others and the pessimism of others, I have the same abiding faith in the future of nut growing, and just the same enthusiasm for it that I had in the beginning, if not greater. (Applause.)

MR. KYNER: Mr. President, I came here to get information on a matter that I am very much interested in. At seventy years of age I have become interested in nut growing—in nut culture. (Applause.) I am not planting

particularly for myself, not that I expect to get any harvest from these trees, but I do want to see them bear fruit—bear nuts. I want to plant the right kind of trees. I have joined this Association; I intend to retain a membership in it as long as the Association lives. (Laughter.)

THE PRESIDENT: My dear sir, that will cost you twenty dollars for a life membership. (Laughter.)

MR. KYNER: And I want to get all the information that the Association has. Now, if I can get it in fifteen or twenty minutes, why, let me have it. (Laughter.) I bought Persian walnuts at a nursery, cultivated them, and watched them, walked around them and looked at them, and along came a winter and killed them. I bought them from a Rochester nursery. Now, they didn't grow them there. They must have grown them somewhere else. If they had been grown in a Rochester nursery they would have withstood the severity of a Maryland winter. Now, there is something wrong there. This Association should take this matter up with that nursery. They should not be allowed to take people's money and give them chaff for it. I am saying this for the benefit of some of our members here who are growing nut trees for sale.

THE PRESIDENT: May I give you a bit of information here? We have a list of accredited nurserymen. This Association has a list of nurserymen in whose trees we think we can place more confidence than in some others.

MR. KYNER: I would like to get that. But now I have set out a whole lot of these Persian walnuts, and pecans, filberts, Japanese walnuts, etc., and I guess every one of them is a seedling, and I don't know what I have, and I don't know how many varieties of Japanese walnuts there are. I supposed a Japanese walnut was a Japanese walnut, and that that was all there was to it. But I get some trees from one nursery, and some from another, and they grow up and aren't alike at all. Now, I haven't so very awfully long to be in

the business of setting out nut growing trees, and I want to get the right kind, and I want this Association's assistance in that matter, and while you are assisting me you are assisting people all over the country. Men and women everywhere are interested in nut growing. They want nut trees, but how are they going to know that they are getting what they want? I believe it is up to this Association to help them get the right kind of stuff. I came in here purposely to get your help.

THE PRESIDENT: You go on the excursion this afternoon and you will find plenty of men there that will take pleasure in explaining some of these things to you. Our plan is to go at one o'clock from the corner of Fourteenth and H streets to the grounds of Mr. Littlepage, who has practically all the good varieties of northern pecans growing there, and on the trip will be men who can answer most every question you want to know. I think that brings us to the point of adjournment.

COL. VAN DUZEE: Mr. President, I move we adjourn.

A MEMBER: Second the motion.

THE PRESIDENT: The meeting stands adjourned.

APPENDIX.

Letter From W. C. Reed, Vice-President of the Association.

FELLOW MEMBERS AND FRIENDS:

It is with the deepest feelings of regret that I am compelled to be absent from what I trust may be one of the most profitable meetings of the Association. It is impossible for me to be present, owing to the fact that I have been summoned on a case in court in Wisconsin.

Having been honored as your Vice-President, I felt it my duty to attend and do what I could to help make this our best meeting, but fate ruled otherwise. Though absent in person, I assure you my thoughts and best wishes will be with you while wandering about the Nation's Capital, viewing its magnificent parks and basking under the shade of its stately Persian walnuts.

The interest in nut culture is widespread. We have had inquiries from many foreign countries, one of the last from near Bombay, British India.

I have arranged with the Indiana Apple Show, which is to be held at West Baden, Indiana, November 14th to 20th, for ample space for a nut exhibit. Anyone having nuts for exhibition should send them to me at Vincennes prior to these dates, or write for information, and I will try and arrange for premiums.

REVIEW OF PAST YEAR.

The present summer has been of extremes, very cold and wet early, followed by extreme heat and drouth. Foliage of all kinds not as good as usual. Nut trees, however, have made a very good growth, not as heavy as last year on younger trees.

Winter, 1915-16, while not extremely cold, was very hard on many kinds of trees, owing to the fact that the previous summer and fall were very wet. Most fruit trees went into winter full of sap, with buds in weakened condition. Pecan buds came through in good shape with a very fair stand in nursery, and one-year trees were not injured a particle. Pecan bloom was very fair, crop, generally seems to be light, in fact such is the case with all kinds of nut trees, generally, and most fruit trees. Pecan trees set in orchard 2 and 3 years ago are making a good growth.

ENGLISH WALNUTS.

Stand of buds in nursery poor; stand of grafts this spring very good where we used good, strong scions of well matured wood, 60 to 75 per cent, and in some cases Mayette was better than that. Where Eastern scions were used from old trees, stand of grafts very poor. All one-year English walnut trees in nursery came through in good shape. Eastern varieties began to vegetate or burst into growth April 15; Mayette and Franquette, May 1; Parisienne, May 5, and one tree from Grenoble, France, grown from scion sent from Department of Agriculture, May 25. These French varieties, I feel, are very promising, owing to the fact that they will escape late frosts. English walnut trees in orchard set 3 years ago, fourth summers growth, doing splendidly, 2 to 4 feet of growth, foliage perfect, varieties, Hall, Rush, Nebo and Burlington. Top-worked trees, 3-year tops doing nicely of Hall, Rush,

Mayette and two or three other Eastern varieties. Grafting in nursery done from May 15 to 25, was best after stocks were in full leaf.

PECAN GRAFTING.

We have usually had best success grafting May 5 to 12, but this year, being a late spring, we did not commence general grafting of pecans until the 12th, and it seems to have been too late. Stand very poor, a few grafts set early in May with old wood, about 40 per cent. stand. We find old wood gives much better stand on pecans, and new wood on English walnuts.

BLACK WALNUTS.

Grafted quite a number of Stabler Black Walnuts, which were almost a failure. Thomas done better, but still poor. However, larger scions gave best results and have made splendid growth, many 5 to 6 feet, very strong. Buds of Thomas set last fall failed to start well. It seems we have something to learn in the propagation of the Black Walnut, as it has proved more difficult than the English.

HARDY ALMOND.

Two years ago we received some buds of the Ridenhauer Almond from Department of Agriculture. Some of these buds were set on a bearing peach tree; these have borne a good crop this summer, and were gathered August 20, some of which are on the exhibition tables. These seem to bear very young, of good quality, a very strong grower and very hardy; do not consider them of any commercial value, but for family use are very good.

BEARING PECANS IN NEBRASKA.

During the past year I have received photographs and description of the pecan trees 12 miles south of Lincoln, Nebraska, and of two trees on the grounds of E. Y. Grupe, of Lincoln. These trees are 20 years old, some having been bearing regular crops for the past 10 years. This season's crop is a failure owing to continuous cold rain at blooming time. The nuts on one of these trees are of fair size and quality.

With kindest regards to the many friends in the Association, and trusting that I may have the pleasure of greeting all at our next annual meeting, I am,

Respectfully yours,

W. C. REED.

THE FOOD VALUE OF NUTS.

DR. J. H. KELLOGG, BATTLE CREEK, MICHIGAN.

Of all really valuable foodstuffs nuts are the least used and the least appreciated. In fact, nuts can hardly be said to constitute a part of the national bill of fare for the reason that when eaten at all they are taken as luxuries or deserts and not as staple foods. But the nut possesses special properties which entitle it to first consideration as a foodstuff, and the writer has no doubt that some time in the future nuts will become a leading constituent of the national bill of fare, and in so doing, will displace certain foodstuffs which today are held in high esteem, but which in the broader light of the next century will be regarded as objectionable and inferior foods and will give place to the products of the various varieties of nut trees which will then be estimated at their true worth, the very choicest of all substances capable of sustaining human life. Botanically, a nut is a fruit, but nuts differ so widely both in composition and appearance from the foods commonly called fruits that they are properly placed in a class by themselves.

In nutritive value the nut far exceeds all other food substances; for example, the average number of food units per pound furnished by half a dozen of the more common varieties of nuts is 3231 calories, while the average of the same number of varieties of cereals is 1654 calories, half the

value of nuts. The average food value of the best vegetables is 300 calories per pound and of the best fresh fruits grown in this country is 278 calories. The average food value of the six principal flesh foods is 810 calories per pound, or one-fourth that of nuts.

The superior nutritive value of nuts is clearly shown by the accompanying tables based upon the analyses of Atwater and other authorities.

TABLE I.

COMPOSITION OF NUTS (C. F. LANGWORTHY).

Composition and Fuel Value of the Edible
Portion.

	Food							
	Edible Carbohy- Value							
	Refuse.	Portion.	Water.	Protein.	Fats.	drates.	Ash.	per lb.
	Per ct.	Per ct.	Per ct.	Per ct.	Per ct.	Per ct.	Per ct.	Cal.
Almonds	64.8	35.2	4.8	21.0	54.9	17.3	2.0	3,030
Brazil nuts	49.6	50.4	5.3	17.0	66.8	7.0	3.9	3,328
Filberts	52.1	47.9	3.7	15.6	65.3	13.0	2.4	3,432
Hickory nuts	62.2	37.8	3.7	15.4	67.4	11.4	2.1	3,495
Pecan nuts	53.2	46.8	3.0	11.0	71.2	13.3	1.5	3,633
English walnuts.	58.0	42.0	2.8	16.7	64.4	14.8	1.3	3,305
Chestnuts, fresh.	16.0	84.0	45.0	6.2	5.4	42.1	1.3	1,125
Chestnuts, dried.	24.0	76.0	5.9	10.7	7.0	74.2	2.2	1,875
Acorns	35.6	64.4	4.1	8.1	37.4	48.0	2.4	2,718
Beechnuts	40.8	59.2	4.0	21.9	57.4	13.2	3.5	3,263
Butternuts	86.4	13.6	4.5	27.9	61.2	3.4	3.0	3,371
Walnuts	74.1	25.9	2.5	27.6	56.3	11.7	1.9	3,105

Cocoanuts	48.8	51.2	14.1	5.7	50.6	27.9	1.7	2,986
Cocoanuts, shredded, ...	100.0	3.5	6.3	57.3	31.6	1.3	3,125	
Pistachios, kernels ...	100.0	4.2	22.6	54.5	15.6	3.1	3,010	
Pine nuts or pinons	40.6	59.4	3.4	14.6	61.9	17.3	2.8	3,364
Peanuts, raw	24.5	75.5	9.2	25.8	38.6	24.4	2.0	2,560
Peanuts, roasted	32.6	67.4	1.6	30.5	49.2	16.2	2.5	3,177
Litchi nuts	41.6	58.4	17.9	2.9	.2	77.5	1.5	1,453

TABLE II.

COMPOSITION OF MEATS (ATWATER AND LANGWORTHY).

	Water.	Protein.	Fat.	Calories per lb.
Beef ribs	43.8	13.9	21.2	1,135
Porterhouse steak	52.4	19.1	16.1	975
Veal cutlet	68.3	20.1	7.5	695
Mutton	51.2	15.1	14.7	890
Mutton chops	42.	13.5	28.3	1,415
Lamb	52.9	15.9	13.6	860
Pork chops	41.8	13.4	24.2	1,245
Ham, smoked	34.8	14.2	33.4	1,635
Bacon, smoked	17.4	9.1	62.2	2,715
Sausage, Frankfort	57.2	19.6	18.9	1,155
Beef soup	92.9	4.4	0.4	120
Chicken (fowl)	47.1	13.7	12.3	765
Goose	38.5	13.4	29.8	1,475
Turkey	42.4	16.1	18.4	1,060
Duck	51.7	14.3	33.4	1,805
Squab	58.	18.6	22.1	1,480

Guinea hen 69.1 23.1 6.5 870

Quail 65.9 25. 6.8 935

TABLE III.

COMPOSITION OF CEREAL FOOD (LANGWORTHY).

	Carbohy- Food				
	Protein.	Fat.	drates.	Ash.	Value
	Per ct.	Per ct.	Per ct.	Per ct.	per lb.
Flour, meal, etc.:					
Entire wheat flour	13.8	1.9	71.9	1.0	1,650
Graham flour	13.3	2.2	71.4	1.8	1,645
Wheat Flour, patent roller process					
—high grade and medium	11.4	1.0	75.1	.5	1,635
Macaroni, vermicelli, etc.	13.4	.9	74.1	1.3	1,645
Wheat breakfast food	12.1	1.8	75.2	1.3	1,680
Buckwheat flour	6.4	1.2	77.9	.9	1,605
Rye flour	6.8	0.9	78.7	.7	1,620
Corn meal	9.2	1.9	75.4	1.0	1,635
Oat breakfast food	16.7	7.3	66.2	2.1	1,800
Rice	8.0	.3	79.0	.4	1,620
Tapioca	.4	.1	88.0	.1	1,650
Starch	90.0	..			1,675

TABLE IV.

COMPOSITION OF VEGETABLES (EDIBLE PORTION).

	Carbohy-				
	Water.	Protein.	Fat.	drates.	Calories
	Per ct.	Per ct.	Per ct.	Per ct.	per lb.

Beans, dried	12.6	22.5	1.8	59.6	1,520
Beans, lima ...	...	...	..	...	
Beans, string	83.0	2.1	.3	6.9	170
Beets	70.0	1.3	.1	7.7	160
Cabbage	77.7	1.4	.2	4.8	115
Celery	75.6	.9	.1	2.6	65
Corn, green (sweet), edible portion	75.4	3.1	1.1	19.7	440
Cucumbers	81.1	.7	.2	2.6	65
Lettuce	80.5	1.0	.2	2.5	65
Mushrooms	88.1	3.5	.4	6.8	185
Onions	78.9	1.4	.3	8.9	190
Parsnips	66.4	1.3	.4	10.8	230
Peas	74.6	7.0	0.5	16.9	440
Potatoes	62.6	1.8	.1	14.7	295
Rhubarb	56.6	.4	.4	2.2	60
Sweet potatoes	55.2	1.4	.6	21.9	440
Spinach	92.3	2.1	.3	3.2	95
Squash	44.2	.7	.2	4.5	100
Tomatoes	94.3	.9	.4	3.9	100

TABLE V.

COMPOSITION OF FRUITS, YEARBOOK OF DEPT. OF AGRICULTURE, 1915.

(C. J. LANGWORTHY).

[illegible]

Apples 84.6 0.4 0.5 13.0 ... 1.2 0.3 290
 Apricots 85.0 1.1 13.45 270
 Avocado 81.1 1.0 10.2 ... 6.89 512
 Bananas 75.3 1.3 .6 21.0 ... 1.0 .8 460
 Blackberries 86.3 1.8 1.0 8.4 ... 2.5 .5 270
 Cactus fruit 79.2 1.4 1.3 11.7 ... 3.7 2.7 375
 Cherries 80.9 1.0 .8 16.52 .6 365
 Cranberries 88.9 .4 .6 8.4 ... 1.5 .2 215
 Currants 85.0 1.5 12.87 265
 Figs 79.1 1.5 18.86 380
 Gooseberries 85.6 1.0 13.13 255
 Grapes 77.4 1.3 1.6 14.9 ... 4.3 .5 450
 Guava 82.9 1.3 .7 8.0 ... 6.6 .5 315
 Huckleberries 81.9 .6 .6 ... 16.63 345
 Lemons 89.3 1.0 .7 7.4 ... 1.1 .5 205
 Mango 87.4 .6 .4 9.9 ... 1.2 .5 220
 Muskmelons 89.5 .6 ... 7.2 ... 2.1 .6 185
 Nectarines 82.9 .6 15.96 305
 Olives 67.0 2.5 17.1 5.7 ... 3.3 4.4 407
 Oranges 86.9 .8 .2 ... 11.65 240
 Peaches 89.4 .7 .1 5.8 ... 3.6 .4 190
 Pears 80.9 1.0 .5 15.7 ... 1.5 .4 163
 Persimmons (Japanese) 80.2 1.4 .6 15.1 ... 2.1 .6 174
 Pineapples 89.3 .4 .3 9.34 .3 200
 Plums 78.4 1.0 20.15 395
 Pomegranates 76.8 1.5 1.6 16.8 ... 2.7 .6 461
 Prunes 79.6 .9 18.96 370
 Raspberries (red) 85.8 1.0 ... 9.7 ... 2.9 .6 255
 Rhubarb stalks 94.4 .6 .7 2.5 ... 1.1 .7 105

Strawberries 90.4 1.0 .6 6.0 ... 1.4 .6 180

Watermelons 92.4 .4 .2 ... 6.73 140

With the exception of smoked bacon, there is no flesh food which even approaches the nut in nutritive value, and bacon owes its high value to the fact that it consists almost exclusively of fat.

That the nut is appreciated as a dainty is attested by the frequency with which it appears as a desert and the extensive use of various nuts as confections. That nuts do not hold a more prominent place in the national bill of fare is due chiefly to two causes; first, the popular idea that nuts are highly indigestible, and second, their comparatively high price.

The notion that nuts are difficult of digestion has really no foundation in fact. The idea is probably the natural outgrowth of the custom of eating nuts at the close of a meal when an abundance, more likely a superabundance, of highly nutritious foods has already been eaten and the equally injurious custom of eating nuts between meals. Neglect of thorough mastication must also be mentioned as a possible cause of indigestion following the use of nuts. Nuts are generally eaten dry and have a firm hard flesh which requires thorough use of the organs of mastication to prepare them for the action of the several digestive juices. Experiments made in Germany showed that nuts are not digested at all but pass through the alimentary canal like foreign bodies unless reduced to a smooth paste in the mouth. Particles of nuts the size of small seeds wholly escaped digestion.

Having been for more than fifty years actively interested in promoting the use of nuts as a staple food, I have given considerable thought and study to their dietetic value and have made many experiments. About twenty-five years ago it occurred to me that one of the above objections to the extensive dietetic use of nuts might be overcome by mechanical preparation of the nut before serving so as to reduce it to a smooth paste and thus insure the

preparation for digestion which the average eater is prone to neglect. The result was a product which I called peanut butter. I was much surprised at the readiness with which the product sprang into public favor. Several years ago I was informed by a wholesale grocer of Chicago that the firm's sales of peanut butter amounted on an average to a carload a week. I think it is safe to estimate that not less than one thousand carloads of this product are annually consumed in this country. The increased demand for peanuts for making peanut butter led to the development of "corners" in the peanut market and more than doubled the price and must have had an equally marked influence upon the annual production.

I am citing my experience with the peanut not for the purpose of recommending this product, for I am obliged to confess that I was soon compelled to abandon the use of peanut butter prepared from roasted nuts, for the reason that the process of roasting renders the nut indigestible to such a degree that it was not adapted to the use of invalids, but simply as an illustration of the readiness with which the public accepts a new dietetic idea when it happens to strike the popular fancy. Ways must be found to render the use of nuts practical by adapting them to our culinary and dietetic customs and to overcome the popular objections to their use by a widespread and efficient campaign of education.

Attention has already been called to the superior nutritive value of the nut. It has other excellencies well worthy of consideration; for example, the protein of nuts is of the very choicest character. Recent investigations by Rubner, Osborne, Mendel, and others have shown that every plant produces its own special varieties of proteins. There is indeed a wide difference even between the proteins of various cereals and the proteins of many vegetables differ so widely in character from those of the human body that it is doubtful whether to any extent they can be utilized for human nutrition. Fortunately the potato is in this regard an exception and furnishes a very

excellent type of protein. This objection does not apply to nuts. The proteins of nuts are in fact so very closely allied to those of the animal body that food chemists of a generation ago referred to the protein of nuts as vegetable casein because of its exceedingly close resemblance to the protein of milk.

The fats of nuts, their leading food principle, are the most digestible of all forms of fat. Having a high melting point, they are far more digestible than animal fats of any sort. The indigestibility of beef and mutton fat has long been recognized. The fat of nuts much more closely resembles human fat than do fats of the sort mentioned. The importance of this will be appreciated when attention is called to the fact that fats entering the body do not undergo the transformation changes which take place in other foodstuffs; for example, protein in the process of digestion is broken into its ultimate molecular units. Starch is transformed into sugar, which serves as fuel to the body, but fats are so slightly modified in the process of digestion and absorption that after reaching the blood and the tissues they are reconstructed into the original form in which they are eaten; that is, beef fat is deposited in the tissues as beef fat without undergoing any chemical change whatever; mutton fat is deposited as mutton fat; lard as pig fat, etc. When the body makes its own fat from starch and sugar, the natural source of this tissue element, the product formed is *sui generis* and must be better adapted to the body uses than the animal fat which was *sui generis* to a pig, a sheep, or a goat. It is certainly a pleasant thought that one who rounds out his figure with the luscious fatness of nuts may felicitate himself upon the fact that his tissues are participating in the sweetness of the nut rather than the relics of the sty and the shambles. It is true that nuts are poor in carbohydrates; that is, they contain no starch and little sugar, but this deficiency can be easily supplied by fruits, as will be readily seen by reference to Table V.

Of the three great food principles required for human nutrition, protein, fats, and carbohydrates, the nut supplies two—protein and fats in rich abundance, and of very finest quality. The amount of protein found in fruits with very few exceptions is so small as to be insignificant; fats are practically wholly absent from fruits, while sugar and dextrine are abundant. Fruits are thus the natural complement of nuts.

The amount of protein contained in nuts is, with two or three exceptions, small as compared with meats, and even some of the cereals; but the studies of nutrition which have been made within the last score of years by Chittenden and numerous other investigators have clearly established the fact that protein which is chiefly represented in the ordinary bill of fare by lean meat, is needed only in very small amount. If the amount of protein eaten equals ten per cent of the total ration the body will receive an abundant supply of material for repairing its nitrogenous tissues, the only function for which protein is essential. Some nuts, as the pine nut and the peanut, are rich in protein. A pound of pine nuts contains as much protein as a pound and a half of lean meat, besides furnishing the equivalent to two-thirds of a pound of butter. The almond is also rich in protein.

But nuts have another special excellence which is worthy of consideration. Recent researches have shown the paramount importance of vitamins—certain subtle elements which are needed to activate or set in operation various processes within the body which are essential to complete nutrition. The vitamins of rice and other cereals are removed with the bran; hence an exclusive diet of polished rice gives rise to *beriberi*. Meat contains vitamins in very small amounts, for vitamins are produced only by plants. The vitamins found in flesh foods represent only the small residue of the supplies which the animal gathered from the grass, corn and other vegetable products which constitute its food.

Twenty years ago, when the diet of sailors consisted chiefly of salt pork, *scurvy* was a dread scourge which often disabled whole ship crews and sent many a poor seaman into "Davy Jones' locker." The cooking of animal foods destroys the vitamins which they contain. Infants suffer from scurvy when fed on sterilized or pasteurized milk. There is good reason for believing that *pellagra* is due to a deficiency of vitamins, which are conspicuously absent from a dietary consisting of "sow belly," molasses, tea, coffee, lard, cornmeal, fine flour and polished rice.

Nuts are rich in vitamins. In fact, the nut consists of the choicest aggregation of all the materials essential for the building of sound human tissues, done up in a hermetically sealed package ready to be delivered by the gracious hand of Nature to those who are fortunate enough to appreciate the value of this choicest of all earth's bounties.

As already noted, nuts consist almost wholly of the two principles, fat and protein. The same is true of meats. Nuts contain more fat and less protein and in this particular as well as others which have been mentioned are better prepared to serve as nutrients to the body than are meats. Besides, nuts have the advantage of being clean, free from the products of disease and putrefaction. Meats of all sorts, as found in the market, with the exception of canned meats, abound with putrefactive bacteria to an astonishing degree. This is true of dried, smoked and salted meats as well as of the fresh meats and game which are displayed upon the walls of the meat shop. An examination of various meats made some time ago by A. W. Nelson, bacteriologist of the Battle Creek Sanitarium, showed the presence of putrefactive bacteria in almost unbelievable numbers, as will be seen by an inspection of the following table:

TABLE VI.

No. Putrefactive
Specimen. Bacteria per ounce.

1. Large sausage 12,600,000,000
2. Small sausage 19,890,000,000
3. Round steak 16,800,000,000
4. Roast beef 16,800,000,000
5. Smoked ham 1,293,600,000
6. Hamburger steak 3,870,000,000
7. Pork 3,781,200,000
8. Porterhouse steak 900,000,000
9. Sirloin steak 11,340,000,000
10. Tenderloin (well done) 756,000,000
11. Tenderloin (rare) 5,040,000,000

Repeated subsequent examinations have given similar results. These results also agree with observations made by various other German and American bacteriologists. Decomposition of animal flesh begins immediately after the animal dies. Within twenty-four hours after killing, even though the carcass is kept in an ice box or refrigerator, the whole mass is permeated with putrefactive bacteria. Refrigeration even to a point close to freezing delays but does not prevent the growth of putrefactive organisms although at lower temperatures the usual volatile products which give notice of the presence of putrefaction by an odor of decay are not produced. Persons whose stomachs manufacture a liberal amount of hydrochloric acid, an essential constituent of healthy gastric juice, are able to disinfect even highly putrescent meat, so that they apparently do not suffer any immediate injury when such meat enters the stomach. In a stomach which produces little or no hydrochloric acid, the process of putrefaction continues all the way through the alimentary canal, giving rise to the same poisonous substances which are present in the putrefying carcass of a dead rat or any

other dead animal, and produces intestinal or alimentary toxemia with the multitude of mischiefs which grow out of this condition, among which may be mentioned all sorts of skin troubles, high blood-pressure, apoplexy, premature senility, Bright's disease, heart failure, gallstones—a list which might be increased by the addition of scores of other common, chronic maladies.

When one recalls the statement made before the congressional committee by the chief of the United States meat inspection service that if all animals, any part of which was diseased, were rejected by inspectors, not more than one in a hundred would pass muster; and when one also reflects upon the wide prevalence of tuberculosis in animals,—at least ten per cent of all the cows in the country are known to be tuberculous,—and the growing prevalence of tapeworm and trichinae, diseases which are exclusively derived from the eating of flesh, and then contemplates the purity and perfection of the choice little food packets which we call nuts, it is easy to be persuaded that a substitution of nuts for flesh foods, even on a very large scale, would be not only a perfectly safe procedure, but one which would be followed by the most desirable results.

The use of nuts as a staple article of food is not an experiment. All the higher apes, man's nearest relatives in the animal world, thrive on nuts. Many savage tribes live almost entirely on nuts. The Indians of the foothills of California gather every fall large quantities of nuts which they store for winter use. The early settlers of California reported also that many tribes of Indians in that part of the United States lived almost wholly upon acorns. Before the great oak forests of this country were cut down for lumber, millions of hogs were fattened on mast, and the price of pork depended more upon the acorn crop than on the corn crop. The peasantry of southern France and northern Italy during half the year make two meals a day on chestnuts.

The objection commonly urged, that nuts are too expensive to enter largely into the ordinary bill of fare, at first sight appears to be valid, but upon examination this objection almost, if not wholly, disappears. For example, a pound of pine nuts which is more than the equivalent in nutritive value to two and a half pounds of the best beefsteak and two-thirds of a pound of butter, can be bought wholesale for twenty-five cents. The cost of the equivalent food value in meat and butter would be at least sixty to seventy cents, or more than double the cost of the nuts. A pound of almonds can be bought at wholesale for forty cents, and has food value equal to that of meat which would cost a dollar or more. A pound of peanuts can be bought at wholesale for seven or eight cents, and furnishes nutritive value equivalent to more than a pound of beefsteak and a half a pound of butter, which would cost forty-five to fifty cents, or seven times as much. No objection can be offered to the fact that we are comparing wholesale with retail prices, for the reason that nuts do not readily spoil as do meat and butter, but will keep in perfect condition for months. Further it is entirely reasonable to suppose that the price of nuts may sometime in the future be considerably reduced when the cultivation of nuts becomes more general, and especially when the United States Forestry Department becomes convinced that it would be a sensible thing to cover with nut trees some of the large areas which have in the last fifty years been laid waste by deforestation. The planting of nut trees along all the public highways of the country would in less than twenty years result in a crop, the food value of which would be greater than that at present produced by the entire livestock industry of the country.

The high price of meat of which so much complaint has been made in recent years is not likely to recede. The high price is not due to manipulations of the market, but to natural causes, the chief of which is the limitation of pasturage and is the consequence of a decrease in the number of livestock. As the country becomes more and more densely settled, the

difficulty of supplying the demand for meat will increase, and in time the necessity for utilizing every foot of ground in the most efficient manner, will necessarily bring about a change in the dietetic habits of the people. Not a single example can be found in the world of a densely populated country dependent upon its own resources in which flesh foods constitute any considerable part of the national bill of fare. Since Germany has been nearly shut off from the outside world by the present war, the government has found it necessary to restrict the consumption of meat to one-half pound per week for each adult. All other European countries are equally dependent on outside sources for their meat supply.

The time will certainly come when nuts and nut trees will become a most important food resource. If a reform in this direction could be effected within the next ten years, the result would be a disappearance to a large extent of the complaint of the high cost of living. Mr. Hill said the basis for complaint was not the high cost of living, but the cost of high living. I should prefer to say that the real cause for complaint was wrong living rather than high living, or necessarily high cost. With right living the cost will be automatically reduced. For example, suppose a person were content to choose the peanut as his source for protein and fat, the elimination of the butcher's bill for meat and the grocer's bill for butter would at once cut out two-thirds of the expense incurred for food.

When a student in college more than forty years ago, I was already making dietetic experiments and lived three months on a diet such as I have suggested, at an average expense of exactly six cents a day. This was the total amount expended for raw foodstuffs. I paid my landlady five times as much for preparing and serving the food, and had reason for believing that some portion of my supplies was utilized by others than myself. As evidence of the fact that the experiment was not dangerous, I may add that I have pursued the same meatless dietary during my entire lifetime since, as I

had done for ten years before, and I am still alive and hard at work. Man is naturally a frugivorous animal. According to Cuvier, the great French naturalist, the natural diet of human beings, like that of those other primates, the orangoutang, the chimpanzee, and the gorilla, consists of fruits, nuts, tender shoots and cereals. A sturdy Scotch highlander informed me that his diet consisted of brose, bannocks, and potatoes, and that he rarely ever tasted meat. When asked what he fed his dogs, he replied, "The same as I eat myself, sir." The high-bred foxhounds of the southern states are fed on cornmeal, oatmeal and bread, and rarely taste flesh of any sort. Dogs thus fed are hardier, healthier, have more endurance, better wind, keener scent, greater intelligence, and are more easily trained than meat-fed dogs. A diet which is safe for carnivorous animals, must certainly be safe for human beings, who belong to a class of animals all representatives of which, with the exception of man are flesh abstainers.

Some years ago I experimented with various sorts of carnivorous animals for the purpose of ascertaining whether nuts could be made a complete substitute for meat. Among the various animals utilized for the experiment was a young wolf from the northwest that had never eaten anything but fresh raw meat. After giving the animal one day to get accustomed to its new surroundings and to acquire a good appetite, I gave him a breakfast of nuts properly prepared and was delighted to find that he took to the new ration without the slightest hesitation and remained in excellent health during the several months of the experiment. I succeeded perfectly in substituting nuts for meat with all the animals experimented upon, including a fish hawk, with the single exception of an old bald-headed eagle, which refused to be converted.

I have a suspicion that the so-called carnivorous animals were all at some remote time nut eaters; the so-called carnivorous teeth would be as useful in

tearing off the husks of cocoanuts and similar fruits, as for tearing and eating flesh.

An economic argument for the general adoption of nuts as a suitable article of food is the enormous increase in food resources which such a change would bring about. Some years ago, an experienced stock-raiser informed the writer that it takes two acres of land and two years to produce a steer weighing 600 pounds when dressed. Fresh meat is three-fourths water; hence the food material actually represented by such an animal would be considerably less than one hundred and fifty pounds, allowing for the weight of the bones. The food value, estimated as dried meat, would be about sixteen hundred calories per pound, or the same as an equal quantity of wheat meal. That is, an acre of land would produce in the form of beef, the food equivalent of seventy-five pounds of wheat in two years, whereas, a single acre of grain would produce on an average, even when poorly cultivated, in two crops not less than thirty-two bushels of more than 1900 pounds of wheat, or more than twenty-five times as much food as the same land would produce in the same length of time in the form of beef. Humboldt showed that the banana would furnish sustenance for twenty-five times as many people as could be nourished by the wheat produced by the same area of land; and according to Hutchinson, the chestnut tree is capable of producing on a given area a still larger amount of nutrient material than the banana. In other words, an acre of ground covered with chestnut trees in full bearing will furnish food for more than six hundred times as many people as could be supported by the same area devoted to meat production.

As a source of protein and fats the nut is vastly superior to the ox and the pig. The nut is sweeter, cleaner, safer, healthier and cheaper than any possible source of animal products.

This choicest product of Nature's laboratory is just beginning to be appreciated. When the Nut Growers' Association celebrates its one hundredth anniversary, it is safe to predict that the descendants of the present generation of nut growers who have followed the example of their forebears, will be living in opulence and will be regarded as the saviors of their country, while the great abattoirs and meat packing establishments will have ceased to exist, and the merry click of the nut cracker will be heard throughout the land.

EXTRACTS FROM A LETTER FROM COLONEL J. C. COOPER, OF
McMINNVILLE,
OREGON, PRESIDENT OF THE WESTERN WALNUT ASSOCIATION.

(Prepared by W. J. SPILLMAN, Chief of the Office of Farm Management U. S. Dept. of Agr., to be read at the 7th Annual Meeting of the N. N. G. A.)

It is probable that the prominence given the walnut growing industry in Oregon and the Northwest is greater than the finished product will justify at present, yet it is growing all the time in spite of the methods in use. I say in spite of the methods rather than because of the methods in use, for the reason that hundreds of thousands of trees have been set out in the last ten or twelve years, a majority of which have failed to meet the expectations of the would-be growers. These expectations, however, have been based largely on the statements of boom literature of those who have trees and lands for sale. We have much land in Western Oregon that is suited to the growing of walnuts, and some trees and orchards that are doing well, but

there are more individual trees that are giving their owners profits than there are orchards.

The industry will continue to grow, I will repeat again, in spite of the cultural methods we use, but we must certainly change our methods or our trees, or both. The excellence of the Oregon walnut is beyond question. The gold and silver medals that we have captured, as well as the testimony of dealers who are bidding for our product for their fancy trade, is evidence of its excellent quality. But there are many things that enter in the making of the perfect nut. Even after the tree has cast down its golden shower of the finest product, the gathering, washing and drying makes for the sweetness of the nut. When I see men who make a success in other lines of horticulture and farming pulling out walnut trees because they have planted a cheap lot or are too impatient for the harvest, and others bringing sackfuls of the finest nuts to market, discolored and dirty from having lain on the wet ground for days and weeks, I sometimes think that it is a long, long way to Tipperary.

But my heart's right there, and our association is doing heroic work, although but two years old; we get our committees together two or three times a year, compare notes and crack the whip for another run. Then when we get together in annual convention there is something doing. We cut out the frills and get at once to business. No welcomes by the mayor and response by Colonel Long Bow with a brass band, but rather like the women at the fish market: "Have yees any nice fish, Mrs. Maloney?" "Indade, I have, Mrs. Flanigan." "They stink." "You lie." And that is the way our fight usually starts, only not so vigorously, of course.

We have one committee that is all important and is doing fine work. The committee on seedling varieties is making a survey of the western states to find a variety or varieties best suited to the soil and climate of the different

localities. This committee includes the best men available for that work; H. M. Williamson, secretary of our state board of horticulture, chairman; C. I. Lewis, chief of division of horticulture, Corvallis; Leon D. Batchelor, experiment station, Riverside, California; A. A. Quarnberg, grower and experimenter, Vancouver, Washington; E. W. Mathews, extensive planter, Portland, and Charles L. McNary, planter, Salem. Mr. McNary told me yesterday that he had made a survey of thirty-five very fine trees, on blank cards similar to the one enclosed. We expect to have the record of at least 200 trees by the time of our convention. Only those that approach the standard wanted are listed.

To give the product of the walnut crop of the state would only be a wild guess. The system and machinery that we have for finding out how much we raise is only in embryo. The estimates reach all the way from 100,000 to 500,000 pounds. There is a good crop this year and the output for the market is growing rapidly. We need education more than we do growers. But we are learning.

I want to give you some facts of things that I find. Yesterday at the orchard of Alex Lafollette, State Senator from Marion county, and peach king of the Willamette Valley, I found seven-year-old walnut trees planted in rows among his peach trees, walnut trees planted sixteen feet apart! He said that his trees were full of little walnuts in the spring, but they all dropped off, and he did not think they would do well there. He said there were no catkins on the little trees, which accounts for the failure of his crop. This he did not know. And he did not know that the trees would produce the catkins in a year or so and remedy the failures. In the famous Dundee orchards I picked up handfuls of little fibrous roots, photo of which I sent you, that had been torn up by the plow and harrow when cultivating the walnut trees. Bales of these roots could be gathered up from the ground under the trees. The owner said that it did the trees good to treat them that

way. Another black walnut tree that I visited in a cultivated field of good deep, rich soil, I found walnut roots protruding from the plowed ground as far away as 108 feet from the tree. The tree was thirty or forty years old.

It would add greatly to the walnut industry of the future if the Forest Service would plant black walnuts in the hills and mountains between here and the coast. You know in that burnt timber section and various localities in the coast mountains there are many places where eight or ten nut trees to the acre would soon give a good account of themselves. If properly planted, in five or ten years they could be topgrafted to a good English variety and add greatly to the value of the public domain as well as the food products of the nation. We have no native walnuts, but almost every variety under the sun will grow here.

WESTERN WALNUT ASSOCIATION.

SEEDLING WALNUT TREE RECORD.

No..... Made..... 191..... by.....
Owner.....
P. O..... State..... Route.....
Exact location.....
NUT—Origin.....
Variety..... planted.....
TREE—Origin..... age now.....
Transplanted 19..... Dia. trunk.....
Height..... spread.....
DATES—of budding out.....
catkin blooms..... nut blooms.....
leaves fall..... nuts fall.....
in 1-lb. kernel wt..... oz. shell wt..... oz.
NUTS—Per tree..... lbs. In cluster..... in lb.....

round,.. oval,.. pointed,.. smooth,.... not well sealed.....
KERNEL—light, dark, not easily removed from shell. Tannin—little
excessive.
Tree vigor..... Blight..... per cent.....

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